To,
The Dean
University of Bern,
Switzerland

Subject: Application for the Post of **Tenure Track Assistant Professor in Inorganic Chemistry**

Dear Sir/Madam,

I am taking this opportunity to introduce myself as Dr. Susanta Hazra, currently working in the group of Prof. Armando J. L. Pombeiro, Técnico Lisboa, Portugal as an Inorganic scientist.

Recently I came across the advertisement (https://www.dcbp.unibe.ch/ueber_uns/team/offene_stellen/index_ger.html) for the post of a **Tenure Track Assistant Professor in Inorganic Chemistry** in the Department of Chemistry, Biochemistry and Pharmaceutical Sciences of the University of Bern. I believe that my research works and interests nicely match with the possible research topics (inorganic clusters, heterogeneous and heterogeneized catalysis, f-block systems, molecular surface functionalization, and photoactive inorganic chemistry) given for the post.

Indeed, my experimental working proficiency (please see attached publications) which includes synthesis of inorganic and organometallic systems (including air-sensitive mixed valence compounds, homo and heterometallic multinuclear systems and MOFs), contributed to more than sixty scientific articles including 4 reviews (*Coord. Chem. Rev.; Inorg. Chem.; Cryst. Growth Des.; Dalton Trans.; CrystEngComm; J. Chem. Phys.; RSC Advances, J. Inorg. Biochem.; Eur. J. Inorg. Chem.*, etc.) of several international journals of high repute. These works are related to all branches of inorganic chemistry including coordination, organometallic, bioinorganic and solid-state inorganic chemistry.

I also have experience in (co)supervising MSc students (at least 6) which would help in future to lead any group and being a core member of the organizing committee of three conferences herein Portugal, I have gained significant experiences to contribute to organize future conferences.

I believe that my long working experience in the group (https://cqe.tecnico.ulisboa.pt/CCC) of researchers with variety of age, sex, personality, mentality, capability, language and nationality have made very experienced in terms of sharing/discussing good practices, and developing strong and productive working relationships even among diverse peoples, which can eventually influence positively the reputation of the department and the university.

Experience of working for a sufficient period in a highly reputed institution in Portugal have helped me to developed/earned/gained/improved my skills, abilities and knowledges which would be very helpful for the post.

Finally, I would like to declare that with full dedication I can perform the following tasks if I get the opportunity to work as an **Assistant Professor in Inorganic Chemistry** in the Department of Chemistry, Biochemistry and Pharmaceutical Sciences of the University of Bern, Switzerland.

- Teaching the educational programs of the university especially within all branches of inorganic chemistry including coordination, organometallic, bioinorganic and solid-state inorganic chemistry.
- Establishing an independent research program, preferably within one or more of the following areas, coordination chemistry, organometallic chemistry, bioinorganic/medicinal inorganic chemistry and catalysis or similar, and with consideration of support for our present research infrastructure.
- Bioinorganic/medicinal inorganic chemistry and catalysis.
- Obtaining external funding for research independent and collaborative research programs.

- Mentor undergraduate and research students.
- Bridge-build between the various disciplines within the interdisciplinary department of Physics, Chemistry and Pharmacy.
- Initiate and participate in public, media, and local school outreach activities.
- Participate in the daily operation, administration and further development of the department's strategy and education programs.

In brief, my long working experience in prestigious institutes of different countries (India and Portugal) under versatile environments, motivated me to apply for the post of the **Tenure Track Assistant Professor in Inorganic Chemistry** in the Department of Chemistry, Biochemistry and Pharmaceutical Sciences of the University of Bern, Switzerland and I will be delighted to acquire experience of working in another beautiful European country such as Switzerland. For your kind perusal, I am attaching all documents requested (please see below). Some personal information along with my academic career, research carrier/profile, and list of referees are included in the CV.

Thanking you,

I'm looking forward to hearing from you!

With Best Regards

Susanta Hazra
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CURRICULUM VITAE

Dr. SUSANTA HAZRA Inorganic Scientist, Centro de Química Estrutural, Técnico Lisboa, Avenida Rovisco Pais, 1049-001 Lisbon, Portugal E-mail: h.susanta@gmail.com Mobile: +351-920181817 (Portugal)	
Research Interest: Synthesis, Characterizations and Applications of Inorganic and Organometallic Metal Complexes; X-ray Crystallography; Crystal Engineering and Supramolecular Chemistry; Catalysis; Magnetic Studies.	**Google Scholar** (**h-index**: 27): https://scholar.google.com/citations?user=-5yoR4sAAAAJ&hl=en **Orcid**: https://orcid.org/0000-0002-8101-2016
Date of Birth: 20th July 1982	**ResearcherID**: F-2295-2010
Nationality: Portuguese	**Scopus ID**: 23481611800

RESEARCH CARRIER

	Supervisor/Mentor	Institution	Year
AUXILIARY RESEARCHER	-	Centro de Química Estrutural, Instituto Superior Técnico, Lisbon, Portugal	2018 (Oct)-Continuing
POST-DOCTORAL FELLOW*	Prof. Armando J. L. Pombeiro	Centro de Química Estrutural, Instituto Superior Técnico, Lisbon, Portugal	2012 (August)–2018 (Sept)
POST-DOCTORAL FELLOW	Prof. Jana Roithova	Charles University in Prague, Faculty of Science, Prague, Czech Republic	2011 (July-October)
DOCTORAL FELLOW**	Prof. Sasankasekhar Mohanta	University of Calcutta, West Bengal, India	2006–2011

*FCT Postdoctoral Fellowship (2011), Portugal; **Qualified CSIR-NET (2005 & 2006), India.

ACADEMIC CARRIER

Degree	Subjects Includes	Board/University	Year	Percentage (%)	Class/ Division
M.Sc.	Inorganic (Special Paper), Physical, Organic and Industrial Chemistry	Vidyasagar University, West Bengal, India	2006	72.4	1st
B.Sc.	Chemistry (Honours), Physics, Mathematics, English	Vidyasagar University, West Bengal, India	2004	64.6	1st
Intermediate (10+2th)	Chemistry, Physics, Mathematics, Biology, English	W. B. C. H. S. E., West Bengal, India	2001	82.3	1st

EXPERIENCE AND EXPERTISE

1. **Experience**: More than 15 years in synthesis and characterization of inorganic compounds (of all type metal ions); More than 15 years in experimental and instrumental working in the field related to Crystallography; More than 10 years in Catalysis.

2. Expertise:

a. ***Experimental***: Syntheses of organic, inorganic, and organometallic compounds (discrete metal complexes to MOF) by several techniques including Schlenk line and hydrothermal processes.

b. ***Instrumental***: Single Crystal (Bruker-APEX II SMART CCD, Bruker AXS-KAPPA APEX II and Bruker D8 Quest) and Powder X-ray diffractometers (Bruker D8 ADVANCE); Electrochemical Analyzer (Bioanalytical Systems); FT-IR (Bruker and Perkin-Elmer); Absorption Spectrophotometer (Shimadzu); Spectrofluorimeter (Perkin-Elmer); NMR 300/400 MHz (Bruker); Thermogravimetric Analyzer (Metler).

c. ***Computations and Softwares***: Many programs related to crystallography (APEX, SHELXTL, XSHELL, PLATON, WINGX and Olex2) and a quantum chemistry program (ORCA).

Selected Publications (*principal corresponding author)

1	Copper tetrazole compounds: Structures, properties and applications Sajjadi, M.; Nasrollahzadeh, M.; Ghafuri, H.; Pombeiro, A. J. L.; **Hazra**, **S**.* *Coord. Chem. Rev*., 2024, *504*, 215604. DOI: 10.1016/j.ccr.2023.215604
2	Theoretical Analysis of $Cu^{II}\cdots Cl$ Semicoordination Bond in Supramolecular Heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ Cocrystals **Hazra**, **S**.*; Kuznetsov, M. L.; Paul, A.; Garazade, I. M.; Pombeiro, A. J. L. *Organometallics* 2023, *42*, 2672–2683. DOI: 10.1021/acs.organomet.3c00274
3	$M^{II}\cdots Cl$ Interaction Supported Heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ and $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ Complex Salts: Possibility of Ion-Pair-Assisted Tetrel Bonds. **Hazra**, **S**.*; Kuznetsov, M. L.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Cryst. Growth Des*., 2022, *22*, 341–355. DOI: 10.1021/acs.cgd.1c00977
4	Urea and thiourea based coordination polymers and metal-organic frameworks: Synthesis, structure and applications. Karmakar, A.; **Hazra**, **S**.; Pombeiro, A. J. L. *Coord. Chem. Rev*., 2022, *453*, 214314. DOI: 10.1016/j.ccr.2021.214314
5	Platinum and palladium complexes with tetrazole ligands: Synthesis, structure and applications. Nasrollahzadeh, M.; Sajjadi, M.; Ghafuri, H.; Bidgoli, N. S. S.; Pombeiro, A. J. L.; **Hazra**, **S**.* *Coord. Chem. Rev*., 2021, *446*, 214132. DOI: 10.1016/j.ccr.2021.214132
6	Alkoxo Bridged Heterobimetallic $Co^{III}Sn^{IV}$ Compounds with Face Shared Coordination Octahedra: Synthesis, Crystal Structure and Cyanosilylation Catalysis. **Hazra**, **S**.*; Karmakar, A.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *J. Organomet. Chem*., 2021, *949*, 121949. DOI: 10.1016/j.jorganchem.2021.121949
7	A Mixed Valence $Co^{II}Co^{III}_2$ Field-Supported Single Molecule Magnet: Solvent-Dependent Structural Variation. **Hazra**, **S**.*; Rajnák, C.; Titiš, J.; Guedes da Silva, M. F. C.; Boča, R.; Pombeiro, A. J. L. *Molecules*, 2021, *26*, 1060. DOI: 10.3390/molecules26041060
8	Metal–Tin Derivatives of Compartmental Schiff Bases: Synthesis, Structure and Application. **Hazra**, **S**.*; Mohanta, S. *Coord. Chem. Rev*., 2019, *395*, 1–24. DOI: 10.1016/j.ccr.2019.05.013
9	A tetranuclear diphenyltin(IV) complex and its catalytic activity in the aerobic Baeyer-Villiger oxidation of cyclohexanone. **Hazra**, **S**.*; Martins, N. M. R.; Mahmudov, K. T.; Zubkov, F. I.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *J. Organomet. Chem*., 2018, *867*, 193–200. DOI: 10.1016/j.jorganchem.2017.12.040
10	Flexibility and Lability of a Phenyl Ligand in Hetero-Organometallic 3d Metal-Sn(IV) Compounds and Their Catalytic Activity in Baeyer–Villiger Oxidation of Cyclohexanone. **Hazra**, **S**.*; Martins, N. M. R.; Kuznetsov, M. L.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Dalton Trans*., 2017, *46*, 13364–13375. DOI: 10.1039/C7DT02534C For Cover Page, see Dalton Trans., 2017, *46*, 13148–13148. DOI: 10.1039/C7DT90183F
11	Unfolding Biological Properties of a Versatile Dicopper(II) Precursor and its Two Mononuclear Copper(II) Derivatives.

	Paul, A.; **Hazra**, **S.***; Sharma, G.; Guedes da Silva, M. F. C.; Koch, B.; Pombeiro, A. J. L. *J. Inorg. Biochem.*, 2017, *174*, 25–36. DOI: 10.1016/j.jinorgbio.2017.05.013
12	N–H···O and N–H···Cl Supported 1D Chains of Heterobimetallic $Cu^{II}/Ni^{II}–Sn^{IV}$ Cocrystals. **Hazra**, **S.***; Meyrelles, R.; Charmier, A. J.; Rijo, P.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Dalton Trans.*, 2016, *45*, 17929–17938. DOI: 10.1039/C6DT03118H
13	Syntheses and Crystal Structures of Benzene-Sulfonate and -Carboxylate Copper Polymers and Their Application to Oxidation of Cyclohexane in Ionic Liquid under Mild Conditions. **Hazra**, **S.**; Ribeiro, A. P. C.; Guedes da Silva, M. F. C.; Nieto de Castro, C. A.; Pombeiro, A. J. L. *Dalton Trans.*, 2016, *45*, 13957–13968. DOI: 10.1039/C6DT02271E
14	Sulfonated Schiff Base Sn(IV) Complexes as Potential Anticancer Agents. **Hazra**, **S.***; Paul, A.; Sharma, G.; Koch, B.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *J. Inorg. Biochem.*, 2016, *162*, 83–95. DOI: 10.1016/j.jinorgbio.2016.06.008
15	Heterometallic Copper(II)–Tin(II/IV) Salts, Cocrystals and Salt Cocrystals: Selectivity and Structural Diversity Depending on Ligand Substitution and Metal Oxidation State. **Hazra**, **S.***; Chakraborty, P.; Mohanta, S. *Cryst. Growth Des.*, 2016, *16*, 3777–3790. DOI: 10.1021/acs.cgd.6b00321
16	Bis-phenoxido and Bis-acetato Bridged Heteronuclear $\{Co^{III}Dy^{III}\}$ Single Molecule Magnets with Two Slow Relaxation Branches. **Hazra**, **S.**; Titiš, J.; Valigura, D.; Boca, R.; Mohanta, S. *Dalton Trans.*, 2016, *45*, 7510–7520. DOI: 10.1039/C6DT00848H
17	Nanoporous Lanthanide Metal-Organic Frameworks as Efficient Heterogeneous Catalysts for Diastereoselective Henry Reaction. Karmakar, A.; **Hazra**, **S.**; Guedes da Silva, M. F. C.; Paul, A., Pombeiro, A. J. L. *CrystEngComm*, 2016, *18*, 1337–1349. DOI: 10.1039/C5CE01456E
18	Metal–Organic Frameworks with Pyridyl-Based Isophthalic Acid and Their Catalytic Applications in Microwave Assisted Peroxidative Oxidation of Alcohols and Henry Reaction. Karmakar, A.; Martins, L. M. D. R. S.; **Hazra**, **S.**; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Cryst. Growth Des.*, 2016, *16*, 1837–1849. DOI: 10.1021/acs.cgd.5b01178
19	Sulfonated Schiff Base Copper(II) Complexes as Efficient Catalysts in Alcohol Oxidation: Syntheses and Crystal Structures. **Hazra**, **S.***; Martins, L. M. D. R. S.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *RSC Adv.*, 2015, *5*, 90079–90088. DOI: 10.1039/C5RA19498A
20	1D Hacksaw Chain Bipyridine-Sulfonate Schiff Base-Dicopper(II) as a Host for Variable Solvent Guests. **Hazra**, **S.***; Guedes da Silva, M. F. C.; Karmakar, A.; Pombeiro, A. J. L. *RSC Adv.*, 2015, *5*, 28070–28079. DOI: 10.1039/C5RA03126E
21	Synthesis, Structure and Catalytic Application of Lead(II) Complexes in Cyanosilylation Reaction. Karmakar, A.; **Hazra**, **S.**; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Dalton Trans.*, 2015, *44*, 268–280. DOI: 10.1039/C4DT02316A
22	A Cyclic Tetranuclear Cuboid Type Copper(II) Complex Doubly Supported by Cyclcohexane-1,4-Dicarboxylate: Molecular and Supramolecular Structure and Cyclohexane Oxidation Activity. **Hazra**, **S.**; Mukherjee, S.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *RSC Adv.*, 2014, *4*, 48449–48457. DOI: 10.1039/C4RA06986B
23	Syntheses, Structures, Magnetic Properties, and Density Functional Theory Magneto-Structural Correlations of Bis(μ-phenoxo) and Bis(μ-phenoxo)-μ-acetate/Bis(μ-phenoxo)-bis(μ-acetate) Dinuclear $Fe^{III}Ni^{II}$ Compounds. **Hazra**, **S.**; Bhattacharya, S.; Singh, M. K.; Carrella, L.; Rentschler, E.; Weyhermueller, T.; Rajaraman, G.; Mohanta, S. *Inorg. Chem.*, 2013, *52*, 12881–12892. DOI: 10.1021/ic400345w
24	A Single Molecule Magnet with Spin 9/2 and Electron Delocalization Between Two Crystallographically Inequivalent Iron Sites in a Binuclear $[Fe_2]^V$ Macrocyclic Complex. **Hazra**, **S.**; Sasmal, S.; Fleck, M.; Grandjean, F.; Sougrati, M. T.; Ghosh, M.; Harris, T. D.; Long, G. J.; Mohanta, S. *J. Chem. Phys.*, 2011, *134*, 174507(1–13). (**Research Highlight**). DOI: 10.1063/1.3581028
25	Magnetic Exchange Interactions and Magneto-Structural Correlations in Heterobridged μ-Phenoxo-μ1,1-Azide Dinickel(II) Compounds: A Combined Experimental and Theoretical Exploration. Sasmal, S.; **Hazra**, **S.**; Kundu, P.; Dutta, S.; Rajaraman, G.; Sañudo, E. C.; Mohanta, S. *Inorg. Chem.*, 2011, *50*, 7257–7267. DOI: 10.1021/ic200833y
26	Syntheses and Crystal Structures of $Cu^{II}Bi^{III}$, $Cu^{II}Ba^{II}Cu^{II}$, $[Cu^{II}Pb^{II}]_2$, and Cocrystallized $(U^{VI}O_2)_2.4Cu^{II}$

	Complexes: Structural Diversity of the Coordination Compounds Derived from N,N'-Ethylenebis(3-ethoxysalicylaldiimine). **Hazra**, **S.**; Sasmal, S.; Nayak, M.; Sparkes, H. A.; Howard, J. A. K.; Mohanta, S. *CrystEngComm*, 2010, *12*, 470–477. DOI: 10.1039/B913408E
27	Magneto-Structural Correlation Studies and Theoretical Calculations of a Unique Family of Single End-to-End Azide Bridged Ni^{II}_4 Cyclic Clusters. Sasmal, S.; **Hazra**, **S.**; Kundu, P.; Majumder, S.; Aliaga-Alcalde, N.; Ruiz, E.; Mohanta, S. *Inorg. Chem.*, 2010, *49*, 9517–9526. DOI: 10.1021/ic101209m
28	Cocrystallized Dinuclear-Mononuclear $Cu^{II}_3Na^I$ and Double-Decker-Triple-Decker $Cu^{II}_5K^I_3$ Complexes Derived from N,N'-Ethylenebis(3-ethoxysalicylaldimine). **Hazra**, **S.**; Koner, R.; Nayak, M.; Sparkes, H. A.; Howard, J. A. K.; Mohanta, S. *Cryst. Growth Des.*, 2009, *9*, 3603–3608. DOI: 10.1021/cg900341r

Additional Experiences:

A. Conferences:

1. A Core Member of the **organizing committee** of the *XXII International Symposium on Homogeneous Catalysis (ISHC-2022)* held in Lisbon in 2022 (July 24-29) and **delivered a lecture** at *ISHC-2022*.
2. A Core Member of the **organizing committee** of the *1st International Conference on Noncovalent Interactions (ICNI)* held in Lisbon in 2019 (September 2-6) and **delivered a lecture** at *ICNI-2019*.
3. A Core Member of the **organizing committee** of the *7th EuCheMS Conference on Nitrogen Ligands* held in Lisbon in 2018 (September 4-7).
4. Attended at the 3rd meeting of Colégio de Química da Universidade de Lisboa, "*A Química na Investigação da ULisboa*" at Universidade de Lisboa, Lisbon, Portugal, 27-28 June, 2018.
5. Attended "*Ciência 2018 - Encontro com a Ciência e Tecnologia em Portugal*" at Centro de Congressos de Lisboa, Portugal, 2-4 July, 2018.
6. Delivered a lecture at the 1st meeting of Colégio de Química da Universidade de Lisboa, "*A Química na Investigação da ULisboa*" at Universidade de Lisboa, Lisbon, Portugal, 20-21 July, 2017.
7. Attended "*Ciência 2017 - Encontro com a Ciência e Tecnologia em Portugal*" at Centro de Congressos de Lisboa, Portugal, 3-5 July, 2017.
8. A Core Member of the **organizing committee** of the "*XXV International Conference on Organometallic Chemistry (XXV-ICOMC)*", Lisbon, Portugal, 2-7 September, 2012.
9. Presented poster in "*Modern Trends in Inorganic Chemistry (MTIC-XIII)*" at IISc, Bangalore, India, 7-10 December, 2009.
10. Attended "*International Conference on Structure and Dynamics: From Micro to Macro*" at Kolkata, India, 14-16 December, 2006.

B. As a Reviewer of the following Journals:

Coordination Chemistry Reviews, Chemistry—A European Journal, Chemical Communications, ACS Applied Materials & Interfaces, Inorganic Chemistry, Crystal Growth & Design, ACS Omega, Dalton Transactions, Applied Catalysis A: General, CrystEngComm, RSC Advances, ChemistrySelect, Arabian Journal of Chemistry, Inorganica Chimica Acta, Inorganic Chemistry Communications, Journal of Molecular Structure, Polyhedron, Natural Product Research, Applied Biochemistry and Biotechnology, Anti-Cancer Agents in Medicinal Chemistry, *etc*.

List of Referees:

1. Prof. Sasankasekhar Mohanta
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2. Prof. Armando J. L. Pombeiro
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3. Prof. Maxim L. Kuznetsov,
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Lisboa, Portugal
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Full List of Publications

67	**S. Hazra** and A. J. L. Pombeiro, *Noncovalent Interactions in Nitro-Aldol (Henry) Reaction* in "Noncovalent Interactions in Catalysis", K. T. Mahmudov, M. N. Kopylovich, M. F. C. Guedes da Silva and A. J. L. Pombeiro (Eds), Chapter 11, 232–252, 2019. ISBN: 978-1-78801-468-7. (Book Chapter)
66	Copper tetrazole compounds: Structures, properties and applications Sajjadi, M.; Nasrollahzadeh, M.; Ghafuri, H.; Pombeiro, A. J. L.; **Hazra**, **S**. *Coord. Chem. Rev.*, 2024, *504*, 215604. DOI: 10.1016/j.ccr.2023.215604
65	Bis(3-aminobenzenesulfonato-*N*)-diaqua-bis(*N,N'*-dimethylformamide-*O*)-copper(II). **Hazra, S.**; Karmakar, A. *Molbank* 2023, *2023*, M1658. DOI: 10.3390/M1658
64	Theoretical Analysis of $Cu^{II}\cdots Cl$ Semicoordination Bond in Supramolecular Heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ Cocrystals **Hazra**, **S.**; Kuznetsov, M. L.; Paul, A.; Garazade, I. M.; Pombeiro, A. J. L. *Organometallics* 2023, *42*, 2672–2683. DOI: 10.1021/acs.organomet.3c00274
63	Electrocatalytic Behavior of an Amide Functionalized Mn(II) Coordination Polymer on ORR, OER and HER. Paul, A.; Radinović, K.; **Hazra**, **S.**; Mladenović, D.; Šljukić, B.; Khan, R. A.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Molecules*, 2022, *27*, 7323 DOI: 10.3390/molecules27217323
62	$M^{II}\cdots Cl$ Interaction Supported Heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ and $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ Complex Salts: Possibility of Ion-Pair-Assisted Tetrel Bonds. **Hazra**, **S.**; Kuznetsov, M. L.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Cryst. Growth Des.*, 2022, *22*, 341–355. DOI: 10.1021/acs.cgd.1c00977
61	Urea and thiourea based coordination polymers and metal-organic frameworks: Synthesis, structure and applications. Karmakar, A.; **Hazra**, **S.**; Pombeiro, A. J. L. *Coord. Chem. Rev.*, 2022, *453*, 214314. DOI: 10.1016/j.ccr.2021.214314
60	Platinum and palladium complexes with tetrazole ligands: Synthesis, structure and applications. Nasrollahzadeh, M.; Sajjadi, M.; Ghafuri, H.; Bidgoli, N. S. S.; Pombeiro, A. J. L.; **Hazra**, **S**. *Coord. Chem. Rev.*, 2021, *446*, 214132. DOI: 10.1016/j.ccr.2021.214132
59	Alkoxo Bridged Heterobimetallic $Co^{III}Sn^{IV}$ Compounds with Face Shared Coordination Octahedra: Synthesis, Crystal Structure and Cyanosilylation Catalysis. **Hazra**, **S.**; Karmakar, A.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *J. Organomet. Chem.*, 2021, *949*, 121949. DOI: 10.1016/j.jorganchem.2021.121949
58	A Mixed Valence $Co^{II}Co^{III}_2$ Field-Supported Single Molecule Magnet: Solvent-Dependent Structural Variation. **Hazra**, **S.**; Rajnák, C.; Titiš, J.; Guedes da Silva, M. F. C.; Boča, R.; Pombeiro, A. J. L. *Molecules*, 2021, *26*, 1060 DOI: 10.3390/molecules26041060
57	Biological Evaluation of Azo-and Imino-Based Carboxylate Triphenyltin(IV) Compounds Paul, A.; **Hazra**, **S.**; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Eur. J. Inorg. Chem.*, 2020, *2020*, 930–941. DOI: 10.1002/ejic.201901177
56	Metal–Tin Derivatives of Compartmental Schiff Bases: Synthesis, Structure and Application. **Hazra**, **S.**; Mohanta, S.

	Coord. Chem. Rev., 2019, *395*, 1–24. DOI: 10.1016/j.ccr.2019.05.013
55	Syntheses, Structures, and Catalytic Hydrocarbon Oxidation Properties of *N*-Heterocycle-Sulfonated Schiff Base Copper(II) Complexes. **Hazra**, **S.**; Rocha, B. G. M.; Guedes da Silva, M. F. C.; Karmakar, A.; Pombeiro, A. J. L. *Inorganics*, 2019, *7*, 17 (1–19). DOI: 10.3390/inorganics7020017
54	Packing polymorphism in 3-amino-2-pyrazinecarboxylate based tin(II) complexes and their catalytic activity towards cyanosilylation of aldehydes. Karmakar, A.; **Hazra**, **S.**; Rúbio, G. M. D. M.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *New J. Chem.*, 2018, *42*, 17513–17523 DOI: 10.1039/C8NJ03805H
53	The Henry reaction catalyzed by Ni^{II} and Cu^{II} complexes bearing arylhydrazones of acetoacetanilide. Gurbanov, A. V.; **Hazra**, **S.**; Maharramov, A. M.; Zubkov, F. I.; Guseinov, F. I.; Pombeiro, A. J. L. *J. Organomet. Chem.*, 2018, *869*, 48–53. DOI: 10.1016/j.jorganchem.2018.05.025
52	A tetranuclear diphenyltin(IV) complex and its catalytic activity in the aerobic Baeyer-Villiger oxidation of cyclohexanone. **Hazra**, **S.**; Martins, N. M. R.; Mahmudov, K. T.; Zubkov, F. I.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *J. Organomet. Chem.*, 2018, *867*, 193–200. DOI: 10.1016/j.jorganchem.2017.12.040
51	Flexibility and Lability of a Phenyl Ligand in Hetero-Organometallic 3d Metal-Sn(IV) Compounds and Their Catalytic Activity in Baeyer–Villiger Oxidation of Cyclohexanone. **Hazra**, **S.**; Martins, N. M. R.; Kuznetsov, M. L.; Guedes da Silva, M. F. C.; Pombeiro, A. J. L. *Dalton Trans.*, 2017, *46*, 13364–13375. DOI: 10.1039/C7DT02534C For **Cover Page**, see Dalton Trans., 2017, *46*, 13148–13148. DOI: 10.1039/C7DT90183F
50	Unfolding Biological Properties of a Versatile Dicopper(II) Precursor and its Two Mononuclear Copper(II) Derivatives. Paul, A.; **Hazra**, **S.**; Sharma, G.; Guedes da Silva, M. F. C.; Koch, B.; Pombeiro, A. J. L. *J. Inorg. Biochem.*, 2017, *174*, 25–36. DOI: 10.1016/j.jinorgbio.2017.05.013
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Theoretical Analysis of Cu^{II}···Cl Semicoordination Bond in Supramolecular Heterometallic {Cu^{II}}{Sn^{IV}} Cocrystals

Susanta Hazra,* Maxim L. Kuznetsov,* Anup Paul, Ismayil M. Garazade, and Armando J. L. Pombeiro*

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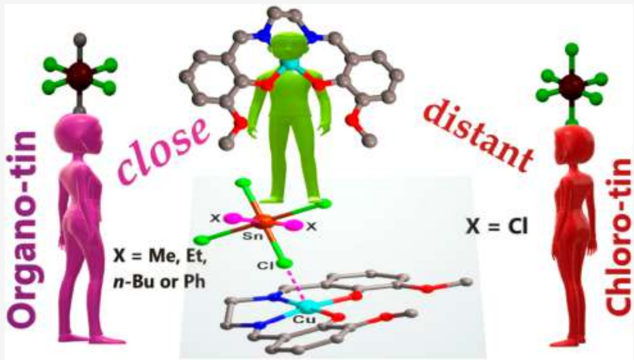

ABSTRACT: Detailed theoretical analysis of the Cu^{II}···Cl interaction nature in a series of three-component supramolecular heterometallic {Cu^{II}}{Sn^{IV}} cocrystals with the general formula of $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ [H_2L = *N,N′*-ethylenebis(3-*O*R-salicylaldimine), R = Me (H_2L^1) or Et (H_2L^2); ED = 1,2-ethylenediamine; X = Me, Et, Bu^n, or Ph; n = 2 or 0] was performed. The neutral metal complex [CuL] and the $[SnX_nCl_{6-n}]^{2-}$ anion in their crystal structures (**1**−**8**) interact via a Cu^{II}···Cl interaction, along with H-bonds. Structural comparison reveals that the Cu^{II}···Cl interaction (2.886−3.247 Å) in $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_2Cl_4]^{2-}$ cocrystals (**1**−**6**, type **A**, X = Me, Et, Bu^n, or Ph) is shorter than that (3.407−3.454 Å) in the inorganic tin(IV) containing versions $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnCl_6]^{2-}$ (**7** and **8**, type **B**), despite both types (**A** and **B**) having similar molecular or supramolecular arrangements. Interaction energies among the components (cation, neutral, and anion) present a triangular energy framework. An attempt has been made to find a correlation between the interaction energies and the concerned noncovalent interactions. Further, the nature of Cu^{II}···Cl interaction in **1** and **7** (as representatives of types **A** and **B**, respectively) was understood as being a semicoordination bond by DFT calculations with application of the QTAIM, ELF, IGM, EDD, and CDF analyses and IBSI calculation. The ranges of the Cu^{II}···Cl interactions in these cocrystals and in other related systems are also discussed. Preliminary catalytic studies on Strecker type cyanation revealed that the catalytic ability of the chloro-tin(IV) precursor, i.e., $[Sn(Me)_2Cl_2]$ or $[SnCl_4]\cdot 5H_2O$, was greatly reduced upon conversion into the corresponding dianion, i.e., $[Sn(Me)_2Cl_4]^{2-}$ or $[SnCl_6]^{2-}$ in **1** or **7**, respectively.

■ INTRODUCTION

Noncovalent interactions have been a fascinating topic in crystal engineering and supramolecular chemistry.[1−7] If the distance between two nonbonded atoms (or ions) fall within the sum of their van der Waals (VDW) radii,[8] then the interaction between the atoms is usually considered as a noncovalent one. In recent years, such interactions are designated with a particular bond name depending on the nature of the atoms involved, for example, triel,[9] tetrel,[10,11] pnictogen,[6,12] chalcogen,[13,14] halogen,[15] regium,[16] spodium,[17] etc.

Lower complicacy (less number of atoms and lower formula weight) has allowed theoretical investigations on noncovalent interactions in the crystal structures of organic species,[18] but their existence in metal containing systems has also been reviewed.[6,7,19−21] However, such interactions in multicomponent metal complexes have been less investigated[11] although such systems are often stabilized by noncovalent interactions and can be easily synthesized.[22−25] For example, the $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ cocrystals [L = *N,N′*-ethylenebis(3-*O*R-salicylaldimine); ED = 1,2-ethylenediamine; X = Me, Et, Bu^n, or Ph; n = 2 or 0; R = Me or Et] were synthesized following either the metalloligand (reaction of [CuL] and $[SnX_2Cl_2]$) or the self-assembly (H_2L, $CuCl_2\cdot 2H_2O$ and $[SnCl_4]\cdot 5H_2O$) synthetic approaches under open atmospheric conditions.[22,23] These cocrystals are mainly aggregated by noncovalent interactions of types N−H···O, N−H···Cl, and Cu···Cl. Although the role of the H-bonding interactions on the stabilization of such cocrystals is fairly understood, the nature of the Cu···Cl interactions remains to be cleared. Theoretical calculations, which are important methods to investigate the nature of any important noncovalent interactions in crystal structures,[11,18,26] could be one of the best logical ways to understand the nature of a Cu···Cl interaction. However, due to the high number of atoms along with the high formula weight of such inorganic multinuclear systems, the theoretical investigation is expected to be complicated. Intricacy further increases as such species are

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salts-cocrystals and possess more than one component of different charge. For instance, $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ cocrystals consist of at least three different components, a cation, an anion, and a neutral species (Figure 1).[22,23]

Figure 1. Type of components (cation, anion, and neutral) present in heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ cocrystals.

On the other hand, the possibility of existence of different types of weak bonds, such as tetrel or halogen, is expected in compounds derived from the chloro-complexes $[SnX_2Cl_2]$ (X = Cl, Me, Et, Bu, Ph, or other organic group).[25] We have recently demonstrated such a possibility in our earlier study concerning the ion pair assisted Sn···Cl tetrel bonds in heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ and $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ complex salts.[11] The presence of an additional Ni^{II}···Cl interaction in these systems inspired us to look at the crystal structures of other cocrystals or salts. Earlier, we reported several three component $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ cocrystals [X = Me, Et, Bu^n, or Ph; n = 2 or 0] which contain Cu^{II}···Cl interaction of different lengths.[22,23] Herein, we present comprehensive theoretical calculations and characterization of such interactions in these cocrystals (Scheme 1 and Table 1). For a better presentation and understanding, the organotin(IV) containing systems $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_2Cl_4]^{2-}$ cocrystals (**1–6**, X = Me, Et, Bu^n, or Ph) are labeled as type **A**, whereas the inorganic tin(IV) containing versions $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnCl_6]^{2-}$ (**7** and **8**) are indicated as type **B**.

Scheme 1. H-Bonds (N−H···O and N−H···Cl) and Cu···Cl Interactions in Supramolecular Heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ Cocrystals $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ (1–8) [R = Me (L = L^1) or Et (L = L^2); ED = 1,2-Ethylenediamine; X = Me, Et, Bu^n, or Ph; n = 2 (A) or 0 (B)][22,23]

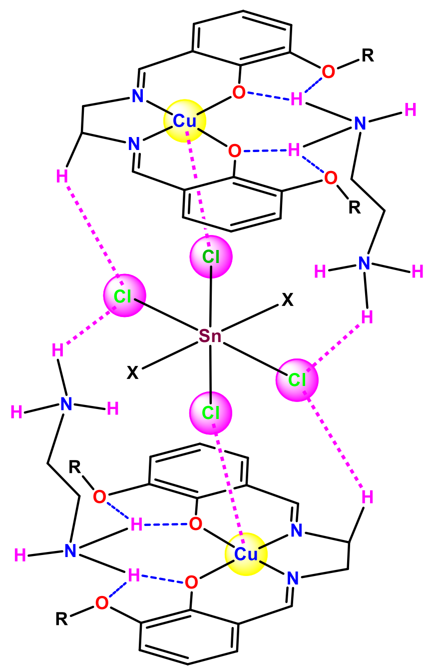

Table 1. List of Compounds $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ (1−8) Selected in This Work

R	X	compounds
Me	Me	$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Me)_2Cl_4]^{2-}$ (**1**)
Me	Et	$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Et)_2Cl_4]^{2-}$ (**2**)
Me	Bu^n	$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Bu^n)_2Cl_4]^{2-}$ (**3**)
Me	Ph	$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}$ (**4**)
Me	Ph	$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}\cdot$2MeOH (**5**)
Et	Ph	$(H_2ED)^{2+}\cdot 2[Cu^{II}L^2]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}$ (**6**)
Me	Cl	$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}Cl_6]^{2-}$ (**7**)
Et	Cl	$(H_2ED)^{2+}\cdot 2[Cu^{II}L^2]\cdot[Sn^{IV}Cl_6]^{2-}\cdot$2MeOH (**8**)

■ RESULTS AND DISCUSSION

Hirshfeld Surface Analysis and Preliminary Interaction Energy Calculations. The crystal structures of the selected series of the cocrystals with the formula $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ [X = Me, Et, Bu^n, or Ph; n = 2 (**A**) or 0 (**B**)] are similar as already discussed in our earlier works.[22,23] Two closest neutral [CuL] complexes are bridged by one $(H_2ED)^{2+}$, supported by N−H···O hydrogen bonds, forming a supramolecular $\{[CuL]\cdots(H_2ED)^{2+}\cdots[CuL]\}$ dimer (Figure S1). As presented in Figure 2, such dimers are interconnected by the $[SnX_nCl_{6-n}]^{2-}$ moiety through Cu···Cl interactions and two types of H-bonds (N−H···Cl and C−H···Cl). One [CuL] unit (presented in the wireframe model in Figure 2) of the supramolecular dimer is not directly connected to the $[SnX_nCl_{6-n}]^{2-}$ species but bridged through a NH_3^+ moiety via O(CuL)···H−N and N−H···Cl−Sn hydrogen bonds.

Although the structural arrangements of the multicomponent cocrystals are quite similar, the Cu···Cl interactions in their crystal structures are of different lengths (2.908 and 3.407 Å for **1** and **7**, respectively). Other noncovalent interactions, such as H-bonds (N−H···O and N−H···Cl) and $\pi\cdots\pi$ interactions, and their role on the crystal packing were already discussed, but the nature of Cu···Cl interaction was not understood properly. Even the presence of any Cu···Cl interaction in the type **B** cocrystals (**7** and **8**) was not recognized since the Cu···Cl contact (∼3.4 Å) is longer than the VDW distance (3.15 Å). However, distances of such interactions in ion pairs or other ionic systems (*e.g.*, adducts) may deviate from the standard values or the common bond-length and bond-strength (BLBS) appraisal.[11,27,28] Thus, to find out any missing interactions and to understand any possible relationships between the relevant noncovalent interactions and stability of these cocrystals, we have analyzed their Hirshfeld surfaces and the interaction energies among the components by The CrystalExplorer17 (CE).[29,30]

The Hirshfeld surface areas for various noncovalent interactions in **1–8** are included in Table 2. The Hirshfeld surface plots of **1** (type **A**) and **7** (type **B**) are presented in Figure 3 while those of the remaining cocrystals (**2–6** and **8**) are shown in Figure S2. The 2D fingerprints for three important intermolecular interactions (Cu···Cl, H···Cl and H···O) in the crystal structure of **1** and **7** are presented in Figure 4

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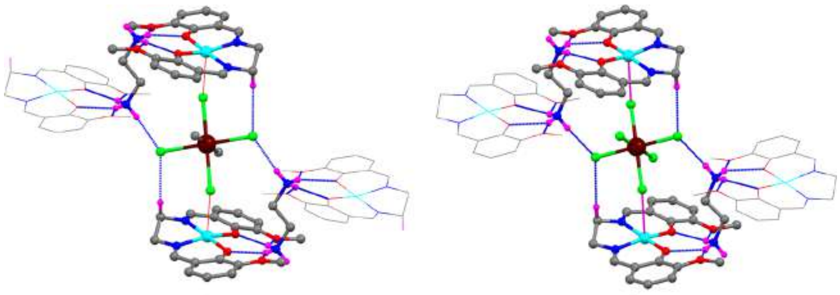

Figure 2. Comparative presentation of all components {dication $(H_2ED)^{2+}$, neutral [CuL], and dianion $[Sn(Me)_2Cl_4]^{2-}$ or $[SnCl_6]^{2-}$} around the Sn(IV) unit and interactions among them in **1** and **7** (as representatives for types **A** and **B**, respectively). Color code: cyan, copper; brown, tin; gray, carbon; green, chlorine; red, oxygen; blue, nitrogen; pink, hydrogen.

Table 2. Hirshfeld Surface Area (%) Involved in Various Noncovalent Interactions in 1−8

compounds	H···Cl	H···O	H···C	H···H	C···C	Cu···Cl
1	18.7	8.9	14.7	51.1	2.1	0.8
2	13.2	3.9	13.3	59.6	3.5	0.9
3	13.2	6.8	11.4	61.7	1.0	0.6
4	14.5	6.3	21.6	51.4	0.8	0.5
5	13.6	8.4	21.9	49.7	0.7	0.5
6	14.1	5.2	27.5	43.4	2.1	0.5
7	35.0	9.3	14.0	34.7	2.1	0.6
8	31.9	10.9	11.7	39.7	1.8	0.3

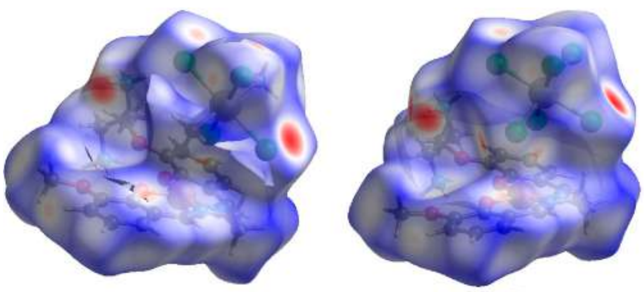

Figure 3. Hirshfeld surface plots of **1** and **7** as representatives for types **A** and **B**, respectively. For others, see Figure S2.

while those of others (**2**−**6** and **8**) can be found in Figure S3. As presented in Table 2, calculations for all cocrystals (**1**−**8**) reveal valuable interactions such as H···Cl, H···O, H···C, C···C, and Cu···Cl. In terms of occupied surface area (34.7−61.7%), the highest contributing interactions are H···H contacts. They concern molecular van der Waals (VDW) interactions, and the slightly lower values for the type **B** cocrystals (**7** and **8**) can be related to the lower number of H atoms present in their crystal structure. Indeed, the highest value (61.7%) was obtained for **3**, which contains two *n*-butyl groups with the highest number of H atoms among all organic groups (Me, Et, Bu^n, or Ph) discussed herein. The second important contribution in the Hirshfeld surface area is observed for the H···Cl contact, which corresponds to the C−H···Cl or N−H···Cl hydrogen bonds. Comparatively lower values (13.2−18.7%) of surface area for the type **A** cocrystals (**1**−**6**) than those (31.9−35.0%) in the type **B** can be accounted for the availability higher number of chloride atoms in latter. The encapsulation of the ammonium moiety (NH_3^+) of the ethylenediammonium cation $(H_2ED)^{2+}$ in the O_4 cavity of the mononuclear complex [CuL] results in common N−H···O hydrogen bonds in all cocrystals (**1**−**8**) and the corresponding surface area appears as the H···O contact (Table 2).

One of the noticeable contributions (11.4−27.5%) in the Hirshfeld surface area of all cocrystals (**1**−**8**) is from the C···H contact, indicating the possibility of C−H···C hydrogen bonds or C−H···π interactions, which were not previously discussed. The higher values (21.6−27.5%) for organo-Ph containing systems **4**−**6** support the C−H···π interactions. As shown in Figure S4, the phenyl groups of the $[SnPh_2Cl_4]^{2-}$ moiety in **6** interact with the ethoxy substituents of vicinal mononuclear $[CuL^2]$ complexes via four C−H···π interactions (C···π = 3.6−3.8 Å).

The calculated surface areas (0.3−0.9%, Table 2) for Cu···Cl interactions are not large but indicate their presence in the crystal structure of **1**−**8**. The lowest value (0.3%) obtained for **8** may be related to the longer Cu···Cl contacts.

Since the crystal structures of **1**−**8** are analogous and the geometrical features around the metal centers are also equivalent, the obtained surface areas are also quite comparative. Due to the higher number of halogens in **7** and **8**, the related surface areas are higher. To understand the cooperative or competitive role of these weak interactions (N−H···O, N−H···Cl or Cu···Cl), we have further calculated interactions energies among the components by the CrystalExplorer program[29] and the corresponding outcomes are shown in Tables 3−5, respectively. Figure 5 shows the interaction among the components in a cocrystal (**1**−**8**). The strength of the interaction is proportional to the radius of the blue cylinders, i.e., the stronger the interaction, the thicker the cylinder. It is evident that the interaction between the ionic species {$(H_2ED)^{2+}$ and $[Sn^{IV}]^{2-}$} is the strongest, while that between the neutral complex $[Cu^{II}L]$ and the anion $[Sn^{IV}]^{2-}$ is the weakest one. The overall solid-state stability of these cocrystals (**1**−**8**) is expected to be dependent on this triangular energy framework.

As shown in Table 3, the interaction energy between the ethylene diammonium cation $(H_2ED)^{2+}$ and the neutral metal complex $[Cu^{II}L]$ is in the range between −129.64 to −139.36 kcal/mol. Negative values suggest attractive interactions. These two moieties are associated via N−H···O H-bonds; therefore, stronger H-bonds can be expected for a higher interaction energy. However, the geometrical parameters of N−H···O H-bonds in types **A** and **B**, particularly the D···A (N···O) distances, are comparable, which are further confirmed by the calculated similar energy parameters. It may be concluded that this interaction and the concerning H-bonds in types **A** and **B** are not dependent on the substituents (organic in **1**−**6** or

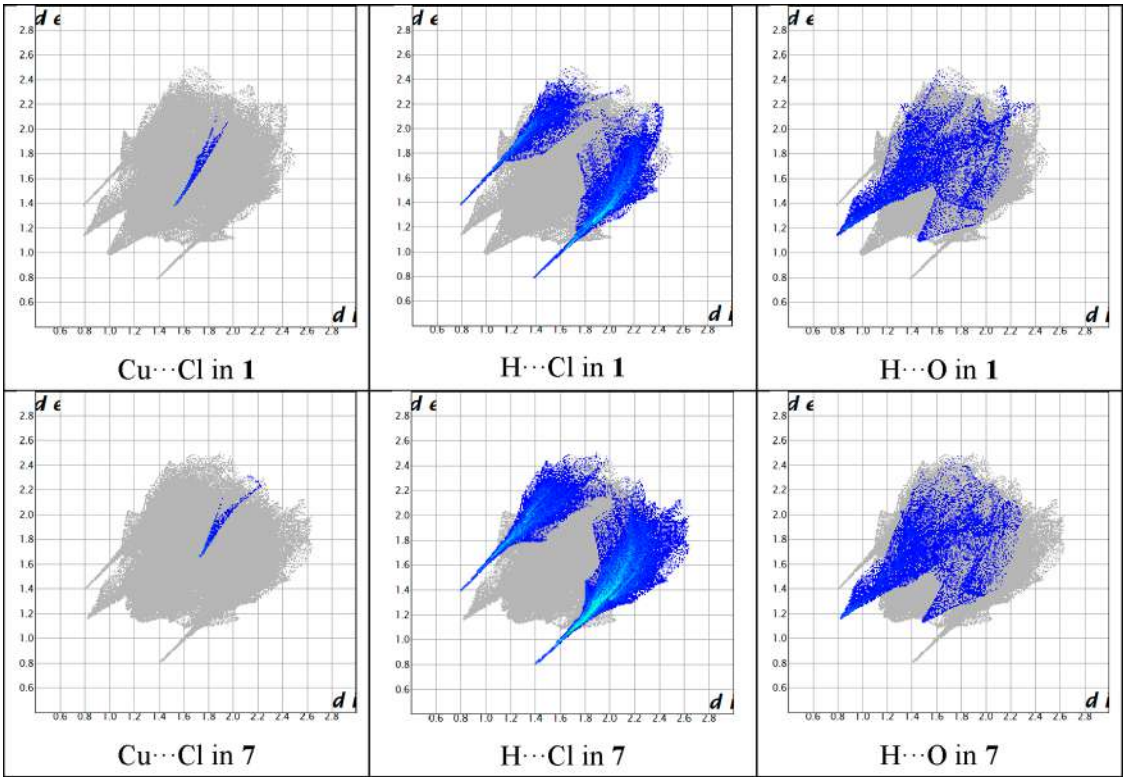

Figure 4. 2D fingerprints for Cu···Cl, H···Cl and H···O contacts in **1** and **7**. For the remaining systems (**2–6** and **8**), see Figure S3.

Table 3. Calculated Interaction Energy (kcal/mol) between $(H_2ED)^{2+}$ and $[Cu^{II}L]$ and Their Comparison with the Geometry of the Concerning N–H···O Bonds in 1–8

	E_{int}^{CE} (kcal/mol)	N–H···O	
		N···O (Å)	<N–H···O (deg)
	Type **A**		
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot [Sn^{IV}(Me)_2Cl_4]^{2-}$ (**1**)	−135.42	2.805–2.896	137.39–145.34
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot [Sn^{IV}(Et)_2Cl_4]^{2-}$ (**2**)	−133.80	2.797–2.917	131.32–150.68
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot [Sn^{IV}(Bu^n)_2Cl_4]^{2-}$ (**3**)	−133.27	2.786–2.888	129.04–155.36
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot [Sn^{IV}(Ph)_2Cl_4]^{2-}$ (**4**)	−136.28	2.776–2.859	121.97–150.19
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot [Sn^{IV}(Ph)_2Cl_4]^{2-}\cdot$ 2MeOH (**5**)	−129.64	2.762–2.914	121.63–143.16
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^2]\cdot [Sn^{IV}(Ph)_2Cl_4]^{2-}$ (**6**)	−139.36	2.761–2.914	115.00–164.77
	Type **B**		
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot [Sn^{IV}Cl_6]^{2-}$ (**7**)	−130.93	2.827–2.921	130.34–150.85
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^2]\cdot [Sn^{IV}Cl_6]^{2-}\cdot$2MeOH (**8**)	−138.86	2.759–2.880	134.73–146.53

halide in **7–8**) either at Sn^{IV} or at the ligand (OMe in L^1 or OEt in L^2).

Interaction energies between the ionic moieties $\{(H_2ED)^{2+}$ and $[SnX_nCl_{6-n}]^{2-}\}$ in **1–8** are very much similar, lying in the range of −253.92 to −275.12 kcal/mol (Table 4) but are indicative of stronger interactions. As these two ionic species are attracted to each other by a N–H···Cl H-bond, the strength of the H-bond is expected to be dependent on the interaction energy values. Each of the three organo-Ph containing systems (**4–6**) has a pair of N–H···Cl interactions (Figure S5), and the concerning interaction energies are in the slightly higher range (−262.76 to −275.12 kcal/mol) while the remaining others participate only in one such H-bond (Figure 2). However, comparable geometrical features (or strength) of the H-bond in all cocrystals (**1–8**) indicate that the nature and the strength of the interaction between the ionic species in these systems are very similar, agreeing satisfactorily with the calculated interaction energy values.

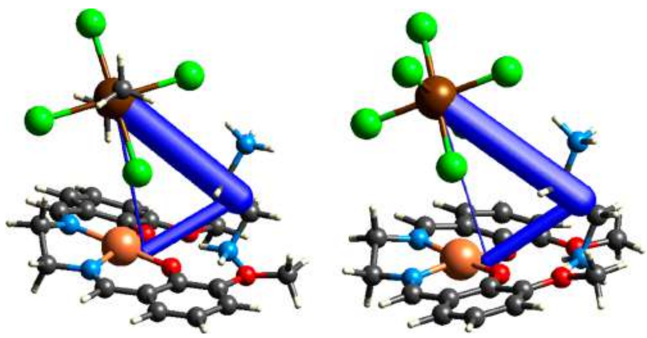

Figure 5. Triangular interaction energy frameworks in **1** and **7** are shown as representatives for types **A** and **B**, respectively.

As shown in Figure 5, the interaction between the neutral species $[Cu^{II}L]$ ($L = L^1$ or L^2) and the anion $[SnX_nCl_{6-n}]^{2-}$ [X = Me, Et, Bu^n, or Ph; n = 2 (**A**) or 0 (**B**)] is the weakest one among all as indicated by the calculated interactions energy which lies in the range of −26.10 to −34.30 kcal/mol (Table 5). These two species are interconnected by one weak Cu···Cl and one C–H···Cl interactions. It would be reasonable to predict that the interaction energy between the neutral and anionic components would be higher for the stronger Cu···Cl and C–H···Cl interactions. However, as can be seen in Table

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Table 4. Calculated Interaction Energy (kcal/mol) between $(H_2ED)^{2+}$ and $[SnX_nCl_{6-n}]^{2-}$ [X = Me, Et, Bu^n, or Ph; n = 2 (A) or 0 (B)] and Their Comparison with the Geometry of the Concerning N−H···Cl Bonds in 1−8

		N−H···Cl	
	E_{int}^{CE} (kcal/mol)	N···Cl (Å)	<N−H···Cl (deg)
Type **A**			
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Me)_2Cl_4]^{2-}$ (**1**)	−264.17	3.173	169.94
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Et)_2Cl_4]^{2-}$ (**2**)	−262.07	3.226	159.82
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Bu^n)_2Cl_4]^{2-}$ (**3**)	−260.73	3.228	152.04
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}$ (**4**)	−262.76	3.345, 3.472	152.21, 120.60
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}\cdot$2MeOH (**5**)	−275.12	3.327, 3.403	130.78, 138.13
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^2]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}$ (**6**)	−266.97	3.332, 3.459	144.58, 127.33
Type **B**			
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}Cl_6]^{2-}$ (**7**)	−260.13	3.201	171.98
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^2]\cdot[Sn^{IV}Cl_6]^{2-}\cdot$2MeOH (**8**)	−253.92	3.260	157.45

Table 5. Calculated Interaction Energy (kcal/mol) between $[Cu^{II}L]$ (L = L^1 or L^2) and $[SnX_nCl_{6-n}]^{2-}$ {X = Me, Et, Bu^n, or Ph; n = 2 (A) or 0 (B)} and Their Comparison with the Geometry of the Concerning Cu···Cl Interaction in 1−8

			C−H···Cl	
	E_{int}^{CE} (kcal/mol)	Cu···Cl (Å)	C···Cl (Å)	<C−H···Cl (deg)
Type **A**				
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Me)_2Cl_4]^{2-}$ (**1**)	−34.20	**2.908**	3.614	148.48
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Et)_2Cl_4]^{2-}$ (**2**)	−34.30	**2.946, 2.996**	3.796, 3.881	146.91, 174.23
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Bu^n)_2Cl_4]^{2-}$ (**3**)	−33.46	**2.886**	3.765	161.02
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}$ (**4**)	−26.15	**3.127**	3.667	119.24
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}\cdot$2MeOH (**5**)	−29.02	**3.071**	3.831	160.23
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^2]\cdot[Sn^{IV}(Ph)_2Cl_4]^{2-}$ (**6**)	−26.96	**3.247**	3.729, 3.900	120.94, 159.81
Type **B**				
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}Cl_6]^{2-}$ (**7**)	−27.39	**3.407**	3.697	154.53
$(H_2ED)^{2+}\cdot 2[Cu^{II}L^2]\cdot[Sn^{IV}Cl_6]^{2-}\cdot$2MeOH (**8**)	−26.10	**3.454**	3.836	125.39

5, similar geometrical parameters (mainly D···A distances) of the C−H···Cl interactions are indicative of their negligible impacts on the interaction energy values, and it appears that these interactions are independent of the substituents at Sn^{IV} or at the ligand.

The interaction energy between $[Cu^{II}L]$ and $[SnX_nCl_{6-n}]^{2-}$ appears to be dependent on the Cu···Cl interaction. Figure 6 presents the graph of the interaction energy vs. Cu···Cl distance, presenting the trend of the higher the interaction energy, the shorter the contact. In fact, longer Cu···Cl distances are observed in the crystal structure of type **B** cocrystals (**7** and **8**), which is in fair concordance with the

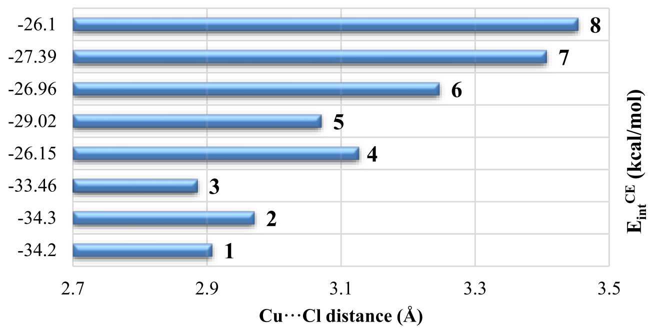

Figure 6. Interaction energy and Cu···Cl distance in cocrystals **1−8**. An average value (2.971 Å) of two Cu···Cl distances (2.946 and 2.996 Å) in **2** is considered. Numerical order of the compounds (**1−8**) from the bottom to top is maintained, and the energy values are indicated on the left side of each bar.

calculated lower energies. It can be mentioned that these two cocrystals have a higher number (6) of Cl-atoms in the inorganic tin(IV) moieties $[Sn^{IV}Cl_6]^{2-}$, compared to the four Cl-atoms in the organotin(IV) species in the type **A** systems. Higher interaction energies are obtained for compounds **1** and **3** as expected from the corresponding shorter Cu···Cl distances (Table 5). Although both Cu···Cl distances in **2** appear to be slightly longer than that in **3**, the presence of two Cu···Cl interactions in its crystal structure compared to the single one in **3** may account for its higher interaction energy. Interestingly, the trend of the interaction energy in type **A** systems (**1−6**) seems to be dependent on the organic ligand at Sn(IV), the larger the ligand, the longer the Cu···Cl distance (Figure 6), which can be correlated to the steric crowding of the ligand. This is in accordance with the plausible elucidation given by Beattie et al.[31] that the bulkier organic ligand (X) results in the *trans*-product $[SnX_2Cl_4]$ as observed in **1−6**, whereas repulsions between the ligands (X−Cl) are of a major importance for any structural distortion in such complexes. To minimize the X−Cl repulsions, all Sn−Cl bonds in **1−6** are appreciably extended[22] and consequently allow the Cl-atom to get closer toward the Cu center of a [CuL] unit, resulting in a shorter Cu···Cl interaction. In contrast, the higher symmetric geometry (octahedral) of the $[SnCl_6]^{2-}$ unit in type **B** cocrystals[23] does not provide any significant distortion that could lead to an appreciably longer Sn−Cl bond and indirectly to a shorter Cu···Cl contact compared to those in type **A** systems (**1−6**).

Cu···Cl Contact in Structural Database. The standard VDW radii for Cu and Cl are 1.40 and 1.75 Å, respectively. Therefore, a standard Cu···Cl contact is believed to be within 3.15 Å. However, for ionic systems, it would be complicated to estimate such a distance. In fact, it can be longer in the crystal structure of an ionic species. We have made a series of searches for Cu···Cl contact, ranging from 2.40 to 3.50 Å, in the Cambridge Structural Database (CSD).[32] As presented in Figure 7, the most common Cu···Cl contact lies in the range 3.00−3.10 Å, found in the crystal structure of 282 compounds. This contact with longer distances (e.g., 3.20−3.50 Å) than the standard VDW contact is also available in the database, revealing a total of 594 compounds. Dividing the range into three different intervals of 3.20−3.30, 3.30−3.40, and 3.40−3.50 Å resulted in 184, 185, and 225 hits, respectively. However, compounds containing such longer Cu···Cl contacts

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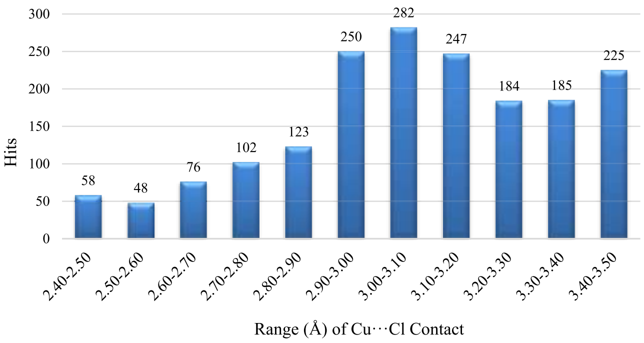

Figure 7. Number of crystal structures containing different ranges of Cu···Cl contact.

are ionic species as expected,[11,27,28,32] which is in line with the present systems (**1**−**8**).

DFT Theoretical Consideration. *Computational Models.* The Cu···Cl distance in complex **1** (2.91 Å) is longer than the sum of the covalent radii of the Cu and Cl atoms (Σcr = 2.34 Å),[33] but it shorter than the sum of the Bondi's van der Waals radii (ΣvdW = 3.15 Å). Meanwhile, this distance in complex 7 (3.41 Å) is larger than both Σcr and ΣvdW. The nature of the Cu···Cl bondings in **1** (as the representative of type **A** cocrystals) and 7 (as the representative of type **B** cocrystals) was analyzed using theoretical methods, i.e., QTAIM, ELF, IGM, NBO, EDD, and CDF (see the "Computational Details" section). The clusters $[H_2NC_2H_4NH_3]_2[CuL^1]_2[Sn(Me)_2Cl_4]$ (**1t**) and $[H_2NC_2H_4NH_3]_2[CuL^1]_2[SnCl_6]$ (**7t**) extracted from the single crystal X-ray diffraction (XRD) structures were used for the calculations (Figure 8). Cartesian atomic coordinates

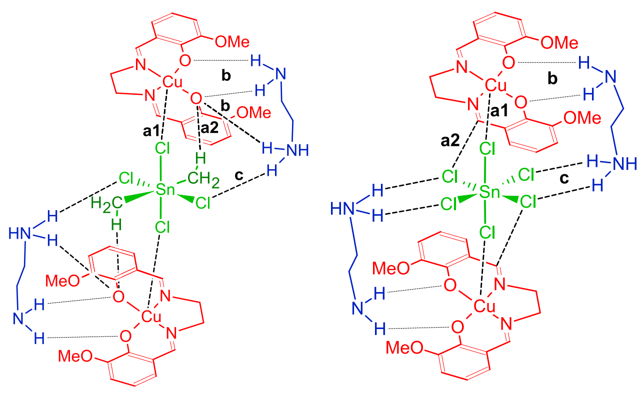

Figure 8. Clusters **1t** and **7t** were used as computational models in theoretical calculations.

(in Å) of the equilibrium structures of **1t** and **7t** are included in the Supporting Information. In **1t** and **7t**, the amino group of the ethylenediammonium counterion interacting with ligand L^1 was deprotonated to provide electroneutrality of the whole cluster molecule.

Geometry Optimization. Since the crystal packing may affect the molecular structure of complexes **1** and **7** and the properties of the Cu···Cl bond, full geometry optimization of **1t** and **7t** was carried out. The optimization resulted in conformational changes of the $H_2NC_2H_4NH_3^+$ counterion and formation of the hydrogen bond between the methyl group of $[Sn(Me)_2Cl_4]^{2-}$ and the O atom of the ligand L^1 in **1t** and a short contact between one of the Cl ligands of $[SnCl_6]^{2-}$ and a C atom of L^1 in **7t**. The Cu···Cl distances decreased from 2.91 (**1**) and 3.41 Å (**7**) (XRD) to 2.73 (**1t**) and 3.06 Å (**7t**), respectively, being longer than Σcr and shorter than ΣvdW in both optimized structures **1t** and **7t**. Thus, crystal packing effects significantly weaken the Cu···Cl bondings. The general structures of the optimized clusters are well preserved compared with the XRD analysis (Figure S6).

Bonding Nature. The QTAIM analysis demonstrates the existence of a bond critical point (BCP) for the Cu···Cl contacts in **1t** and **7t** (Figure 9,A). The quite low calculated values of the electron density (ρ_b) and the potential and kinetic energy densities (V_b and G_b) at BCP(Cu···Cl), the positive Laplacian values ($\nabla^2\rho_b$), and small ratio of the curvatures of the electron density distribution ($|\lambda_1/\lambda_3| \ll 1$) and typical for noncovalent interactions rather than for normal coordination bonds (Table 6). This is confirmed by the ELF analysis, which does not reveal a valence disynaptic basin for the Cu···Cl interaction. Instead, two monosynaptic basins V(Cu) and V(Cl) were found with the ELF value at the corresponding BCP of 0.077 a.u. (Figure 9B). The AIM parameters for the Cu···Cl contacts of all structures are typical for interactions with the predominant ionic contribution (Table 6). Meanwhile, the small negative value of the total energy density ($H_b = V_b + G_b$) and the $-G_b/V_b < 1$ ratio indicate a small covalent contribution in these bondings for **1t** (both optimized and XRD) and **7t** (optimized).

The IGM analysis confirms the attractive nature of the Cu···Cl interactions in **1t** and **7t** showing a region of the negative $\mathrm{sign}(\lambda_2)\rho(r)$ function on the isosurface of the g^{inter} descriptor around BCP(Cu···Cl) (Figure 10). The intrinsic bond strength indices (IBSI) correlating with the bond strength (in terms of the stretching force constants)[34] are 0.034 and 0.015 for the Cu···Cl interactions in **1t** and **7t**, respectively. These values are typical for the noncovalent or ionic interactions (IBSI < 0.15) rather than for the normal coordination bonds (0.15 < IBSI< 0.60).[33] The QTAIM, IGM, and IBSI analyses demonstrate that the Cu···Cl bond is stronger in **1t** compared to **7t**.

Electrostatic forces play a significant role in the interaction between the $[CuL^1]$ and $[Sn(Me)_2Cl_4]^{2-}/[SnCl_6]^{2-}$ complexes due to the ionic character of the latter. The electrostatic potential (ESP) distribution in $[CuL^1]$ reveals the positive potential area above the Cu atom on the 0.002 a.u. electron density isosurface with the maximum value, $V_{s,max}$, of 37 kcal/mol (Figure 11). In the tin complex anions, the most negative ESP is expectedly concentrated near the Cl atoms. Thus, the Cu···Cl interactions are electrostatically favorable.

On the other hand, charge transfer (CT) is also important in the formation of these bonds. In accord with the NBO analysis, CT between complex $[CuL^1]$ and anions $[Sn(Me)_2Cl_4]^{2-}$ or $[SnCl_6]^{2-}$ is associated with the LP(Cl) → LP*(Cu) transitions. The total second-order perturbation energy, $E(2)$, for each Cu···Cl bond in **1t** is 34.5 kcal/mol. This CT in complex **7t** is significantly lower with an $E(2)$ value of only 7.6 kcal/mol. The back Cu → Cl transfers are not efficient. The net negative NBO charge of the $\{Sn(Me)_2Cl_4\}$ moiety in **1t** is lower than the charge of the $\{SnCl_6\}$ moiety in **7t** (−1.79 e vs. −1.82 e) also demonstrating more efficient CT in the former complex.

The NBO data correlate with the results of the EDD and CDF analyses. The EDD map shows significant increase of the electron density between the Cl and Cu atoms because of charge transfer upon interaction between two $[CuL^1]$ and one

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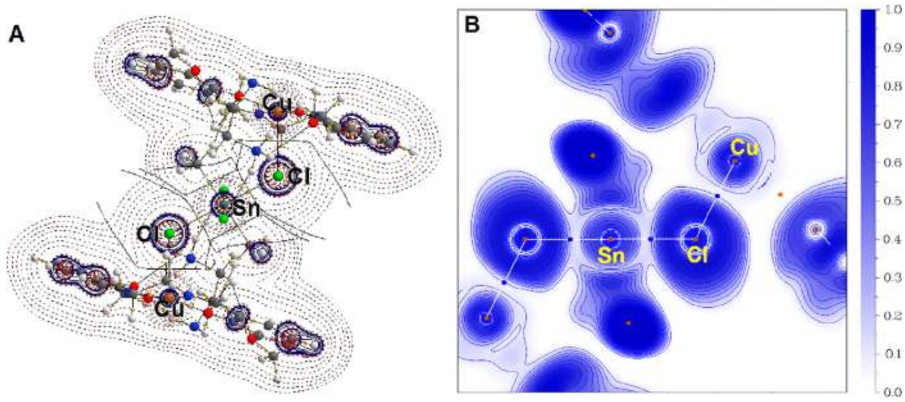


Figure 9. Contour line diagram of the Laplacian distribution, $\nabla^2\rho(r)$, selected zero flux surfaces and bond paths (dashed lines indicate charge depletion ($\nabla^2\rho(r) > 0$); solid lines indicate charge concentration ($\nabla^2\rho(r) < 0$) (A), and ELF (B) in the Sn−Cl···Cu plane for **1t**.

Table 6. Electron Density (ρ_b), Its Laplacian ($\nabla^2\rho_b$), Potential and Kinetic Energy Densities (V_b and G_b), Eigenvalues of the Hessian Matrix and Their Ratio (λ_2, λ_1/λ_3) at BCPs (Cu···Cl), and the Integrated Populations of the Monosynaptic Basins V_{Cl} and V_{Cu} (in a.u.) Calculated for 1t and 7t in Their X-ray Geometries and for the Optimized Structures

	$\mathbf{1t}_{\text{X-ray}}$	$\mathbf{1t}_{\text{opt}}$	$\mathbf{7t}_{\text{X-ray}}$	$\mathbf{7t}_{\text{opt}}$
ρ_b	0.0185	0.0261	0.0071	0.0135
$\nabla^2\rho_b$	0.0445	0.0687	0.0217	0.0354
V_b	−0.0159	−0.0238 −0.0085[b]	−0.0045	−0.0108 −0.0056[a]
G_b	0.0135	0.0205	0.0050	0.0098
$-G_b/V_b$	0.85	0.86	1.11	0.91
λ_2	−0.0126	−0.0192	−0.0041	−0.0087
λ_1/λ_3	−0.187	−0.182	−0.143	−0.173
$N(V_{Cl})$	6.96	7.10	6.92	7.03
$N(V_{Cu})$	16.75	17.26	16.95	17.09

[a]For the Cl···C contact (**a2**). [b]For the H···O contact (**a2**).

$[NH_2C_2H_4NH_3]_2[Sn(Me)_2Cl_4]$ molecules (Figure 12). The integrated CDF calculated along one of the Cl···Cu bonds reveals the Cl → Cu net CT of 29 me at the isoboundary between these two atoms.

Interaction Energies. In complex **1t**, the following types of intermolecular interactions are recognized (Figures 8 and S7), i.e., those (i) between complexes [CuL] and $[Sn(Me)_2Cl_4]^{2-}$, Cu···Cl (**a1**) and CH···O (**a2**), (ii) between cations $[NH_2C_2H_4NH_3]^+$ and complex $[CuL^1]$ (**b**), and (iii) between cations $[NH_2C_2H_4NH_3]^+$ and complex $[Sn(Me)_2Cl_4]^{2-}$ (**c**). In

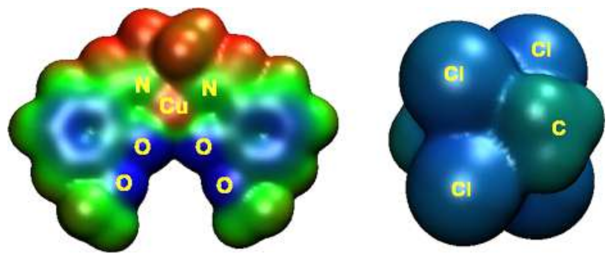


Figure 11. ESP distribution mapped on the 0.002 a.u. isosurface of electron density for $[CuL^1]$ (left) and $[Sn(Me)_2Cl_4]^{2-}$ (right) with the blue−cyan−green−yellow−red color scale −0.05 < ESP < 0.05 (left) and −0.45 < ESP < 0.45 (right).

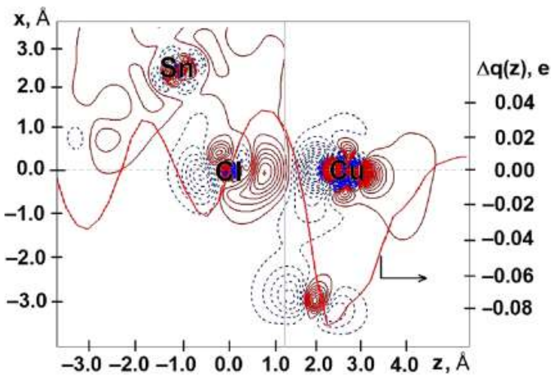
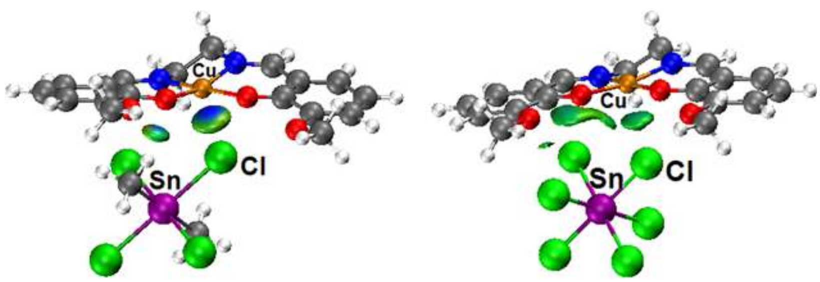


Figure 12. EDD contour plot (step 0.0005 a.u., red solid contours, charge concentration; blue dashed contours, charge depletion) and CDF function along the Cl···Cu bond in **1t** (the gray vertical line identifies the boundary between two molecules.



1t **7t**

Figure 10. Sign(λ_2)$\rho(r)$ function mapped on the δg^{inter} isosurface for the interactions between $[CuL^1]$ and $[SnMe_nCl_{6-n}]^{2-}$ (n = 2 or 0) in **1t** and **7t** (δg^{inter} = 0.01 a.u. and blue−cyan−green−yellow−red color scale $-0.05 < \text{sign}(\lambda_2)\,\rho(r) < 0.05$).

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complex **7t**, the **a2** interaction is that between the Cl atom of $[SnCl_6]^{2-}$ and a carbon atom of the L ligand in $[CuL^1]$, while the other interactions are of the same type as in **1t**.

The calculated BSSE corrected interaction energies, E_{int}, between molecules/ions are given in Table 7 (see the Supporting Information for details of calculations). The E_{int}(**a** + **b**) values for the interaction between each $[CuL^1]$ and the neutral trimer $[NH_2C_2H_4NH_3]_2[SnMe_nCl_{6-n}]$ are −56.4 and −32.5 kcal/mol for **1t** (n = 2) and **7t** (n = 0), respectively. The E_{int}(**a** + **c**) and E_{int}(**b** + **c**) values for the $[CuL^1][NH_2C_2H_4NH_3]^+\cdots[SnMe_nCl_{6-n}]^{2-}$ and $[NH_2C_2H_4NH_3]^+\cdots[CuL^1][SnMe_nCl_{6-n}]^{2-}$ interactions, respectively, are significantly higher due to the ionic nature for the interacting species.

Table 7. Calculated Interaction Energies (kcal/mol)[a]

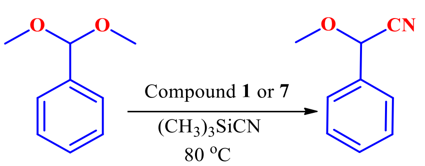
interaction	1t	7t
a + **b**	−56.4	−32.5
a + **c**	−134.2	−134.1
b + **c**	−170.2	−153.6
a	−10.2	−6.5
a1	−7.5	−4.8
a2	−2.7	−1.7
b	−46.2	−26.0
c	−124.0	−127.7

[a]Interactions are labelled as follows: (i) between the anion $[SnMe_nCl_{6-n}]^{2-}$ and the neutral complex $[CuL^1]$ (**a**), (ii) between the cation $[NH_2C_2H_4NH_3]^+$ and the neutral complex $[CuL^1]$ (**b**), (iii) between the cation $[NH_2C_2H_4NH_3]^+$ and the anion $[SnMe_nCl_{6-n}]^{2-}$ (**c**), (iv) the noncovalent Cu···Cl interaction (**a1**), and (v) the C−H···O or Cl···C bondings (**a2**). Schematic presentations are depicted in Figures 8 and S7.

Among the energies of paired intermolecular interactions, the highest values were found for the $[NH_2C_2H_4NH_3]^+\cdots[SnMe_nCl_{6-n}]^{2-}$ one, E_{int}(**c**), whereas the energy of the $[CuL^1]\cdots[SnMe_nCl_{6-n}]^{2-}$ binding, E_{int}(**a**), is only −10.2 and −6.5 kcal/mol. The latter value includes two interatomic interactions, **a1** and **a2**. The contribution of the second component **a2** may be estimated using the empiric formula by Espinosa, Molins, and Lecomte,[35] $E_{int} \sim 0.5\ V_b$, and it is −2.7 and −1.7 kcal/mol for **1t** and **7t**, respectively. Thus, the energies of the separate Cu···Cl interaction, **a1**, in these complexes are −7.5 and −4.8 kcal/mol. All these theoretical data allow the characterization of the Cu···Cl interactions as semicoordination bonds.[36]

Catalytic Study. The Strecker-type cyanation of acetals, a C−C bond forming reaction, is typically catalyzed by acidic materials, such as organic acids or Lewis acidic metal species including tin source.[37−39] Our systems contain two types of potential Lewis acid metal centers, i.e., based on Sn^{IV} and Cu^{II}, and thus, it would be of interest to study their catalytic activity for such a reaction (Scheme 2). The heterometallic cocrystal derivatives $(H_2ED)^{2+}\cdot2[Cu^{II}L^1]\cdot[Sn^{IV}(Me)_2Cl_4]^{2-}$ (**1**) and $(H_2ED)^{2+}\cdot2[Cu^{II}L^1]\cdot[Sn^{IV}Cl_6]^{2-}$ (**7**) were taken as representative systems. Moreover, this study can also provide an opportunity for comparison with the starting Sn^{IV} compounds, i.e., $[Sn(Me)_2Cl_2]$ and $[SnCl_4]\cdot5H_2O$.

Scheme 2. Catalytic Cyanation of Benzaldehyde Dimethyl Acetal



Based on preliminary findings, the best yield was obtained in CH_3CN at 80 °C for 2 h. By using benzaldehyde dimethyl acetal as the test acetal compound (Scheme 2), we found that **7** marginally led to a higher product yield, as compared to **1**, for the same reaction time and temperature (*e.g.*, 25 or 27% yield for **1** or **7**, respectively; entry 1 and 2, Table 8). The product yield was calculated by using ^1H NMR spectroscopy (Figures S8−S11).

Table 8. Cyanation of Benzaldehyde Dimethyl Acetal with Trimethylsilyl Cyanide (TMSCN) Catalyzed by Tin(IV) Compounds 1 and 7 and Related Ones[a]

entry	catalyst	amount of catalyst (mol %)	temp (°C)	solvent	time (h)	yield (%)[b]
1	**1**	1	80	CH_3CN	2	25
2	**7**	1	80	CH_3CN	2	27
3	blank		80	CH_3CN	2	5
4	H_2L1	1	80	CH_3CN	2	13
5	$CuCl_2\cdot2H_2O$	1	80	CH_3CN	2	15
6	$[Sn(Me)_2Cl_2]$	1	80	CH_3CN	2	84
7	$[SnCl_4]\cdot5H_2O$	1	80	CH_3CN	2	100
8	$[CuL^1]\cdot H_2O$	1	80	CH_3CN	2	21

[a]Reaction conditions: 1 mol % of catalyst, benzaldehyde dimethyl acetal (152 mg, 1 mmol) and trimethylsilyl cyanide (250 μL, 2 mmol) in CH_3CN (2 mL). [b]Calculated by ^1H NMR as the number of moles of final product 2-methoxy-2-phenylacetonitrile per mole of benzaldehyde dimethyl acetal x 100.

A blank reaction (without any catalyst) was also performed, leading to an yield of only 5% after 2 h (entry 3, Table 8). By using the free ligand precursor H_2L1 and the concerning copper(II) salt ($CuCl_2\cdot2H_2O$), the yields are also quite low (13 and 15%; entries 4 and 5, Table 8). In contrast, the use of the tin(IV) compounds $[Sn(Me)_2Cl_2]$ (a reactant used in the synthesis of **1**) and $[SnCl_4]\cdot5H_2O$ (a reactant used in the synthesis of **7**) led to yields of 84% and 100%, respectively (entries 6 and 7, Table 8), which are much higher than those obtained with their heterometallic derivatives, **1** and **7**. These results may be related to the conversion of the starting neutral tin(IV) compounds into their stannate versions $\{[Sn^{IV}(Me)_2Cl_4]^{2-}$ in **1** and $[Sn^{IV}Cl_6]^{2-}$ in **7**}, with a decrease in their Lewis acidity. Moreover, surrounding the tin(IV) metal site by the [CuL] complexes via formation of Cu···Cl interactions reduces the access of the substrate to the Sn^{IV} center with a resulting decrease in the catalytic performances of **1** and **7**, in comparison to those of the free $[Sn(Me)_2Cl_2]$ and $[SnCl_4]\cdot5H_2O$.

Absorption Spectroscopy. The UV−vis studies of heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ cocrystals **1**−**8** and the concerned metalloligands ($[Cu^{II}L^1]$ or $[Cu^{II}L^2]$) were carried out in methanol, and the related spectra are shown in Figures 13 and S12, respectively. The UV−vis spectra of the metalloligands (Figure S12) show one sharp band at ∼347 nm and a broad signal at ∼595 nm, which arise from the Cu^{II}−O(phenolate) charge transfer and d → d transitions,[40] respectively, whereas for the heterometallic cocrystals **1**−**8**, the sharp and broad bands appear in the 344−358 nm and 593−597 nm ranges

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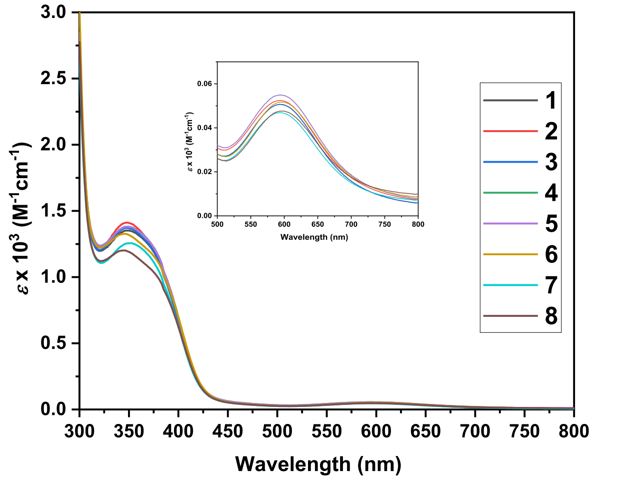

Figure 13. UV−vis spectra (300−800 nm range) of the heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ cocrystals $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ (**1**−**8**) along with an inset graph presenting a part (500−800 nm) of it.

(Figure 13), respectively, were thus with no significant shifting of peaks. Moreover, as presented in Table S1, the molar absorption coefficient (ε) values for **1**−**8** are in the 140.8−165.3 and 3604−4235 $M^{-1}cm^{-1}$ ranges for the broad and sharp bands, respectively, being similar to those (124.1−144.9 and 4164−4176 $M^{-1}cm^{-1}$, respectively) observed for the metalloligands. These results indicate that the copper environments of the metalloligands ($[Cu^{II}L^1]$ or $[Cu^{II}L^2]$) are not significantly affected upon formation of heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ cocrystals **1**−**8**, which is supported by the weak Cu···Cl semicoordination bonds.

■ CONCLUSIONS

The nature of the Cu···Cl interaction in the crystal structures of the series of related three-component supramolecular heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ cocrystals $(H_2ED)^{2+}\cdot 2[CuL]\cdot[SnX_nCl_{6-n}]^{2-}$ [H_2L = *N*,*N*′-ethylenebis(3-*O*R-salicylaldimine), R = Me (H_2L^1) or Et (H_2L^2); ED = 1,2-ethylenediamine; X = Me, Et, Bu^n, or Ph; n = 2 (type **A**) or 0 (type **B**)] is evaluated by theoretical calculations.

There are three types of interactions (cation−anion, cation−neutral, and neutral−anion) as expected in a three-component structure containing $(H_2ED)^{2+}$, $[Cu^{II}L]$, and $[SnX_nCl_{6-n}]^{2-}$. Interactions between the ionic moieties $\{(H_2ED)^{2+}$ and $[SnX_nCl_{6-n}]^{2-}\}$ in **1**−**8** are the strongest, as revealed by the calculated energies. Stronger interactions in the Ph_2Sn-containing species (**4**−**6**) than in the remaining ones may tentatively be attributed to the presence of two N−H···Cl hydrogen bonds in comparison to only one in the others. The interactions between the ethylene diammonium cation $(H_2ED)^{2+}$ and the neutral metal complex $[Cu^{II}L]$ which are principally supported by two N−H···O hydrogen bonds involving the ligand O_4 cavity and ammonium-H are weak and not considerably affected by a variation of the X or R group at Sn(IV) or at the ligand (L^1 or L^2), respectively. The weakest interaction is observed between the neutral species $[Cu^{II}L]$ (L = L^1 or L^2) and the anion $[SnX_nCl_{6-n}]^{2-}$ which is connected by a Cu···Cl interaction and a H-bond (C−H···Cl). However, the distance of the Cu···Cl interaction, which shortens with the increase of the concerned interaction energy, is dependent on the X ligand at the $[SnX_2Cl_4]^{2-}$ moiety in type **A** systems (**1**−**6**). Moreover, the presence of X−Cl repulsions in **1**−**6** triggering a Sn−Cl elongation and subsequent shortening of the Cu···Cl interaction can be an additional factor for this outcome. In contrast, the absence of a second type of ligand (such as X in **1**−**6**) and the higher symmetric geometry of the $[SnCl_6]^{2-}$ unit in type **B** cocrystals (**7** and **8**) cannot produce any significant distortion or extension of a Sn−Cl bond (toward the Cu-center) to result in a shorter Cu···Cl contact in comparison to those in all type **A** systems (**1**−**6**).

DFT calculations of the models of **1** and **7** revealed that electrostatic forces play a predominant role in the Cu···Cl interaction in complex **7t**, while Cl → Cu charge transfer is significant in **1t**. The calculated values of the interaction energies for the Cu···Cl interactions are −7.5 and −4.8 kcal/mol in **1t** and **7t**, respectively, indicating the semicoordination nature of these bonds.

Preliminary studies on a C−C bond forming catalytic reaction (Strecker type cyanation) revealed that the catalytic ability of the tin(IV) precursor $[Sn(Me)_2Cl_2]$ or $[SnCl_4]\cdot 5H_2O$ is almost lost upon its conversion into the corresponding dianion, i.e., $[Sn(Me)_2Cl_4]^{2-}$ or $[SnCl_6]^{2-}$ in **1** or **7**, respectively.

In summary, interaction energies among the different components in a series of heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ cocrystals were calculated, revealing a triangle energy framework. The Cu···Cl contacts among the heterometallic species are identified by the DFT calculations as semicoordination bonds.

■ EXPERIMENTAL SECTION

Hirshfeld Surface Analysis and Preliminary Interaction Energy Calculations. CrystalExplorer17,[29,30] which is a useful tool for finding all relevant noncovalent interactions in a crystal structure,[11,41−43] is utilized for the Hirshfeld surface analysis and the preliminary interaction energy calculations among the components of the selected cocrystals (**1**−**8**). Each component was selected separately to find out the interaction with the remaining ones and then the energies were calculated using the MP2/DGDZVP model.[44]

DFT Computational Details. The models of two representative cocrystals of types **A** and **B**, i.e., $(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Me)_2Cl_4]^{2-}$ (**1**) and $(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}Cl_6]^{2-}$ (**7**), respectively, were selected for comprehensive analysis at the density functional theory (DFT) level of theory. All DFT calculations were carried out by using the M06-2X functional[45] with the help of the Gaussian 09 program package.[46] This functional demonstrated good performance in the considerations of noncovalent interactions with noticeable electrostatic components.[11,47,48] The full geometry optimization was performed using a relativistic Stuttgart pseudopotential with 10 core electrons (MDF10) and a quasi-relativistic Stuttgart pseudopotential with 46 core electrons (MWB46) for the Cu and Sn atoms, respectively, together with appropriate contracted basis sets (8s7p6d)/[6s5p3d] and (4s4p)/[2s2p] for Cu and Sn, respectively, and the 6-31G* basis set for other atoms. Single-point calculations were carried out for the equilibrium and X-ray geometries using the DZP-DKH basis set taken from the Basis Set Exchange library.[49−51] The Douglas−Kroll−Hess second-order scalar relativistic method (DKH)[49−52] was applied. An ultrafine integration grid was used for the numerical integrations. The basis set superposition error (BSSE) was estimated using the counterpoise method[53] at the M06-2X/DZP-DKH level. The atomic charges and bonding nature were analyzed by using the natural bond orbital (NBO) partitioning scheme.[54,55] A topological analysis of the electron density distribution was performed with the help of the quantum theory of atoms in molecules (QTAIM) developed by Bader[56,57] using the AIMAll program.[58] The electron localization function (ELF),[59,60] independent gradient model (IGM),[61,62] electron density difference (EDD)

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and charge displacement function (CDF)[63] analyses, and calculations of the intrinsic bond strength index (IBSI)[64] were performed using the Multiwfn 3.8 software.[65] Such studies presenting a toolbox of different bonding descriptors have been termed as complementary bonding analysis.[66] Geometry optimized structures of the complexes $([CuL^1]_2[Sn(Me)_2Cl_4])^{2-}$ and $([CuL^1]_2[SnCl_6])^{2-}$ without cations are presented with their parent structures $[H_2NC_2H_4NH_3]_2[CuL^1]_2[Sn(Me)_2Cl_4]$ (**1t**) and $[H_2NC_2H_4NH_3]_2[CuL^1]_2[SnCl_6]$ (**7t**), respectively, in Figure S13.

Catalytic C–C Bond Forming Reactions. The tin(IV) reactants $[Sn(Me)_2Cl_2]$ and $[SnCl_4]\cdot 5H_2O$ and their corresponding supramolecular heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ cocrystal derivatives $(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}(Me)_2Cl_4]^{2-}$ (**1**) and $(H_2ED)^{2+}\cdot 2[Cu^{II}L^1]\cdot[Sn^{IV}Cl_6]^{2-}$ (**7**) were tested in a C–C bond forming reaction, the Strecker type cyanation of benzaldehyde diacetal.

A mixture of benzaldehyde dimethyl acetal (0.152 g, 1.0 mmol), trimethylsilyl cyanide (250 μL, 2.0 mmol), and 1.0 mol % catalyst (2.2 mg of $[Sn(Me)_2Cl_2]$, 3.5 mg of $[SnCl_4]\cdot 5H_2O$, 11.3 mg of **1** or 11.7 mg of **7**) in acetonitrile (2 mL), contained in a capped glass vessel, was stirred at 80 °C. After the desired time (2 h), the reaction mixture was taken to dryness. Organic products were then extracted from the crude solid mixture by using dichloromethane (DCM) and subsequent removal of the solvent (DCM) gave the crude product 2-methoxy-2-phenylacetonitrile which was analyzed by ^{1}H NMR in $CDCl_3$.

Absorption Spectroscopy. Absorbance spectra of the metalloligands ($[Cu^{II}L^1]$ and $[Cu^{II}L^2]$) and the heterometallic cocrystals **1**–**8** in methanol (0.001 M) were recorded in the range 1000–200 nm at a scan rate of 480 nm/min with a PerkinElmer Lambda 35 UV/vis spectrometer.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.organomet.3c00274.

Results of UV–vis measurements; additional computational details; product yield calculation of the cyanation reaction (PDF)

Cartesian coordinates of the equilibrium structures of **1t** and **7t** (XYZ)

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Notes

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CRYSTAL GROWTH & DESIGN

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$M^{II}\cdots Cl$ Interaction Supported Heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ and $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ Complex Salts: Possibility of Ion-Pair-Assisted Tetrel Bonds

Susanta Hazra,* Maxim L. Kuznetsov, M. Fátima C. Guedes da Silva, and Armando J. L. Pombeiro*

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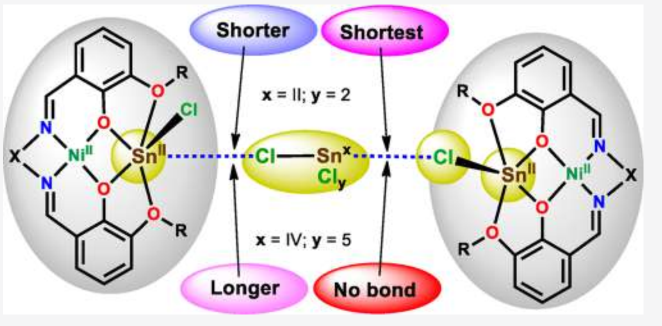
ABSTRACT: Crystal structures and theoretical calculations on the ion pairs of two types of heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ (type **I**) and $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ (type **II**) complex salts synthesized from the self-assembly reaction of the Schiff base *N,N′*-ethylenebis(3-methoxysalicylaldimine) (H_2L^1), *N,N′*-ethylenebis(3-ethoxysalicylaldimine) (H_2L^2), or *N,N′*-bis(3-methoxysalicylidene)-1,2-cyclohexanediamine (H_2L^3) with an adequate mixture of $NiCl_2 \cdot 6H_2O$, $SnCl_2$, and $SnCl_4 \cdot 5H_2O$ are presented. The heterometallic cation $[Ni^{II}Sn^{II}L(Cl)]^+$ is neutralized and stabilized by half of the hexachlorostannate(IV) $[Sn^{IV}Cl_6]^{2-}$ [in type **I**: L = L^1 (**1**), L^2 (**2**)] or by the trichlorostannate(II) $[Sn^{II}Cl_3]^-$ [in type **II**: L = L^1 (**3**), L^2 (**4**), L^3 (**5**)]. Ion pairs in the crystal structures of the **1**−**7** interact via a variety of $M^{II}\cdots Cl$ interactions (M = Ni, Cu, Sn) along with H bonds. Hirshfeld surface analysis was carried out to discover any other weak interactions such as $\pi\cdots\pi$. Detailed geometrical analysis reveals that $Sn^{II}\cdots Cl$ interactions are dependent on the oxidation state of the metal halide donor, being lengthened with a higher oxidation state (Sn^{II} to Sn^{IV}). Structural comparison of **4** with the previously reported $\{Cu^{II}Sn^{II}\}\{Sn^{II}\}$ derivatives (**6** and **7**) of H_2L^1 and H_2L^2 reveals that the length of other $M^{II}\cdots Cl$ interactions (M^{II} = Ni, Cu), being dependent on the halide acceptor [Ni^{II} (d^8) or Cu^{II} (d^9) system], decreases from the d^8 metal ion system to the d^9 system and hence that interaction ($Ni^{II}\cdots Cl$) in **4** converts virtually to a covalent bond (Cu^{II}−Cl) in **6** and **7**. The nature of ion-pair interactions in **1** and **3** was further investigated by DFT calculations with application of the QTAIM, ESP, NBO, and NCI methods. Significant charge transfer from the Cl to Sn atoms allows the characterization of the Sn⋯Cl interactions in these complex salts as an uncommon ion-pair-assisted Sn⋯Cl tetrel bond.



INTRODUCTION

Noncovalent interactions constitute a fascinating topic in crystal engineering and supramolecular chemistry.[1−7] They are usually considered when the distance between two atoms (or ions) is within the sum of their van der Waals radii but longer than that of a covalent bond. With the exception of the H bond,[8,9] they have been commonly indicated as "interactions" and not as bonds. In the last few decades, the study of such interactions has increased markedly from both experimental and theoretical points of view, revealing their differences and significant roles in applications (e.g., polymer sciences, liquid crystals, conductive materials, medicinal chemistry, etc.) other than crystal packing.[10−19] In fact, several such interactions have now been differentiated and recognized with a particular bond name: for example, triel,[12] tetrel,[13] pnictogen,[14] chalcogen,[15−17] halogen,[10,11] regium,[18] spodium,[19] etc. Review articles have mainly been based on the structures that are already available in the crystal structure database (CSD).[20] Although the majority of such weak interactions concern organic molecules, their existence in metal-containing systems has also been reviewed.[6,7,14,15,21,22] However, the investigation of systems produced from some metal ions that have both accepting and donating abilities is still underdeveloped. For example, the possibility of the existence of different types of weak bonds, such as tetrel or halogen, can be expected in systems derived from the $SnCl_2$ or $SnCl_4$ salts, which are easily converted into their stannate versions with a higher number of chlorides: i.e., $[SnCl_3]^-$, $[SnCl_4]^{2-}$, $[SnCl_5]^-$, and $[SnCl_6]^{2-}$.

On the other hand, acyclic compartmental Schiff bases have been extensively utilized to design heterometallic systems[23−36] that bring valuable contributions to coordination chemistry. For instance, their heterometallic 3d−4f derivatives (Cu^{II}−Gd^{III}, Co^{II}−Dy^{III}, Zn^{II}−Dy^{III}, etc.)[30,31] are important in single-molecule magnetism, while others (3d−s,[32] 3d−p,[33−35] and 3d−5f[36] systems) can exhibit uncommon solid-state structures

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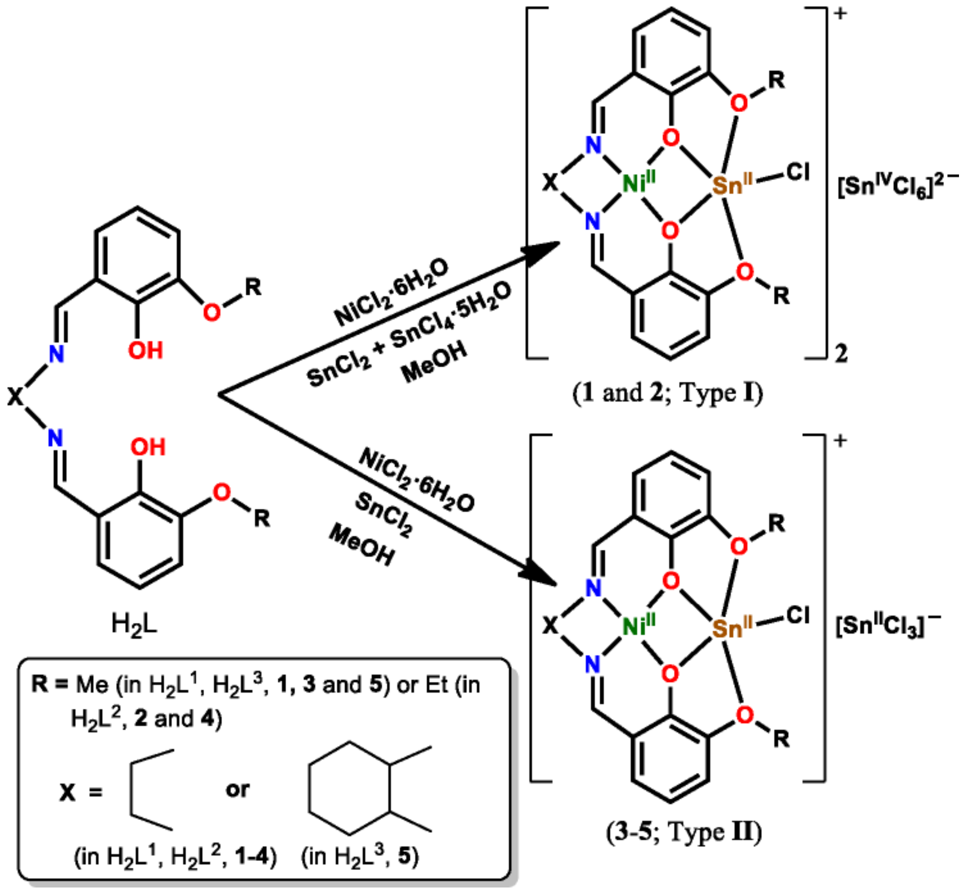
Scheme 1. Synthesis of Heterometallic {Ni^{II}Sn^{II}}{Sn^{IV}} (1 and 2) and {Ni^{II}Sn^{II}}{Sn^{II}} (3−5) Complex Salts



that are valuable in supramolecular chemistry and crystal engineering. Specifically, the combination of a transition ion with a p-block metal (e.g., Sn) ion in the presence of such a ligand can produce a diversity of possible products for a number of reasons, such as (i) tin has two stable oxidation states (2+ and 4+) with different sizes, (ii) it has variable coordination numbers, (iii) it can donate a lone pair in the 2+ state while it can accept one in the 4+ state, (iv) it can form organometallic compounds in a high oxidation state (Sn^{4+}), (v) its inorganic or organometallic versions can effortlessly form adducts, (vi) the halide(s) coordinated to a tin metal ion (Sn^{2+} or Sn^{4+}) can easily participate in noncovalent interactions, etc. In addition, the tin cations (Sn^{2+} and Sn^{4+}) do not usually prefer the N_2O_2 cavity of a compartmental Schiff base ligand (Scheme 1), but a tin moiety (organic or inorganic) can interact with such a ligand or metalloligand (a transition metal ion already in the N_2O_2 cavity of a ligand) in various ways. As anticipated, several heterometallic 3d-Sn^{IV} or 3d-Sn^{II} products[37−45] were obtained under inert conditions that have been usual for the aerobically unstable tin systems. However, the expected high versatility of products was not observed, as most of the isolated species were simple adducts or heterobimetallic covalent compounds. With the aim of discovering possible changes in complexation/composition under different conditions (other than the usual inert conditions), we have carried out related reactions but in an open atmosphere.[46] Consequently, several heterometallic 3d-Sn^{IV} or 3d-Sn^{II} products have been isolated and many of them have shown intriguing solid-state molecular structures.[33−35,46−48] A number of interesting structural phenomena were also observed, such as the aerobic oxidation of a metal center forming the corresponding mixed-valence compound, a Sn to Co phenyl ligand transfer, etc.[35,47,48] Moreover, the obtained products were uncommon in terms of their compositions, being identified as simple adducts, cocrystals (neutral), salt cocrystals (ionic + neutral), or metal complex salts (ionic).[46] In most cases, one or more noncovalent interactions (O−H···O/N−H···O/N−H···Cl/M···Cl/Sn···Cl/M···π/π···π) play an important role to stabilize their structures, as well as their crystal packings.[33−35,46−48]

Interestingly, a Sn^{II} ion sits on the O_4 cavity (second cavity) of a compartmental ligand of the L^{2-} type [H_2L = *N,N′*-ethylenebis(3-methoxysalicylaldimine) (H_2L^1), *N,N′*-ethylenebis(3-ethoxysalicylaldimine) (H_2L^2), *N,N′*-bis(3-methoxysalicylidene)-1,2-cyclohexanediamine (H_2L^3)] (Scheme 1), as observed in the metal complex salts[33] obtained from the reaction of [CuL(H_2O)] and $SnCl_2$. However, salt cocrystals are synthesized from the reactions of H_2L, $CuCl_2$/$NiCl_2 \cdot xH_2O$, and $SnCl_4 \cdot 5H_2O$, revealing that Sn^{IV} remains outside the ligand/metalloligand units,[34] in a contrasting choice of two different oxidation states of tin. Within this context and after the successful synthesis of isolated M−Sn^{IV} and M−Sn^{II} systems, we anticipated that the reaction of such a Schiff base type with both Sn^{II} and Sn^{IV} salts together with a transition-metal ion (Ni^{II}) could produce heterometallic products with interesting noncovalent interactions. Accordingly, we have reacted the Schiff base H_2L^1 or H_2L^2 with $NiCl_2 \cdot 6H_2O$, $SnCl_2$, and $SnCl_4 \cdot 5H_2O$ and obtained the new heterometallic {Ni^{II}Sn^{II}}{Sn^{IV}} complex salt (type **I**) [Ni^{II}Sn^{II}L^1(Cl)]$_2$[Sn^{IV}Cl$_6$] (**1**) or [Ni^{II}Sn^{II}L^2(Cl)]$_2$[Sn^{IV}Cl$_6$]

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(**2**), respectively (Scheme 1). For comparison, we have also synthesized related $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ salts (type **II**), i.e. $[Ni^{II}Sn^{II}L^1(Cl)][Sn^{II}Cl_3]\cdot0.25H_2O$ (**3**), $[Ni^{II}Sn^{II}L^2(Cl)][Sn^{II}Cl_3]$ (**4**), and $[Ni^{II}Sn^{II}L^3(Cl)][Sn^{II}Cl_3]$ (**5**) (Scheme 1), from the reaction of the corresponding Schiff base (H_2L^1, H_2L^2, or H_2L^2), $NiCl_2\cdot6H_2O$, and $SnCl_2$. Herein, we discuss the syntheses and crystal structures of these two types (**I** and **II**) of salts and analyze the involvement of noncovalent interactions in their structures. The Sn···Cl interactions in **1** and **3**, as representatives of types **I** and **II**, respectively, have also been subjected to theoretical calculations to understand the nature of their bondings. Detailed structural comparisons among them and with the reported copper(II) analogues[33] of **3** and **4**, i.e., $[Cu^{II}Sn^{II}L^1(Cl)][Sn^{II}Cl_3]$ (**6**) and $[Cu^{II}Sn^{II}L^2(Cl)][Sn^{II}Cl_3]$ (**7**), respectively, are also discussed.

EXPERIMENTAL SECTION

Synthesis. The Schiff bases *N,N'*-ethylenebis(3-methoxysalicylaldimine) (H_2L^1), *N,N'*-ethylenebis(3-ethoxysalicylaldimine) (H_2L^2), and *N,N'*-bis(3-methoxysalicylidene)-1,2-cyclohexanediamine (H_2L^3) (Scheme 1) were synthesized and characterized by following our previously reported procedures.[46] The heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ (**1** and **2**, type **I**) and $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ (**3**−**5**, type **II**) complex salts were prepared under open atmospheric conditions by following the procedures given below for **1** and **3**, respectively.

$[Ni^{II}Sn^{II}L^1(Cl)]_2[Sn^{IV}Cl_6]$ (1). To a methanol suspension (20 mL) of H_2L^1 (0.082 g, 0.25 mmol) was added a methanol solution (5 mL) of $NiCl_2\cdot6H_2O$ (0.060 g, 0.25 mmol). To the resulting red solution was added a methanol solution (10 mL) of $SnCl_2$ (0.048 g, 0.25 mmol) and $SnCl_4\cdot5H_2O$ (0.088 g, 0.25 mmol) at room temperature to obtain a red solution. Within 10 min, orange crystals suitable for an X-ray diffraction analysis were formed. They were collected by filtration and washed with cold methanol. Yield: 0.143 g (81%). Anal. Calcd for $C_{36}H_{36}Cl_8N_4O_8Ni_2Sn_3$ (1409.78): C, 30.67; H, 2.57; N, 3.97. Found: C, 30.99; H, 2.65; N, 3.89. IR data (KBr, cm^{-1}): ν(C═N), 1633s.

$[Ni^{II}Sn^{II}L^2(Cl)]_2[Sn^{IV}Cl_6]$ (2). Yield: 0.156 g (85%). Anal. Calcd for $C_{40}H_{44}Cl_8N_4O_8Ni_2Sn_3$ (1465.88): C, 32.77; H, 3.03; N, 3.82. Found: C, 32.89; H, 3.09; N, 3.77. IR data (KBr, cm^{-1}): ν(C═N), 1632s.

$[Ni^{II}Sn^{II}L^1(Cl)][Sn^{II}Cl_3]\cdot0.25H_2O$ (3). To a methanol suspension (20 mL) of H_2L^1 (0.082 g, 0.25 mmol) was added a methanol solution (5 mL) of $NiCl_2\cdot6H_2O$ (0.060 g, 0.25 mmol). To the resulting red solution was added a methanol solution (20 mL) of $SnCl_2$ (0.096 g, 0.50 mmol) at room temperature to obtain a red solution. Within 10 min, red crystals suitable for an X-ray diffraction analysis were obtained. They were collected by filtration and washed with cold methanol. Yield: 0.176 g (92%). Anal. Calcd for $C_{18}H_{18.75}Cl_4N_2O_{4.25}NiSn_2$ (769.03): C, 28.12; H, 2.43; N, 3.64. Found: C, 28.20; H, 2.37; N, 3.67. IR data (KBr, cm^{-1}): ν(C═N), 1628s.

$[Ni^{II}Sn^{II}L^2(Cl)][Sn^{II}Cl_3]$ (4). Yield: 0.174 g (88%). Anal. Calcd for $C_{20}H_{22}Cl_4N_2O_4NiSn_2$ (792.28): C, 30.32; H, 2.80; N, 3.54. Found: C, 30.50; H, 2.78; N, 3.51. IR data (KBr, cm^{-1}): ν(C═N), 1630s.

$[Ni^{II}Sn^{II}L^3(Cl)][Sn^{II}Cl_3]$ (5). Yield: 0.184 g (90%). Anal. Calcd for $C_{22}H_{24}Cl_4N_2O_4NiSn_2$ (818.32): C, 32.29; H, 2.96; N, 3.42. Found: C, 32.42; H, 2.94; N, 3.39. IR data (KBr, cm^{-1}): ν(C═N), 1626s.

Crystal Structure Determination. X-ray-quality crystals of **1**−**5** were immersed in cryo-oil, mounted on a Nylon loop, and measured at 298 K. Intensity data were collected using a Bruker APEX II SMART CCD diffractometer with graphite-monochromated Mo Kα (λ = 0.71073 Å) radiation. Cell parameters were obtained with Bruker SMART[49] software and refined with Bruker SAINT[49] on all of the observed reflections. Absorption corrections were made by the multiscan method (SADABS).[49] Structures were solved by direct methods by using the SHELXS-2014 package[50] and refined with SHELXL-2014/7.[50] Calculations were performed using the WinGX System-Version 2014.1.[51] The hydrogen atoms attached to carbon atoms were inserted at geometrically calculated positions and included in the refinement using the riding-model approximation. The corresponding hydrogen atoms were then inserted at geometrically calculated positions and included in the refinement using the riding-model approximation, but they came close to other H atoms, as suitable positions were not available for them. Least-squares refinements with anisotropic thermal motion parameters for all of the non-hydrogen atoms were employed. Crystallographic data are summarized in Table S1 in the Supporting Information.

Computational Details. All calculations were carried out at the density functional theory (DFT) level by using the M06-2X functional[52] with the help of the Gaussian 09 program package.[53] This functional performs well in the analysis of noncovalent interactions,[54] including those with significant electrostatic components.[55] The DZP-DKH basis set taken from the Basis Set Exchange library[56] and the Douglas−Kroll−Hess second-order scalar relativistic method (DKH)[56,57] were applied. An ultrafine integration grid was used for numerical integrations. The basis set superposition error (BSSE) was estimated using the counterpoise method[58] at the M06-2X/DZP-DKH level. The BSSE-corrected E_{int} values for the X···Y interaction were calculated as $E_{int}(X\cdots Y) = E(X\cdots Y) - E(X) - E(Y) + \mathrm{BSSE}$, where $E(X)$ and $E(Y)$ are the total energies of the fragments X and Y with unrelaxed geometries. The atomic charges and bonding nature were analyzed by using the natural bond orbital (NBO) partitioning scheme.[59] A topological analysis of the electron density distribution was performed with the help of the atoms in molecules (AIM) method developed by Bader[60] using the AIMAll program.[61] The IGM analysis[62−66] and calculations of the intrinsic bond strength index (IBSI) were performed using the Multiwfn 3.8 software.[67]

Within the hard−soft acid−base (HSAB) formalism, the chemical potential (μ) and global softness (S) of fully optimized fragments $SnCl_3^-$ and $[NiL^1(SnCl)]^+$ were calculated using eqs 1 and 2

$$\mu \approx -(I + A)/2 \tag{1}$$

$$S \approx 1/(\mathrm{I} - A) \tag{2}$$

where I and A are the ionization potential and electron affinity, respectively:

$$I \approx -E(\mathrm{HOMO})$$

$$A \approx -E(\mathrm{LUMO})$$

Fukui functions of atom x in a molecule with N electrons, f_x^+ and f_x^- (the former representing nucleophilic attack and the latter representing electrophilic attack) were defined in a finite-difference approximation using eqs 3 and 4

$$f_x^+ = [q_x(N + 1) - q_x(N)] \tag{3}$$

$$f_x^- = [q_x(N) - q_x(N - 1)] \tag{4}$$

where q_x is the NPA electronic population of atom x in a molecule. The local atomic softnesses s_x^+ and s_x^- were calculated by eq 5:

$$s_x^+ = f_x^+ S \text{ and } s_x^- = f_x^- S \tag{5}$$

RESULTS AND DISCUSSION

Synthesis and Characterization. The self-assembly reaction of the Schiff base H_2L^1 or H_2L^2 with $NiCl_2\cdot6H_2O$, $SnCl_2$, and $SnCl_4\cdot5H_2O$ (1:1:1:1) in methanol under an open atmosphere and at room temperature produces the heteronuclear $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ complex salts $[Ni^{II}Sn^{II}L(Cl)]_2[Sn^{IV}Cl_6]$ [L = L^1 (**1**), L^2 (**2**)] (Scheme 1). The reaction without $SnCl_4\cdot5H_2O$ under the same conditions resulted in the $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ complex salts $[Ni^{II}Sn^{II}L(Cl)][Sn^{II}Cl_3]$ [L = L^1 (**3**), L^2 (**4**), L^3 (**5**)]. Their synthesis is very simple (only mixing), is fast (10 min), and does not consume electrical or heat energy (neither stirring nor heating) (Scheme

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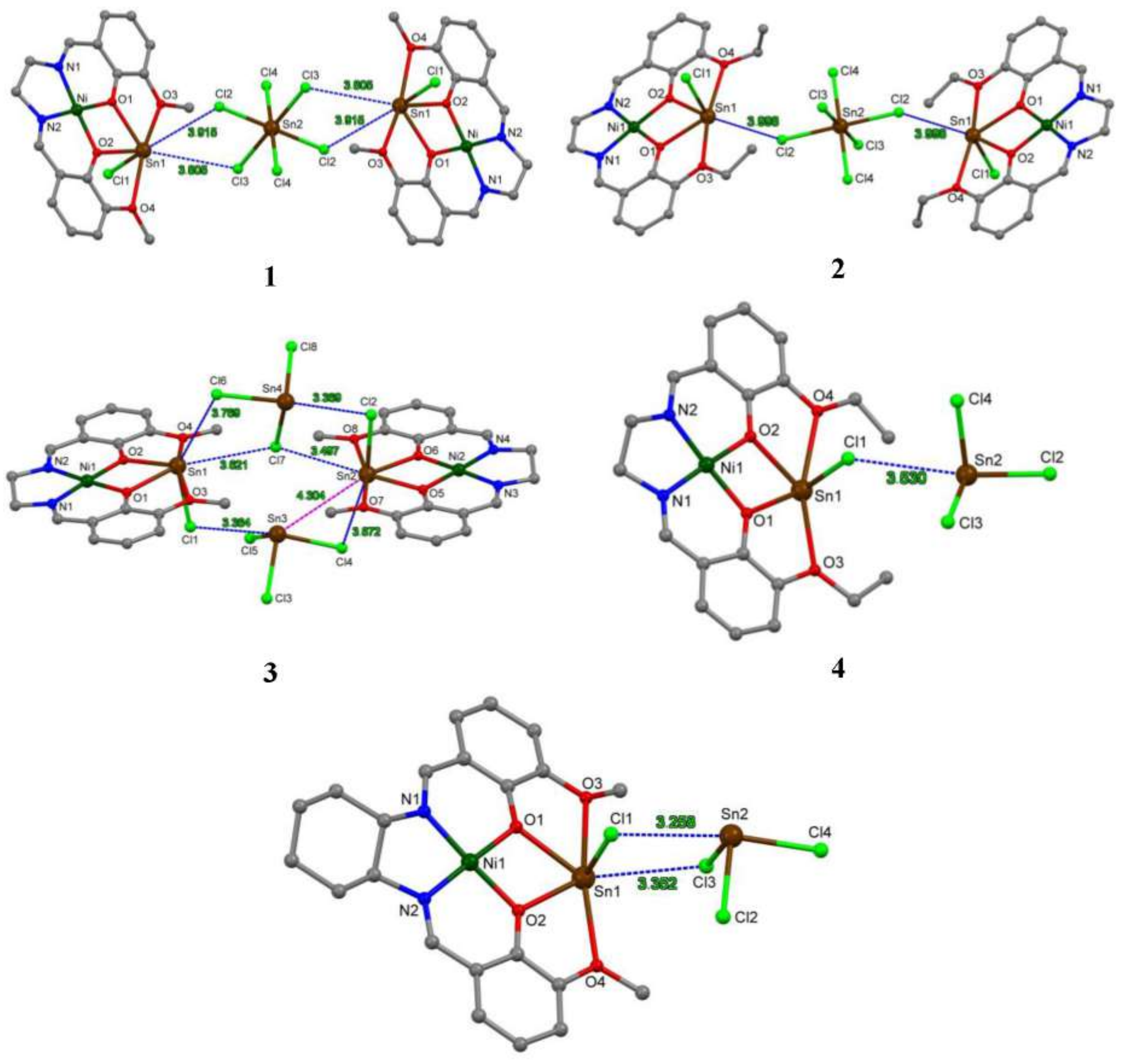


Figure 1. Molecular structures of type **I** {$Ni^{II}Sn^{II}$}{Sn^{IV}} (**1** and **2**) and type **II** {$Ni^{II}Sn^{II}$}{Sn^{II}} complex salts (**3**−**5**). Sn···Cl intermolecular contacts are indicated by dashed blue lines (values in Å). Hydrogen atoms are omitted for clarity.

1). In addition, these compounds were isolated in very good yields (81−92%).

Their IR spectra exhibit a medium-intensity absorption band in the 1626−1633 cm^{-1} range due to ν(C═N). They were also characterized by elemental analyses and single crystal X-ray diffraction.

Description of Crystal Structures. The crystal structures of the two types (**I** and **II**) of the heterobimetallic salts $[Ni^{II}Sn^{II}L^1(Cl)]_2[Sn^{IV}Cl_6]$ (**1**), $[Ni^{II}Sn^{II}L^2(Cl)]_2[Sn^{IV}Cl_6]$ (**2**), $[Ni^{II}Sn^{II}L^1(Cl)][Sn^{II}Cl_3]\cdot 0.25H_2O$ (**3**), $[Ni^{II}Sn^{II}L^2(Cl)]$-$[Sn^{II}Cl_3]$ (**4**), and $[Ni^{II}Sn^{II}L^3(Cl)][Sn^{II}Cl_3]$ (**5**) are depicted in Figure 1, and their selected geometrical parameters are included in Table 1. The compounds crystallize in the space group $P\overline{1}$ except for **5** ($P2_1/n$). The heterometallic complex cations $[Ni^{II}Sn^{II}L(Cl)]^+$ are neutralized either by hexachlorostannate(IV) $[Sn^{IV}Cl_6]^{2-}$ (in **1** and **2**) or by trichlorostannate(II) $[Sn^{II}Cl_3]^-$ (in **3**−**5**). The coordination environments of Ni^{II} (square-planar, τ_4 = 0.03−0.05; Table 1) are fulfilled by the two N_{imine} and the two $O_{phenolate}$ atoms of the ligand. The Sn^{II} cations in both complexes are bound to L^{2-} through the two $O_{phenolate}$ and the two $O_{methoxido/ethoxido}$ atoms, their highly distorted 5-coordinated sphere (τ_5 = 0.51−0.62) being fulfilled with a chloride ligand in the apical site. All structural parameters involving the metal ions (Ni^{II}, Sn^{II}, and Sn^{IV}) are comparable to those of our previously reported heterometallic {$Cu^{II}Sn^{II}$}{Sn^{II}} (**6** and **7**)[33] and {Cu^{II}}{Sn^{IV}}[34] systems.

One or more Sn^{II}···Cl contacts between the heterometallic cation and the stannate anion are found in their crystal structures. For instance, in **1** two Cl atoms of the hexachlorostannate anion contact with two tin atoms of the complex cations, therefore achieving a pentanuclear organization [$d_{Sn\cdots Cl}$ values of 3.915(3) and 3.805(3) Å, respectively] (Figure 1); a related interaction is observed in the analogous compound **2**, although it is derived from just one Sn^{II}···Cl contact [$d_{Sn\cdots Cl}$ value of 3.998(4) Å] (Figure 1). The probability of such Sn^{II}···Cl interactions increases when $[SnCl_6]^{2-}$ is replaced by $[SnCl_3]^-$. For example, there are six Sn···Cl contacts in **3**, forming two four-membered Sn_2Cl_2 cycles and one six-membered Sn_3Cl_3 cycle (Figure 1).

Table 1. Selected Structural Parameters [Distances (Å) and Angles (deg)] in the Crystal Structures of 1−5[a]

	1	2	3	4	5
			In Heterometallic Cation		
C=N	1.247(10), 1.280(10)	1.267(15), 1.294(15)	1.274(7), 1.281(6)	1.25(2), 1.27(2)	1.270(8), 1.286(8)
∠between the ls planes of the aromatic rings	2.74	1.99	0.60, 3.66	4.88	10.77
$Ni-N_{imino}$	1.842(6), 1.842(7)	1.833(8), 1.837(10)	1.838(4), 1.839(4), 1.840(4), 1.844(4)	1.834(13), 1.842(13)	1.848(5), 1.852(5)
$Ni-O_{phenoxido}$	1.862(5), 1.869(5)	1.861(7), 1.863(7)	1.855(3), 1.855(3), 1.857(3), 1.866(3)	1.861(11), 1.862(9)	1.859(4), 1.868(4)
$Sn^{II}-O_{phenoxido}$	2.269(5), 2.292(5)	2.258(7), 2.290(7)	2.290(3), 2.291(3), 2.291(3), 2.292(3)	2.235(10), 2.244(10)	2.294(4), 2.307(4)
$Sn^{II}-O_{alkoxido}$	2.599, 2.640	2.554, 2.706	2.585, 2.612, 2.619, 2.638	2.637, 2.671	2.601, 2.608
$Sn^{II}-Cl$	2.474(3)	2.449(3)	2.4665(13), 2.4718(15)	2.457(5)	2.5505(18)
$\angle N_{imino}-Ni-O_{phenoxido}$	177.6(3), 177.9(3)	177.9(4), 178.0(4)	177.48(17), 177.97(17)	176.1(6), 176.7(6)	177.1(2), 178.1(2)
τ_4 for Ni	0.032	0.029	0.032	0.051	0.034
$\angle O_{phenoxido}-Sn^{II}-O_{phenoxido}$	66.11(18)	65.7(2)	65.15(11), 65.26(11)	67.1(3)	64.98(13)
$\angle O_{alkoxido}-Sn^{II}-O_{alkoxido}$	164.85	165.47	164.65, 166.60	161.62	165.92
τ_5 for Sn^{II}	0.59	0.58	0.62,0.59	0.51	0.62
displacement of Sn^{II} from O_4 basal plane	0.154	0.108	0.180, 0.053	0.221	0.121
			In $[Sn^{IV}Cl_6]^{2-}$ in **1** and **2**		
$Sn^{IV}-Cl$	2.437(3), 2.448(2), 2.408(2)	2.489(3), 2.407(3), 2.415(5)			
			In $[Sn^{II}Cl_3]^-$ in **3**−**5**		
$Sn^{II}-Cl$			2.433(2), 2.4538(19), 2.5237(16), 2.4643(19), 2.5135(15), 2.5181(18)	2.444(7), 2.454(6), 2.495(5)	2.469(2), 2.4874(19), 2.530(2)

[a]For related structural parameters of **6** and **7**, see our article.[33]

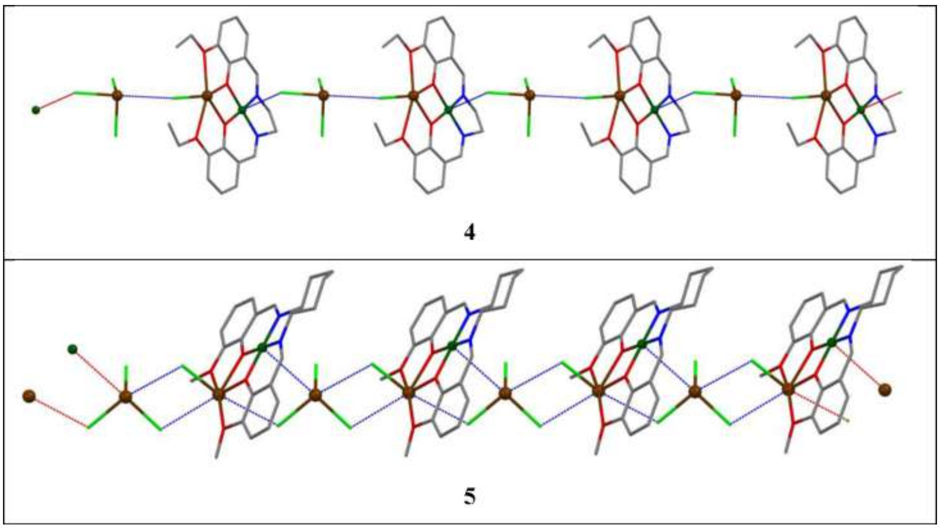


Figure 2. 1D chains in **4** and **5** supported by $Sn^{II}\cdots Cl$, $Ni^{II}\cdots Cl$ (only in **4**) ,and $Ni^{II}\cdots Sn^{II}$ (only in **5**) interactions. Color codes as labeled in Figure 1: dark green, nickel; brown, tin; red, oxygen; blue, nitrogen; green, chlorine; gray, carbon.

Altogether, an eight-membered Sn_4Cl_4 macrocycle is formed in the hexanuclear Ni_2Sn_4 structure of **3**. Two $M^{II}\cdots Cl$ interactions (one $Sn^{II}\cdots Cl$ and one $Ni^{II}\cdots Cl$) in the crystal structure of compound **4** interconnect the heterometallic cations $[Ni^{II}Sn^{II}L^{I}(Cl)]^+$ and the stannates $[SnCl_3]^-$ of the vicinal molecules (Figure 2), forming a one-dimensional chain along the crystallographic *c* axis. In **5**, two $Sn^{II}\cdots Cl$ contacts create a four-membered Sn_2Cl_2 cycle and a third contact generates a one-dimensional chain along the *b* axis, which is reinforced with a $Ni^{II}\cdots Sn^{II}$ interaction [3.5327(9) Å] (Figure 2).

Type and Dependence of $M^{II}\cdots Cl$ Interactions. As presented in Table 2, there are two types of $Sn^{II}\cdots Cl$ interactions observed in the crystal structures of **1**−**5** and the relevant reported systems **6** and **7**. The first concerns the $O_4Sn^{II}\cdots Cl-Sn^{IV}$ or $O_4Sn^{II}\cdots Cl-Sn^{II}$ interactions, and the second involves $O_4Sn^{II}-Cl\cdots Sn^{II}$. The participating Sn^{II} and Cl atoms are from the heterometallic species and the tri- or hexahalostannate. In **1** and **2** the first types of $Sn^{II}\cdots Cl$ contacts [$O_4Sn^{II}\cdots Cl-Sn^{IV}$, 3.805(3)−3.998(4) Å] are comparatively longer, while the analogous $Sn^{II}\cdots Cl$ interactions ($O_4Sn^{II}\cdots Cl-Sn^{II}$) are shorter [3.301(2)−3.872(2) Å] in **3**, **5** and **6**,

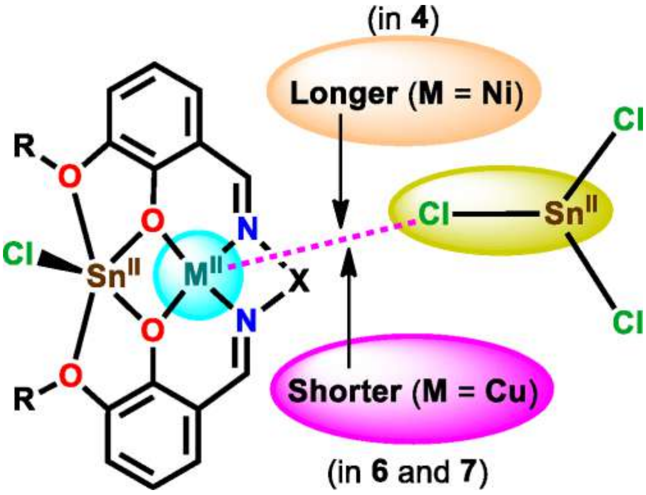
Table 2. Distances (Å) of $Sn^{II}\cdots Cl$ and $Ni^{II}/Cu^{II}\cdots Cl$ Interactions in 1−7

compound	$O_4Sn^{II}\cdots Cl{-}Sn^{IV/II}$	$O_4Sn^{II}{-}Cl\cdots Sn^{II}$	$Ni^{II}/Cu^{II}\cdots Cl{-}SnII$
1	3.805(3), 3.915(3)		
2	3.998(4)		
3	3.497(2), 3.621(2), 3.769(2), 3.872(2)	3.364(2), 3.369(1)	
4		3.530(5)	3.293(6)
5	3.301(2), 3.352(2)	3.258(2)	
6	3.805(1)	3.4594(9)	2.856(1)
7		3.471(5)	2.916(6)

indicating that the length of a such an interaction depends on the oxidation state of the chloride donor (Sn^{IV} in **1** and **2** or Sn^{II} in **3**, **5**, and **6**), being longer for Sn^{IV} (Scheme 2). However, the reverse $Sn^{II}\cdots Cl$ interactions in **3**−**7** are the shortest [3.258(2)−3.530(5) Å] among all $Sn^{II}\cdots Cl$ interactions. The pair of crystallographically inequivalent anions $[SnCl_3]^-$ in **3** has a $Sn^{II}\cdots Cl$ interaction [3.956(2) Å]. Overall, the number of $Sn^{II}\cdots Cl$ interactions increases with the lowering of the oxidation state of the chloride donor (Sn), revealing a higher potentiality of Sn^{II} in comparison to Sn^{IV} toward participating in a Sn···Cl interaction. As shown in Table 2, the $M^{II}\cdots Cl$ contacts (M = Sn, Ni, Cu) in **1**−**7** are of variable lengths. However, except for that (Table 2) in **2**, all of them are shorter than the VDW sum (3.92 Å) of Sn and Cl atoms. Relevant bond angle features are presented in Table S2 in the Supporting Information.

As already mentioned and presented in Figure 2, there is another visible interaction in the crystal structure of **4**, in which the metalloligand cation $[Ni^{II}Sn^{II}L^2(Cl)]^+$ interacts with the $[SnCl_3]^-$ anion via a $Ni^{II}\cdots Cl$ interaction. A close examination of the crystal structures of our previously reported copper(II) analogues (**6** and **7**, respectively) of **3** and **4** reveals that they are also stabilized by a similar interaction ($Cu^{II}\cdots Cl{-}Sn^{II}$) but much shorter [2.856(1) and 2.916(6) Å in **6** and **7**, respectively] than that [3.293(6) Å] in **4** (Scheme 3). In fact, they appear as covalent $Cu^{II}{-}Cl$ bonds in common crystal

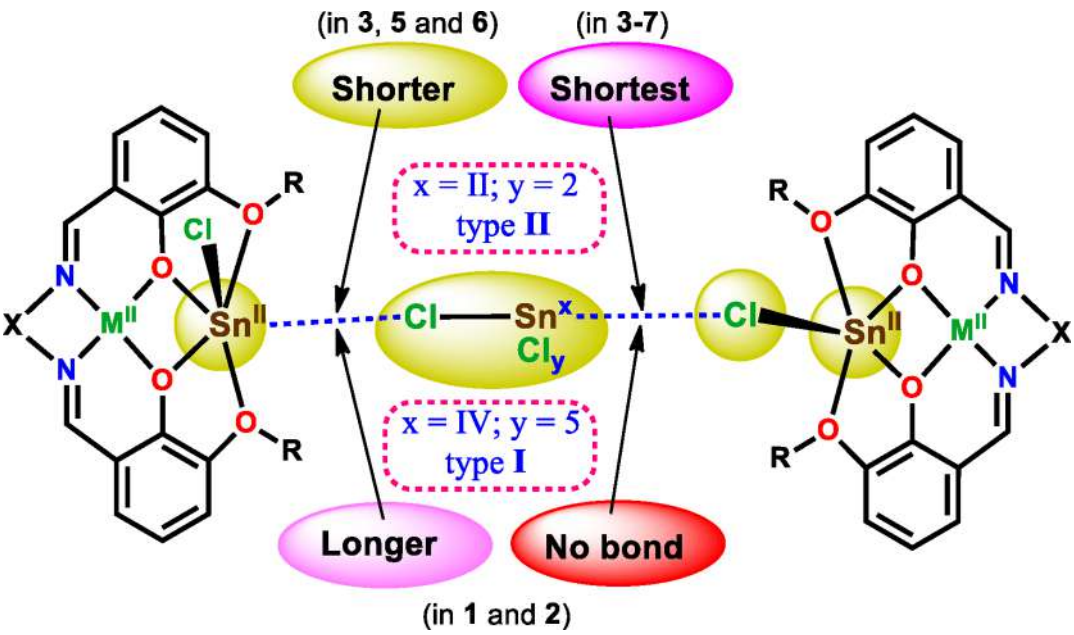
Scheme 3. Dependence of $M^{II}\cdots Cl$ Interaction on the Chloride Acceptor (M = Cu or Ni) in 4, 6, and 7



structure view programs, such as Mercury,[68] even though the distances are reasonably higher than the sum of the covalent radii of copper and chlorine. This comparison indicates that such an interaction ($Ni^{II}/Cu^{II}\cdots Cl{-}Sn^{II}$) shortens from a Ni^{II} (d^8) to a Cu^{II} (d^9) system, being dependent on the transition-metal ion, which acts as a halide acceptor.

Effect of $M^{II}\cdots Cl$ Interactions. The $M^{II}\cdots Cl$ interactions in the crystal structures of **1**−**7** significantly affect the bond distances around tin(IV) orand tin(II) in the corresponding anions $[Sn^{IV}Cl_6]^{2-}$ and $[Sn^{II}Cl_3]^-$, respectively. The Sn−Cl bonds that are involved in $M^{II}\cdots Cl$ interactions are longer than the others; for example, in the $[Sn^{IV}Cl_6]^{2-}$ type **I** systems (**1** and **2**) they are in the range of 2.437−2.489 Å, while the

Scheme 2. Dependence of $Sn^{II}\cdots Cl$ Bond on the Chloride Donor Center (Sn^{IV} or Sn^{II}) in Heterometallic Complex Salts 1−7[a]



[a]M = Ni (in **1**−**5**) or M = Cu (in **6** and **7**).

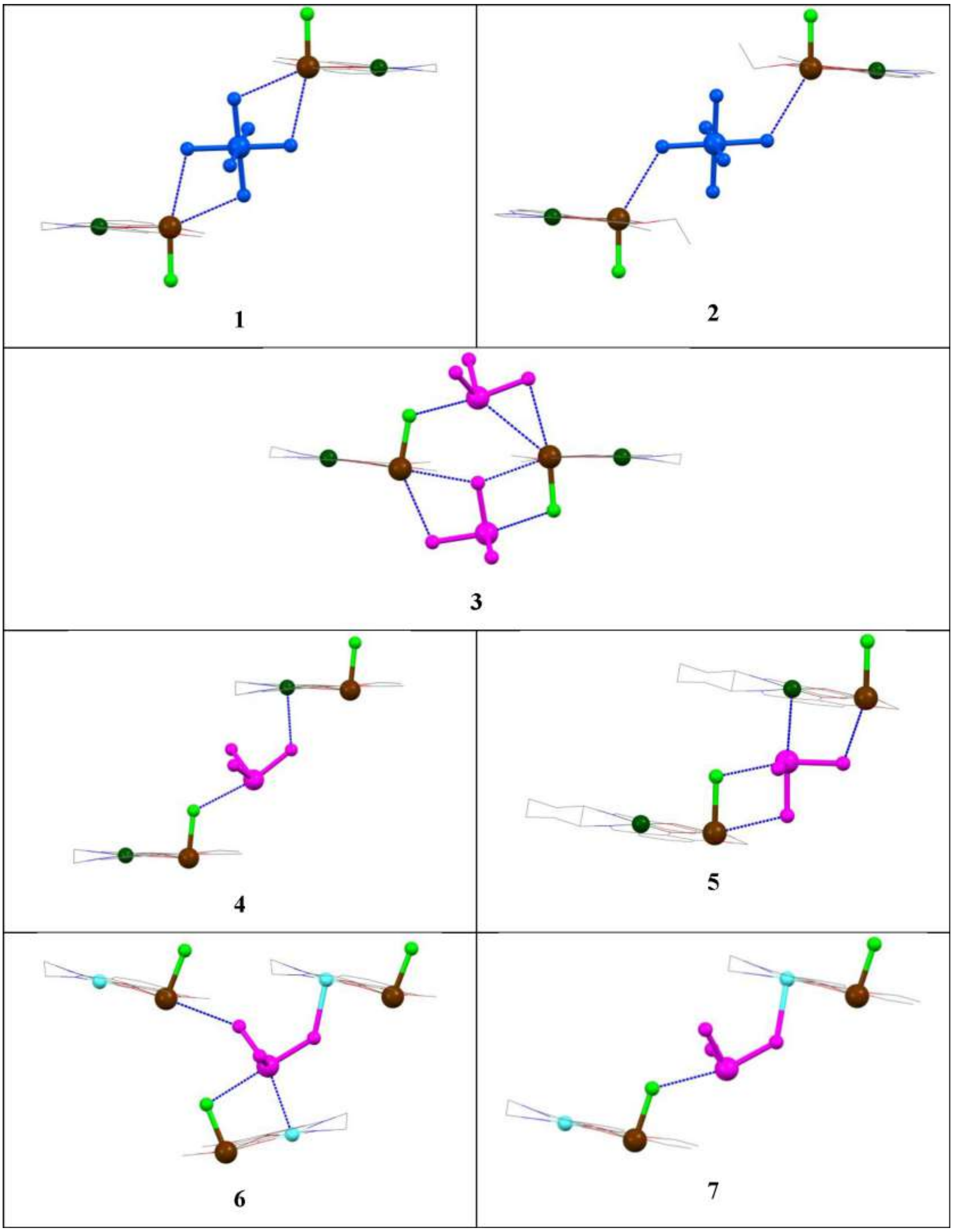


Figure 3. *Anti* and *syn* arrangements of Sn^{II}−Cl bonds in the heterometallic cation $[M^{II}Sn^{II}L(Cl)]^{+}$ in **1**−**7** (M = Ni in **1**−**5** and M = Cu in **6** and **7**). The anionic moieties $[Sn^{IV}Cl_6]^{2-}$ and $[Sn^{II}Cl_3]^{-}$ are presented in blue and pink, respectively, for clarity.

remaining interactions fall in the 2.408−2.415 Å range. A similar trend is observed in type **II** systems (**3**−**7**), where the Sn−Cl bonds that participate in $M^{II}\cdots Cl$ interactions are longer (2.465−2.565 Å) in comparison with the others (2.424−2.469 Å). However, all Sn−Cl bonds in the $[Sn^{IV}Cl_6]^{2-}$ anions of type **I** systems are shorter than those in the $[Sn^{II}Cl_3]^{-}$ anions in type **II** salts, therefore being dependent on the oxidation state of tin.

The geometrical arrangements of the heterometallic cations interconnected by a hexachlorostannate or a trichlorostannate are also affected by specific $M^{II}\cdots Cl$ interactions. As presented in Figure 3, the chloride ligands in the cations of **1** or **2** are in an *anti* arrangement interconnected by $[Sn^{IV}Cl_6]^{2-}$. In contrast, in **4**−**7** they prefer a *syn* arrangement reinforced by the trichlorostannate $Cl\cdots Sn^{II}$ interactions, partially supported by $Ni^{II}\cdots Cl$ (in **4**), $Cu^{II}\cdots Cl$ (in **6** and **7**), $Ni^{II}\cdots Sn^{II}$ (in **5**), and $Cu^{II}\cdots Sn^{II}$ (in **6**) contacts. Despite having $[Sn^{II}Cl_3]^{-}$ in its structure type **II**, complex salt **3** exhibits an *anti* arrangement of the chloride ligands in the heterometallic moieties.

Hirshfeld Surface Analysis and Crystal Packing. A Hirshfeld surface analysis[69] constitutes an effective tool to analyze the differences in crystal packing between structurally related compounds and is especially effective in studying polymorphs,[69] multicomponent crystals,[70] and supramolecular self-assemblies[71] with variable counterions. It provides a useful means for visualizing three-dimensional electron density as well as for finding all relevant noncovalent interactions in a crystal structure. We used the Crystal Explorer program[72] for this task.

The Hirshfeld surface plots are presented in Figure S1, while the 2D fingerprints of most important intermolecular interactions (Sn···Cl and H···Cl) are shown in Figure S2. Calculations for all complex salts (**1**−**7**) reveal the valuable interactions H···Cl and $\pi\cdots\pi$ (reflected by C···C in Table 3), along with the already discussed M···Cl or M···Sn interactions, and the usual H···H contacts. The calculated surface areas involved in different interactions are presented in Table 3, which show that one of the highest contributing interactions, in terms of occupied surface area, concerns H···H (25.2−35.4%) contacts. It corresponds to the molecular van der Waals interactions and does not vary significantly with the type of salt (**I** and **II**). The other valuable interaction is the H···Cl contact, covering a significant surface area in the crystal structures of all compounds (**1**−**7**). This contact corresponds to the C−H···Cl or O−H···Cl hydrogen-bonding interaction. However, the surface areas (∼45%) for the H···Cl contact in type **I** salts (**1** and **2**) are larger than those (29.1−38.7%) observed for the same contact in type **II** salts (**3**−**7**), which can be attributed to the larger size and three additional chlorides of the hexachlorostannate in comparison to the trichlorostannate. The surface area (7.7%) observed for the H···O contact in **3** is almost twice those (2.4−4.2%) found in the remaining compounds (**1**, **2**, and **4**−**7**), indicating that only **3** has the possibility of having any O−H···O/C−H···O hydrogen bonds, which is consistent with the fact that its crystal structure contains a water molecule as a solvent of crystallization. All structures have C···C contacts, although with significantly lower surface areas (2.2−9.9%), indicating the possibility of $\pi\cdots\pi$ interactions, which must be among the heterometallic cationic moieties in the crystal packing.

Table 3. Hirshfeld Surface Area (in Percent) Involved in Various Noncovalent Interactions in 1−7

compound	Sn···Cl	H···Cl	H···O	C···C	M···Cl[a]	M···Sn[a]	H···H
1	2.9	45	2.9	7.4	0	0	25.4
2	0.7	44.5	4.2	6.7	0	0	30.8
3	0.5	38.7	7.7	9.9	0	0	26.4
4	1.4	34.1	3.5	6.4	0	0	30.7
5	1.3	29.1	4.0	3.0	0.3	1	35.4
6	3.0	34.7	2.4	2.2	0	1.3	25.2
7	1.5	32.6	3.4	6.1	0	0	31.7

[a]M = Ni (in **1**−**5**) or M = Cu (in **6** and **7**).

The calculated surface areas (Table 3) obtained for Sn···Cl, M···Cl, or M···Sn interactions are not significantly large but confirm their presence or absence in a crystal structure. For example, all structures have the most important Sn···Cl interactions, while **5** contains both Ni···Cl and Ni···Sn interactions. The presence of a similar intermetallic interaction (Cu···Sn) in **6**, which was already presented in our earlier study,[33] is further attested by the Hirshfeld analysis. However, the surface areas related to the Cu···Cl interactions in **6** and **7** were not obtained by these calculations. Probably the interaction, being significantly shorter (2.856 and 2.916 Å in **6** and **7**, respectively), was automatically selected as a covalent Cu−Cl bond in the calculation by the Crystal Explorer program,[72] leading to the unavailability of the related surface area.

After evaluating the possibility of the presence of noncovalent interactions by a Hirshfeld surface analysis, we have verified their effects on the crystal packing of each complex salt (**1**−**7**). The H bonds along with other noncovalent interactions are presented in Figure 4 (and Figure S3), while their crystal packings with the corresponding anions ($[SnCl_6]^{2-}$ in **1** and **2**; $[SnCl_3]^-$ in **3**−**7**) in the interstitial positions are shown in Figure 5. The hexachlorostannate in type **I** salts (**1** and **2**) interconnects eight or six heterometallic cations $[Ni^{II}Sn^{II}L^1(Cl)]^+$ or $[Ni^{II}Sn^{II}L^2(Cl)]^+$ through four or three C−H···Cl bonds, respectively, along with the Sn···Cl interactions, leading to a two-dimensional architecture (Figure S4). Similarly to the type **I** complex salts (**1** and **2**), there are several H bonds interconnecting cations and anions in type **II** systems (**3**−**7**). For example, a three-dimensional extended network (Figure S4) is formed in the crystal packing of **3** (a type **II** salt) with the help of five C−H···Cl and two O−H···Cl hydrogen bonds interlinking several cations and anions. The crystal structure of **4** contains three C−H···Cl hydrogen bonds along with the Sn···Cl and Ni···Cl interactions, extending its 1D chain (Figure 2) into a double-layered two-dimensional motif in its crystal packing (Figure S4). However, the cutoff value of 130° selected for the D−H···A angle reveals no such H-bonding interaction in the crystal structure of **5**.

The crystal structure of the reported copper analogue (**6**) of **3** contains only one C−H···Cl hydrogen bond, but additional supports from the Sn···Cl, Cu···Cl, and Cu···Sn interactions result in a three-dimensional network in its crystal packing (Figure S4). Crystal packing of the copper analogue (**7**) of **4**

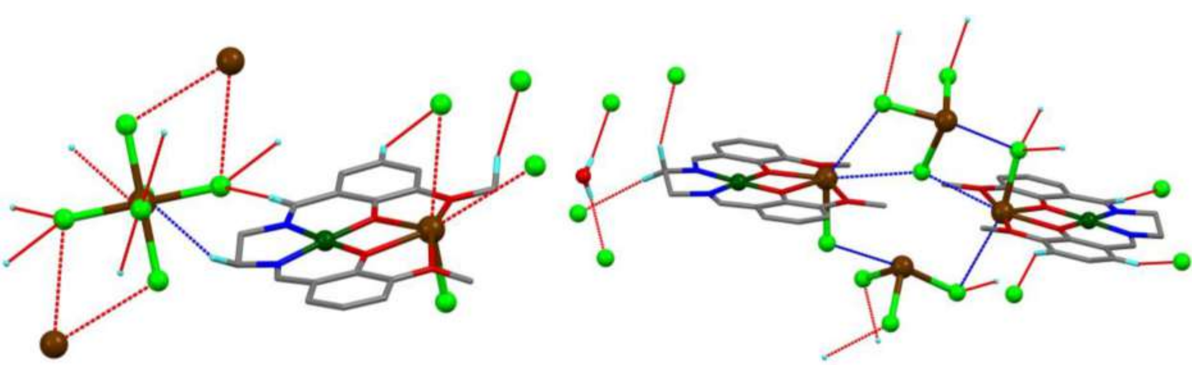


Figure 4. H bonds along with the Sn···Cl contacts in **1** (left) and **3** (right) as representatives of types **I** and **II**, respectively.

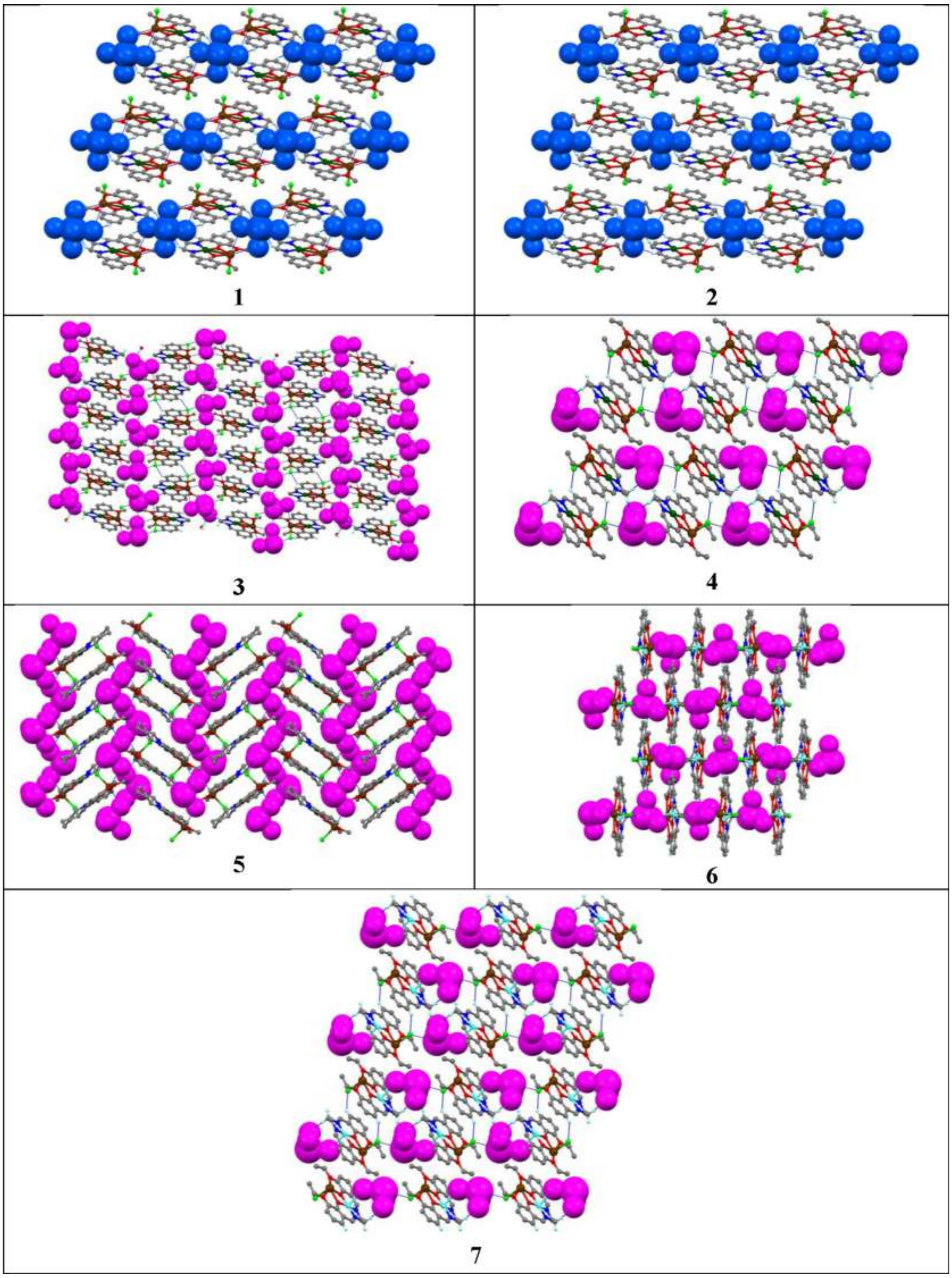


Figure 5. Crystal packings of **1**–**7**. $[SnCl_6]^{2-}$ (in **1** and **2**) and $[SnCl_3]^-$ (in **3**–**7**) are indicated by blue and pink space-filling models, respectively.

reveals a double-layered two-dimensional sheet (Figure S4), which is similar to that in **4**.

Crystal packings after removing the corresponding heterometallic cationic moieties show that hexachlorostannates in the type **I** complex salts align in a symmetrical fashion in the interstitial position while trichlorostannates in the type **II** salts prefer a random arrangement (Figure S5), presenting a striking difference in the crystal packings of the two different salt types. In addition to all of these noncovalent interactions, crystal packings of all structures have $\pi\cdots\pi$ interactions of variable strength (3.483–3.950 Å, Table S3), which agrees reasonably with the Hirshfeld surface analysis concerning the C⋯C contacts.

Theoretical Study. In order to understand the nature of bonding in the two different types of complex salts, compounds **1** and **3** as representatives of types **I** and **II**, respectively, were additionally investigated by theoretical methods: namely, the quantum theory of atoms in molecules (QTAIM), electrostatic potential (ESP), natural bond orbital (NBO), noncovalent interaction (NCI), and independent gradient model (IGM) analyses. The calculations were performed for the clusters **1′** and **3′** (Figure 6). The first is

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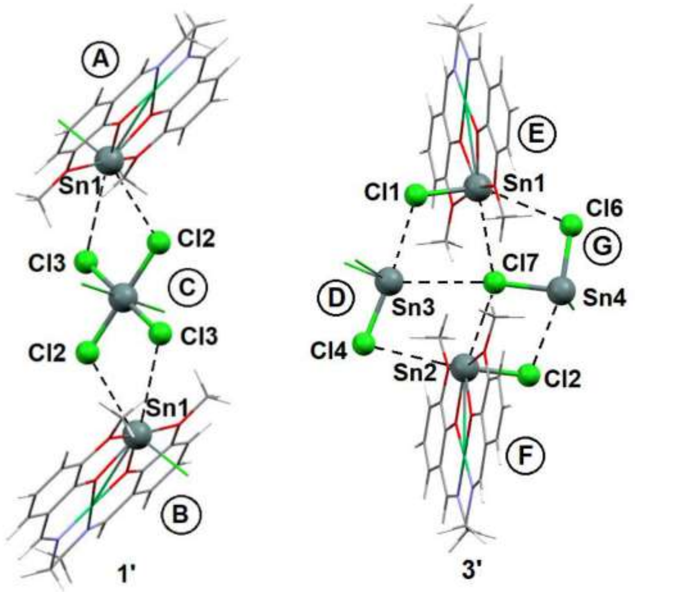

Figure 6. Calculated clusters $\mathbf{1'}$ and $\mathbf{3'}$.

composed of two $[NiL^1(SnCl)]^+$ units (A and B) and one $[SnCl_6]^{2-}$ dianion (C) which links the cationic moieties, while the second is formed by two $[SnCl_3]^-$ units (D and G) and by two $[NiL^1(SnCl)]^+$ units (E and F).

The QTAIM analysis of $\mathbf{1'}$ and $\mathbf{3'}$ indicates the presence of the bond critical points (BCPs) for four and seven interactions of the Sn···Cl type, respectively (Figure 7A,C, Table 4, and Table S3). Quite low values of the electron density at BCPs, ρ_b (0.0050−0.0106 au), positive values of its Laplacian, $\nabla^2\rho_b$, slightly positive and close to zero total energy densities, H_b, and very small ratios of the curvatures of the electron density distribution, $\lambda_1/\lambda_3 \ll 1$, indicate closed-shell type interactions of ionic character.[73,74] The NCI and IGM analyses demonstrate the regions of the negative sign$(\lambda_2)\rho(r)$ function mapped on the reduced density gradient isosurfaces around the Sn···Cl BCPs and the IGM-δg^{pair} bond signatures for the Sn···Cl interactions (Figure 7B,D and Figures S8 and S9).

Clusters $\mathbf{1'}$ and $\mathbf{3'}$ are both stabilized by the ionic interactions between the cation $[NiL^1(SnCl)]^+$ and the anion $[SnCl_6]^{2-}$ or $[SnCl_3]^-$. The calculated overall NBO and AIM net (in parentheses) charges on the fragments A, B, and C in $\mathbf{1'}$ are 0.93 (0.97), 0.93 (0.97), and −1.86 (−1.94) e, respectively, whereas those on the fragments D, E, F, and G

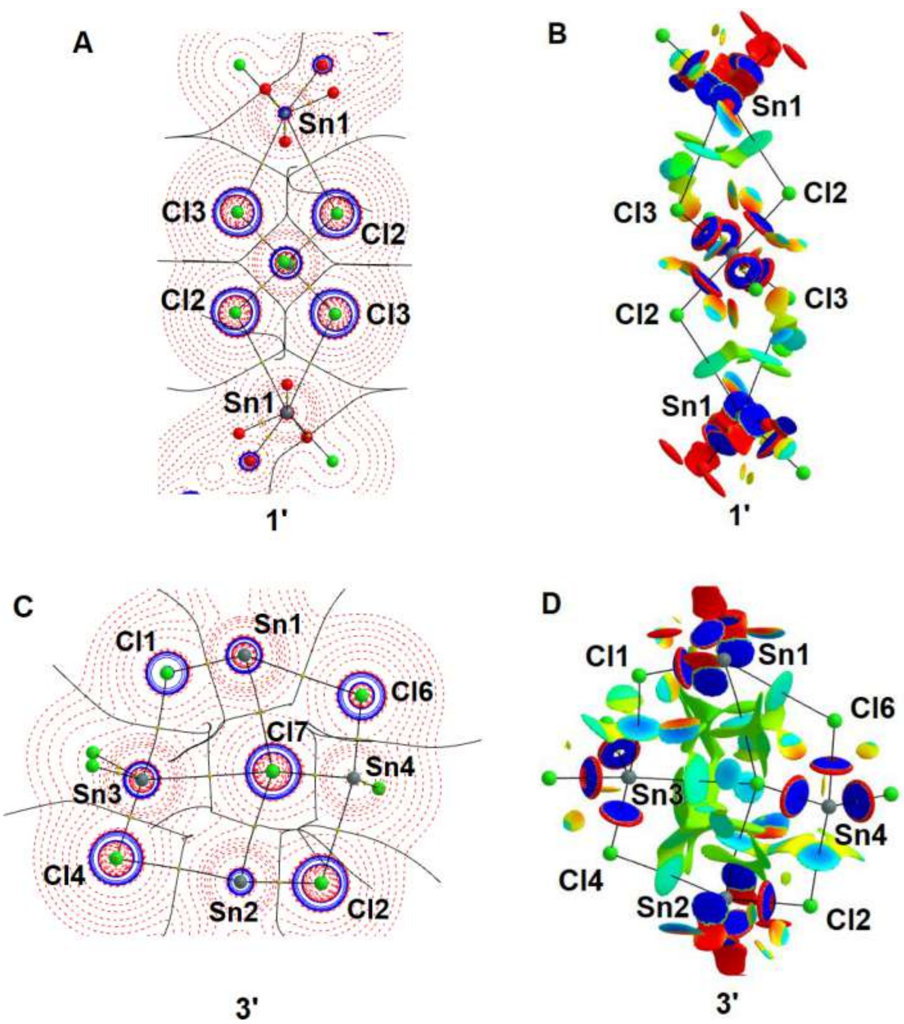

Figure 7. Contour line diagram of the Laplacian distribution $\nabla^2\rho(r)$ in the plane formed by the Sn1, Sn3, and Cl7 atoms, bond paths, zero flux surfaces (A, C) and the sign$(\lambda_2)\rho(r)$ function mapped on the reduced density gradient (s) isosurface with s = 0.5 au, cutoff of the maximum electron density 0.07 au, and blue−cyan−green−yellow−red color scale −0.015 < sign$(\lambda_2)\rho(r)$ < 0.015 (B, D) for $\mathbf{1'}$ (A, B) and $\mathbf{3'}$ (C, D).

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Table 4. Electron Density, ρ_b, its Laplacian, $\nabla^2\rho_b$, Potential and Kinetic Energy Densities, V_b and G_b, Second Eigenvalue of the Hessian Matrix, λ_2, at BCPs (in au), the Sn···Cl Internuclear Distance, d(Sn···Cl), (in Å), and IBSI Index Calculated for the X-ray Structures of 1 and 3 at M06-2X/DZP-DKH

contact	ρ_b	$\nabla^2\rho_b$	V_b	G_b	λ_2	d(Sn···Cl)	IBSI
			compound **1**				
Sn1···Cl3	0.0060	0.0144	−0.0022	0.0029	−0.0029	3.805	0.006
Sn1···Cl2	0.0050	0.0118	−0.0018	0.0024	−0.0022	3.915	0.005
			compound **3**				
Sn1···Cl7	0.0077	0.0197	−0.0032	0.0041	−0.0040	3.621	0.008
Sn1···Cl6	0.0064	0.0161	−0.0025	0.0033	−0.0030	3.769	0.006
Sn3···Cl1	0.0106	0.0311	−0.0053	0.0065	−0.0058	3.364	0.013
Sn2···Cl7	0.0093	0.0248	−0.0042	0.0052	−0.0047	3.497	0.011
Sn2···Cl4	0.0059	0.0130	−0.0021	0.0027	−0.0025	3.872	0.006
Sn4···Cl2	0.0106	0.0307	−0.0053	0.0065	−0.0058	3.369	0.013
Sn3···Cl7	0.0059	0.0115	−0.0022	0.0025	−0.0023	3.956	0.007

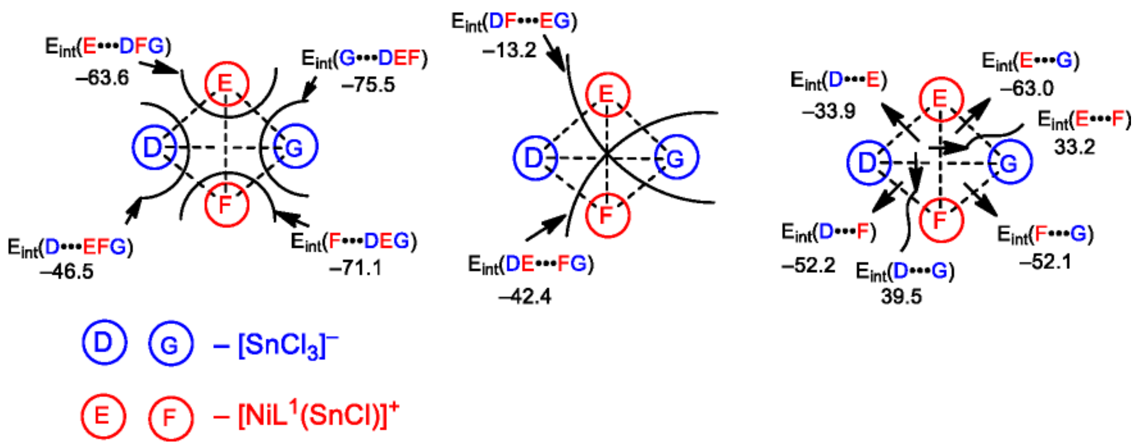

Figure 8. Schematic representation of the fragments in **3′** and corresponding interaction energies (in kcal/mol).

in **3′** are −0.99 (−0.94), 0.96 (0.96), 0.94 (0.94), and −0.90 (−0.96) e, correspondingly.

The interactions between fragments in cluster **3′** are schematically represented in Figure 8. Three types of interaction energies (E_{int}) between fragments in **3′** were calculated using the supermolecular approach (Figure 8) (see the Supporting Information for details). The first type of E_{int} corresponds to the interaction of an isolated ionic fragment D, E, F, or G with the ionic rest of the cluster (D···EFG, E···DFG, F···DEG, or G···DEF). The second type of E_{int} is associated with the interactions between two neutral fragments DE···FG and DF···EG. The third type of E_{int} corresponds to the isolated paired interactions between ionic molecules in **3′** (D···E, E···G, F···G, D···F, D···G, and E···F). All E_{int} values of the first and second types are strongly negative, which correlates with the ionic nature of the bondings. The least negative value of E_{int} was obtained for the DF···EG interaction (−13.2 kcal/mol). The paired E_{int} values of the third type were also strongly negative for the cation−anion interactions (D···E, E···G, F···G, and D···F), whereas they were positive for the anion−anion and cation−cation interactions (D···G and E···F). The paired E_{int} values between fragments A···C (or B···C) in **1′** is −93.5 kcal/mol, which is significantly higher, in absolute value, than E_{int} in **3′** due to the dianionic nature of $[SnCl_6]^{2-}$.

Despite the fragments which compose the clusters **1′** and **3′** having close to integer nonzero overall charges, the distribution of the effective atomic charges, electron densities, and ESPs, around these fragments is highly anisotropic. This anisotropy controls the relative orientations of the fragments in the clusters. In the $[SnCl_3]^-$ anions, regions of the most negative ESP are situated around the Cl atoms, while the least negative ESP is around the Sn atom (Figure 9A). On the other hand,

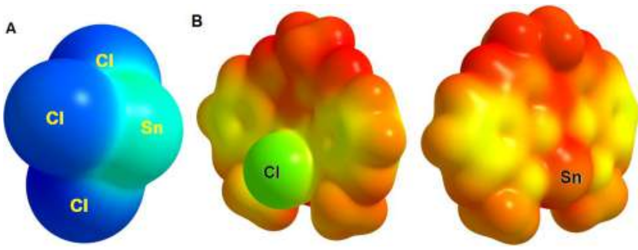

Figure 9. ESP distribution mapped on the 0.002 au isosurface of electron density for $[SnCl_3]^-$ (A) and $[NiL^1(SnCl)]^+$ (B) with the blue−cyan−green−yellow−red color scale −0.2 < ESP < 0.2 au (A) and −0.6 < ESP < 0.6 au (B).

the least positive ESP in the cation $[NiL^1(SnCl)]^+$ was found around the Cl atom, whereas one of the regions of the most positive ESP is near the Sn atom Figure 9B). Such a distribution dictates the existence of the paired Sn···Cl interactions between the fragments.

Despite the ionic nature of the interfragment interactions, the charge transfer (CT) also plays a certain role in the final charge distribution. The NBO analysis demonstrates that the principal contribution to the CT between fragments comes from the Cl → Sn transitions (Table 5), while the back Sn → Cl CT is negligible. The CT in **1′** is relatively smaller than that

Table 5. Second-Order NBO Perturbation Energies, $E(2)$ (in kcal/mol), Calculated for 1′ and 3′

fragment	transition[a]	$E(2)$
cluster 1′		
A⋯C, B⋯C	LP(Cl3) → σ*(Sn1−Cl)	8.9
	LP(Cl2) → σ*(Sn1−Cl)	6.0
cluster 3′		
D⋯E	LP(Cl1) → σ*(Sn3−Cl)	25.9
D⋯F	LP(Cl4) → σ*(Sn2−Cl)	12.5
E⋯G	LP(Cl6) → σ*(Sn1−Cl)	13.6
	LP(Cl7) → σ*(Sn1−Cl)	16.9
F⋯G	LP(Cl7) → σ*(Sn2−Cl)	25.7
	LP(Cl2) → σ*(Sn4−Cl)	23.5
D⋯G	LP(Cl7) → σ*(Sn3−Cl)	4.9

[a]LP = lone pair.

in **3′**. Thus, the Cl atoms serve as nucleophilic centers in the Sn⋯Cl interactions, whereas the Sn atoms are electrophilic centers. The electron density difference (EDD) map for the structure **3′** in the Cl1Sn3Cl7 plane is shown in Figure S10. The calculations also indicate that the negative net charge on the interacting Cl atoms decreases by 19 e upon bonding between $SnCl_6^{2-}$ and two $[NiL^1(SnCl)]^+$ fragments. The total net charge on the $SnCl_6^{2-}$ fragment in **1′** is −1.94 e. Following the definition of a halogen bond,[75] a tetrel bond may be defined as an attractive interaction between an electrophilic center associated with the tetrel atom (C, Si, Ge, Sn, Pb) and a nucleophilic center. Therefore, the computational results *allow the characterization of the interfragment interactions in* **1′** *and* **3′** *as ion-pair tetrel bonds.* Ion-pair-assisted halogen bonds[76−79] have already been reported, but we are not aware of an ion-pair-based tetrel bond.

An interesting feature of this system is the fact that in two of the seven interfragment interactions in **3′** (i.e., Cl1⋯Sn3 and Cl2⋯Sn4, Figure 6) the charge transfer occurs in the direction from the cationic fragment to the anionic fragment. This effect is possible because the NBO atomic charges on the Sn atom in $SnCl_3^-$ and on the Cl atom in $[NiL^1(SnCl)]^+$ are positive and negative, respectively (1.16 and −0.68 e). Additionally, this phenomenon can be interpreted in terms of the hard−soft acid−base (HSAB) formalism by an analysis of the ratios of the local softnesses s^+/s^- and s^-/s^+, the former called "relative electrophilicity" and the latter called "relative nucleophilicity" (see Computational Details). As was shown previously,[80] the site of a molecule with the highest s^+/s^- value is the most susceptible to the nucleophilic attack, whereas the site with the highest s^-/s^+ parameter is the most probable for the electrophilic attack. The calculated values of s^+/s^- and s^-/s^+ are the highest for the Sn atom of the $SnCl_3^-$ fragment and for the Cl atom of the $[NiL^1(SnCl)]^+$ fragment, respectively (Table 6 and Table S5), thus explaining the predominant charge transfer between these two atoms.

Meanwhile, the main contribution to the interaction energies comes from the electrostatic component (see the Supporting Information for details). Correspondingly, the strengths of the Cl1⋯Sn3 and Cl2⋯Sn4 interactions are lower than those for the Cl4⋯Sn2, Cl7⋯Sn2, Cl6⋯Sn1, and Cl7⋯Sn1 interactions due to the lower difference of ESP near the interacting atoms.

Table 6. Relative Electrophilicities (s^+/s^-) and Relative Nucleophilicities (s^-/s^+) (in au) Calculated for the Sn and Cl Atoms of the Optimized Fragments $SnCl_3^-$ and $[NiL^1(SnCl)]^+$

fragment	atom	s^+/s^-	s^-/s^+
$SnCl_3^-$	Sn	2.203	
	Cl		4.080
$[NiL^1(SnCl)]^+$	Sn	0.453	
	Cl		7.810

CONCLUSIONS

Due to the heterometallic complex formation ability of a compartmental Schiff base and the possible existence of a tetrel or halogen bond in $SnCl_2$- or $SnCl_4$-derived systems, the Schiff bases *N*,*N′*-ethylenebis(3-methoxysalicylaldimine) (H_2L^1), *N*,*N′*-ethylenebis(3-ethoxysalicylaldimine) (H_2L^2) and *N*,*N′*-bis(3-methoxysalicylidene)-1,2-cyclohexanediamine (H_2L^3) were reacted with $NiCl_2{\cdot}6H_2O$, $SnCl_2$, and $SnCl_4{\cdot}5H_2O$. Another purpose was also to investigate the competitive effects between two tin salts with different metal oxidation states on the product formation. The reaction without the fourth component ($SnCl_4{\cdot}5H_2O$) was also carried out for a comparative structural investigation. While the first set of reactions produced heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ complex salts (type **I**), the latter three-component reactions formed the $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ salts (type **II**).

In the first type (**I**), formulated as $[Ni^{II}Sn^{II}L(Cl)]_2[Sn^{IV}Cl_6]$ (L = L^1 (**1**), L^2 (**2**)), the heterobimetallic cations $[Ni^{II}Sn^{II}L(Cl)]^+$ are neutralized by hexachlorostannate via weak Sn^{II}⋯Cl interactions (3.80−3.99 Å). These contacts shorten (3.25−3.87 Å in **3**−**5**) appreciably when $[Sn^{IV}Cl_6]^{2-}$ is replaced by trichlorostannate $[Sn^{II}Cl_3]^-$ in the latter type (**II**) of complex salts: i.e., $[Ni^{II}Sn^{II}L(Cl)][Sn^{II}Cl_3]$ (L = L^1 (**3**), L^2 (**4**), L^3 (**5**)). However, theoretical calculations reveal that the strength of these interactions decreases from **1** (type **I**) to **3** (type **II**), as evidenced from the estimated interaction energies of the corresponding Sn^{II}⋯Cl contacts. The exception to the common bond length and bond strength (BLBS) appraisal[81,82] could be attributed to the crystal packing, charge transfers, bond ionicity (i.e., ionic interactions), and steric effects, which are different in type **I** and **II** systems.

The calculations demonstrate that the Cl → Sn charge transfer plays a significant role, despite the ionic nature of the interfragment interactions, allowing the interpretation of the Sn⋯Cl interactions in **1** and **3** as ion-pair-assisted tetrel bonds that, to our knowledge, have not been reported in a heterometallic inorganic complex.

In addition to the Sn^{II}⋯Cl contacts, a Ni^{II}⋯Cl interaction between the cation $[Ni^{II}Sn^{II}L^2(Cl)]^+$ and the anion $[Sn^{II}Cl_3]^-$ is found in the crystal structure of compound **4**. A structural comparison with our previously reported Cu^{II} analogues **6** and **7** $[Cu^{II}Sn^{II}L(Cl)][Sn^{II}Cl_3]$ (L = L^1 or L^2), which also belong to the type **II** salts, reveals that such a M^{II}⋯Cl contact shortens from a d^8 (Ni^{II}) to a d^9 (Cu^{II}) analogue, being dependent on the transition-metal ion while it acts as a halide acceptor.

In summary, a simple, quick, and energy-efficient synthesis, detailed crystal structures, and a theoretical evaluation on the interactions of two different types (**I** and **II**) of heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ and $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ salts are presented. This study evidences an unprecedented ion-pair-assisted tetrel bond in heterometallic inorganic systems and is of significance in crystal engineering and supramolecular chemistry: namely, to

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understand the stability of multimetallic components with different metal ions in various oxidation states and to provide ideas for designing other metal complexes with tetrel bonds. Further investigations related to the stabilities and activities in the solid state or solution of the compounds of this study and the synthesis of related systems are underway.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.cgd.1c00977.

Additional figures and tables as described in the text and additional computational details (PDF)

Accession Codes

CCDC 2094939–2094943 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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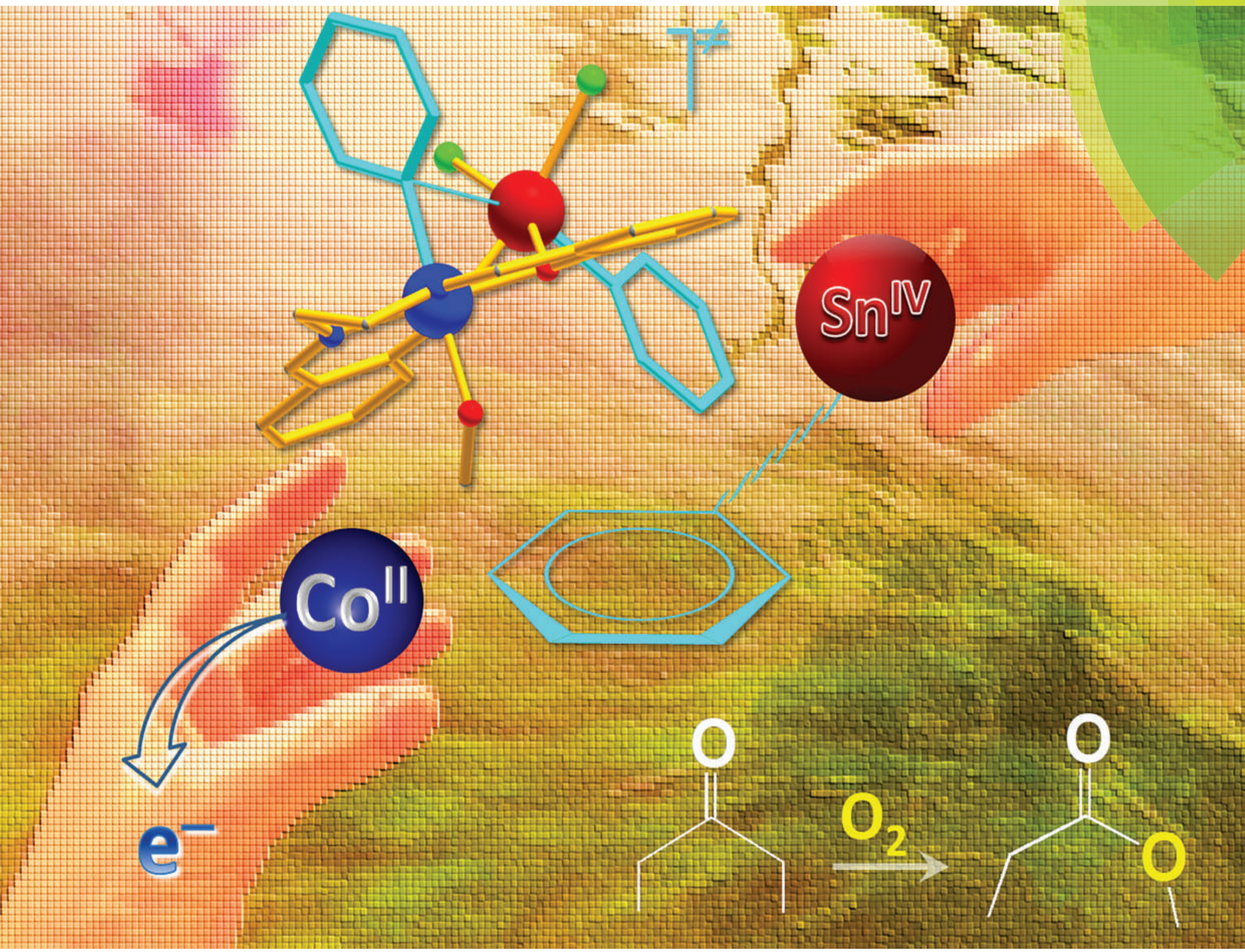

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Flexibility and lability of a phenyl ligand in hetero-organometallic 3d metal–Sn(IV) compounds and their catalytic activity in Baeyer–Villiger oxidation of cyclohexanone

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Flexibility and lability of a phenyl ligand in hetero-organometallic 3d metal–Sn(IV) compounds and their catalytic activity in Baeyer–Villiger oxidation of cyclohexanone†

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The single compartmental Schiff base *N*,*N*′-ethylenebis(salicylaldimine) (H_2L) and [$SnPh_2Cl_2$] were utilized to synthesize heterobimetallic 3d metal–Sn complexes, the $Co^{III}Sn^{IV}$ compound [{$SnPhCl_2$}($1\kappa O^2N^2,2\kappa O^2$-μ-L)(μ-OMe){CoPh}] (**1**), the $Ni^{II}Sn^{IV}$ compound [{$SnPh_2Cl_2$}($1\kappa O^2N^2,2\kappa O^2$-μ-L)Ni] (**2**) and the $Cu^{II}Sn^{IV}$ compound [{$SnPh_2Cl_2$}($1\kappa O^2N^2,2\kappa O^2$-μ-L)Cu] (**3**). Attempting to prepare the ethoxido bridged compound analogous to **1** (in ethanol) gives the phenylcobalt(III) complex [Co(κO^2N^2)Ph(H_2O)] (**1A**). Single crystal X-ray structure analyses reveal that **1** is derived from an intermetallic (Sn to Co) phenyl shift and that **1A** is a transmetallated product; in compounds **2** and **3**, the phenyl groups remain coordinated to Sn^{IV} but one of the π rings interacts with the 3d-metal. Thus, while systems **1** and **1A** show the lability of the phenyl ligand, **2** and **3** reveal its flexible nature. Theoretical DFT calculations demonstrate that the conceivable Ph group shift occurs in the oxidized Co^{III} intermediate [{$Sn^{IV}Ph_2Cl_2$}(κO_2N_2-μ-L){Co^{III}(MeO)}] (**5**) rather than in the corresponding Co^{II} species [{$Sn^{IV}Ph_2Cl_2$}(κO_2N_2-μ-L){Co^{II}(MeOH)}] (**4**). Their catalytic studies in the Baeyer–Villiger oxidation of cyclohexanone into ε-caprolactone with two different oxidants reveal that the sacrificial aldehyde method (with dioxygen/benzaldehyde) is better than that with aqueous H_2O_2 (30%). The effects of various reaction parameters such as solvent, catalyst amount, temperature, time and heating method were studied allowing the achievement of yields up to 83% with 89% selectivity.

Introduction

Among the huge diversity of heterometallic compounds,[1] those of the 3d–p blocks are within the less investigated systems, particularly the 3d metal–Sn ones,[2] in comparison with the highly explored heterometallic 3d–3d[3] and 3d–4f[4] systems. Usually, compartmental Schiff bases are highly suitable pro-ligands for the synthesis of numerous heterometallic compounds[5,6] including 3d metal–Sn ones.[5i–o,6] In the 1980–2002 period, utilizing a few of these Schiff bases, Cunningham *et al.* have prepared several heterobimetallic 3d metal–tin(IV) compounds[5i–o,6a–e] and investigated the ^{119}Sn Mössbauer spectroscopic properties in some cases[6a–e] to understand the coordination geometry. However, the formulations in most of the compounds were based only on elemental analysis and IR spectroscopy and only a few of them[5l–o,6d] were structurally characterized. Recently, we have reported a few Schiff base heterobimetallic $Cu^{II}/Ni^{II}–Sn^{IV}$ systems,[6j,k] for which non-covalent interactions (O–H⋯O/O–H⋯Cl/N–H⋯O/N–H⋯Cl/Cu⋯Cl/π⋯π) play a key role in their solid state stabilization.[6j] An inspiring aspect of such systems is the formation of uncommon products depending on the ligand, metal oxidation state and solvent.[5m–o,6j,k] For example, the reaction of dichlorodimethyltin(IV) with the copper metalloligand of *N*,*N*′-ethylenebis(3-methoxysalicylaldimine) produces three component salt cocrystals (charged species), whereas the related reaction with an analogous system containing the ethoxido substituent in the ligand instead of methoxido gives two component cocrystals (neutral species).[6j] Thus, the synthesis and structural characterization of new heterometallic 3d metal–Sn compounds, which can provide new insight into crystal engineering and organometallic chemistry, deserve further attention. In addition, the possibility, in principle, of a σ-bonded phenyl to form a π interaction with another metal should also be tested.

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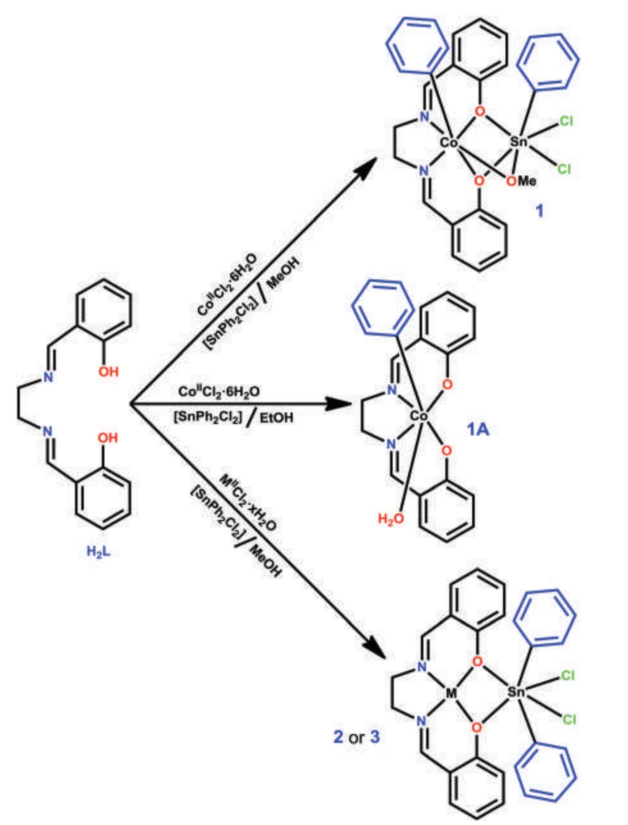
† Electronic supplementary information (ESI) available: Discussion on the structural scarcity of related organometallic compounds, Tables S1–S5 and crystallographic information in CIF format. CCDC 1540201–1540204 for **1–3** and **1A**. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c7dt02534c

In addition, the structural diversity of Schiff base metal complexes has inspired theoretical studies addressed not only to understand structure–property relationships[3a,7] but also to predict and explain several structural phenomena including the formation of unusual compounds.[8] In the present study, we have also investigated the stability of the newly synthesized compounds, as well as of the hypothetical related ones, toward the understanding of their formation.

Inspired by our previous studies[6j,k] and with the aim to explore the structural role of different 3d-metal ions in heterometallic phenyl–Sn^{IV} systems, we have now varied the 3d-metal salt (Co^{II} to Cu^{II}) keeping the starting organotin(IV) compound. Accordingly, the simplest and single compartmental Schiff base *N*,*N*′-ethylenebis(salicylaldimine) (H_2L)[9] is reacted with the d-block salt $MCl_2 \cdot xH_2O$ (M = Co, Ni or Cu; x = 2 or 6) and the diphenyldichlorotin(IV) compound $[SnPh_2Cl_2]$, producing the new heterobimetallic 3d metal–Sn^{IV} compounds $[\{SnPhCl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})(\mu\text{-OMe})\{CoPh\}]$ (**1**), $[\{SnPh_2Cl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})Ni]$ (**2**) and $[\{SnPh_2Cl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})Cu]$ (**3**) and the mononuclear phenylcobalt complex $[Co(\kappa O^2N^2)Ph(H_2O)]$ (**1A**) (Scheme 1).

On the other hand, the Baeyer–Villiger (B–V) oxidation, named after A. Baeyer and V. Villiger who first reported the reaction in 1899, provides an important method to synthesize valuable esters and lactones by the oxidation of linear and cyclic ketones.[10] Peracids (mainly in earlier stages) or strong oxidant peroxides are applied in this reaction. However, the use of peracids (*e.g.*, trifluoroperacetic acid, perbenzoic acid and *m*-chloroperbenzoic acid) has limitations, such as waste treatment, reactant instability, shock sensitivity and expensive protection and storage.[11] In recent years, several approaches to more atom-efficient and eco-friendly oxidants like dioxygen (O_2) and hydrogen peroxide (H_2O_2), following sustainable chemistry principles, have been proposed.[12] However, milder conditions and more atom-efficient oxidants require a catalyst. Thus, a wide range of metal catalysts[13] including organotin (IV),[12i,14] have been reported, as well as a few Schiff base transition metal complexes.[10f–h] Thus, the newly synthesized Schiff base heterometallic complexes **1–3** containing both transition and tin(IV) metal ions could also be useful catalysts for this reaction. Accordingly, we have investigated their catalytic activity in the B–V reaction and, for comparison, that of a cobalt complex (**1A**) containing the same ligand but without the organotin(IV) group.



Scheme 1 Synthesis of **1–3** and **1A**. M = Ni, x = 6 (**2**); M = Cu, x = 2 (**3**) (for the correct ligand arrangements, see Fig. 1 and 2).

Results and discussion

Synthesis and characterization

The single compartmental Schiff base *N*,*N*′-ethylenebis(salicylaldimine) (H_2L) (Scheme 1) was synthesized by condensing salicylaldehyde with 1,2-ethelenediamine (2 : 1) in methanol following an earlier reported procedure,[9] and it was characterized by NMR spectroscopy and elemental analysis which matched the reported data.[9] The reactions of H_2L with $MCl_2 \cdot xH_2O$ (M = Co, Ni or Cu; x = 2 or 6) and $[SnPh_2Cl_2]$ in methanol, in an open atmosphere and at room temperature, produce the heterobimetallic compounds $[\{SnPhCl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})(\mu\text{-OMe})\{CoPh\}]$ (**1**), $[\{SnPh_2Cl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})Ni]$ (**2**) and $[\{SnPh_2Cl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})Cu]$ (**3**) (Scheme 1). Attempts to isolate the ethoxido bridged analogue of **1** by performing the reaction in ethanol resulted in the formation of the mononuclear phenylcobalt(III) compound $[Co(\kappa O^2N^2)Ph(H_2O)]$ (**1A**) (Scheme 1). The H_2L ligand remained intact along the reactions, unlike the related Schiff bases *N*,*N*′-ethylenebis(3-methoxysalicylaldimine) and *N*,*N*′-ethylenebis(3-ethoxysalicylaldimine) which, under similar experimental conditions, could break affording the 1,2-ethylenediammonium cation.[6j,k]

In the formation of **1** and **1A**, an unusual phenyl shift from Sn^{IV} to Co^{III} has occurred, whereas in the cases of **2** and **3** the phenyl ligands remained bound to Sn^{IV} but approach the 3d-metal ion though a metal⋯π interaction (see below). Compound **1A** can be a transmetallated product of the reaction between the hypothetical intermediate $[Co^{III}L(H_2O)_2]Cl$[15] (produced in the reaction of H_2L with $CoCl_2 \cdot 6H_2O$) and $[SnPh_2Cl_2]$. However, it was not possible to characterize the Sn product of the reaction.

Compounds **1** and **2** were isolated in good yields (69 and 62%, respectively) from the reactions of the Schiff base, the metal salt $MCl_2 \cdot xH_2O$ and $[SnPh_2Cl_2]$ in 1 : 1 : 1 molar ratios. A lower yield was obtained for **3** in view of the concomitant formation of the already known tetracopper[16] compound $[Cu_2(\kappa O_2N_2\text{-L})Cl_2]_2$. Complex **1A** was found in higher yield (76%). The open air formation of the OMe^- bridged $Co^{III}Sn^{IV}$ compound (**1**) resulted from a Co^{II}/Co^{III} oxidation and phenyl shift, in a process that is somewhat related to the previously reported analogue $[\{Sn^{IV\,n}BuCl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})(\mu\text{-OMe})\{Co^{III}Cl\}]$,[5m] where no Sn-to-Co ^{n}Bu shift was observed but then the cobalt center ended with a Cl^- ligand.

In the IR spectra, all compounds exhibit a medium intense absorption peak in the 1624–1658 cm^{-1} range due to ν(C=N). An additional broad peak at 3429 cm^{-1} is observed for the co-ordinated water in the IR spectrum of **1A**. All compounds were also characterized by elemental analyses and single crystal X-ray diffraction studies.

Crystal structures

The crystal structures of the heterobimetallic compounds $[\{SnPhCl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})(\mu\text{-OMe})\{CoPh\}]$ (**1**), $[\{SnPh_2Cl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})Ni]$ (**2**) and $[\{SnPh_2Cl_2\}(1\kappa O^2N^2,2\kappa O^2\text{-}\mu\text{-L})Cu]$ (**3**), and of the mononuclear phenylcobalt(III) $[Co(\kappa O^2N^2)Ph(H_2O)]$ (**1A**) are shown in Fig. 1. Selected bond distances and angles are included in Table 1.

Compounds **2** and **3** are isostructural and crystallize in the monoclinic *P*21/*n* space group, **1** in the orthorhombic *Pbca* and **1A** in the non-centrosymmetric monoclinic *Cc*. In all the compounds the L^{2-} Schiff base ligand chelates (in the *OO′NN′* mode) the transition metal cations, and *OO′*-bridges to a $\{SnPhCl_2\}$ (in **1**) or to a $\{SnPh_2Cl_2\}$ set (in **2** and **3**). A bridging methoxide anion in **1** fulfills the Sn(IV) octahedral coordination environment as well as, together with a phenyl group, the octahedral geometry of Co(III). The nickel (in **2**) and copper cations (in **3**) adopt a perfect square planar geometry (τ_4 = 0.03 and 0.01, respectively).[17] The cobalt(III) metal center in **1A** accomplishes its six-coordinate cluster with a phenyl group and a water molecule.

The octahedral volume around the tin(IV) cation is much shorter in **1** (14.280 Å^3) than that in **2** or **3** (16.086 or 16.196 Å^3, respectively; see Table 1)[18] and the angle variances vary considerably (Table 1) in the order: **1** (143.43°^{2}) > **2** (128.30°^{2}) > **3** (112.55°^{2}). Such observations can be accounted for by the presence (in **1**) of the methoxide anion bridging the two metals which led to shorter Sn–O$_{phenoxido}$ bonds in **1** (avg.

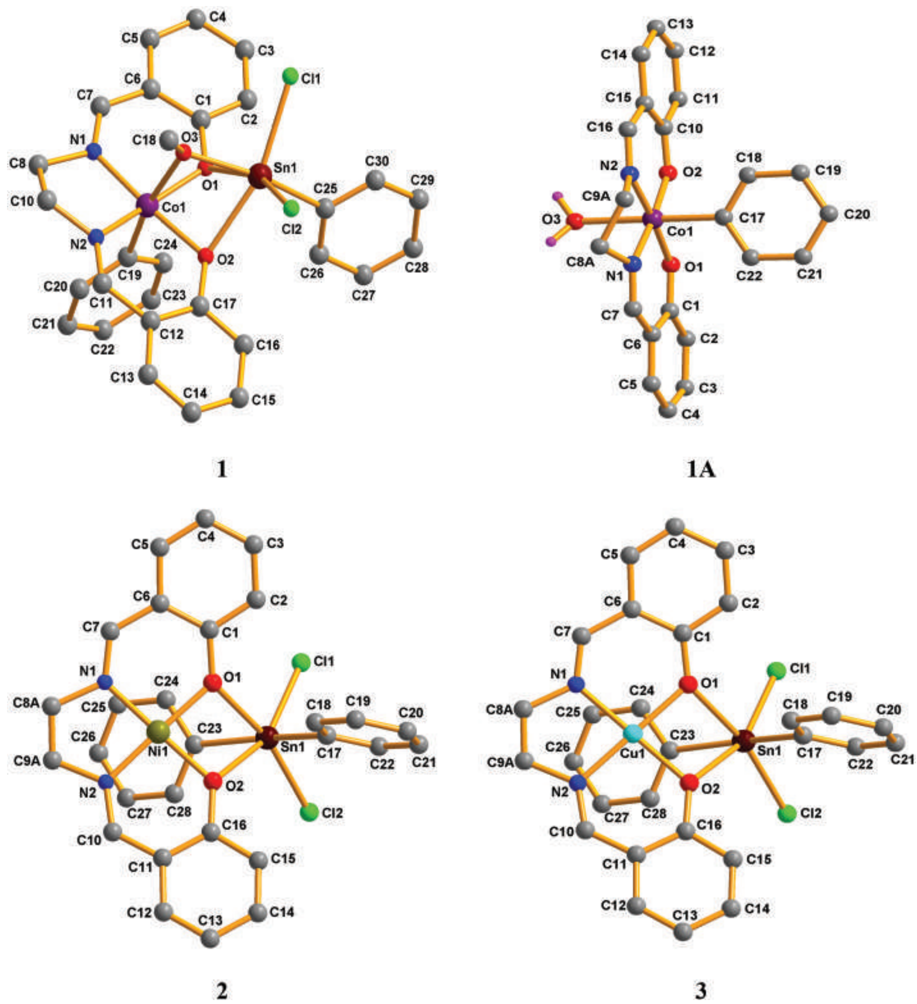


Fig. 1 Idealized ball and stick presentations with the atom labeling scheme of the crystal structures of **1–3** and **1A**.

Table 1 Comparison of selected bond distances (Å) and angles (°) in complexes **1–3** and **1A**. M = Co (**1** and **1A**), Ni (**2**) and Cu (**3**)

	1	**2**	**3**	**1A**
In the L^{-2} ligand				
C=N	1.280(5), 1.278(5)	1.278(3), 1.284(3)	1.266(4), 1.281(4)	1.276(10)
∠ between the l.s. planes of the aromatic rings (°)	8.46	23.40	23.32	3.37
Around metal ions				
Sn octahedral volume ($Å^3$)	14.280	16.086	16.196	—
Sn angle variance ($degree^2$)	143.43	128.30	112.55	—
$M–N_{imine}$	1.855(3), 1.856(3)	1.8354(18), 1.8380 (19)	1.913(2), 1.905(2)	1.872(7), 1.870 (7)
$M–O_{phenoxido}$	1.910(2), 1.906(3)	1.8654(13), 1.8517 (14)	1.9161(16), 1.9067 (16)	1.893(6), 1.899 (5)
$M–O_{methoxido/water}$	2.093(3)	—	—	2.156(5)
$M–C_{phenyl}$	1.945(4)	—	—	1.941(6)
$Sn–O_{phenoxido}$	2.275(2), 2.281(2)	2.4240(13), 2.4426 (14)	2.4218(15), 2.4545 (16)	—
$Sn–O_{methoxido}$	2.018(3)	—	—	—
$Sn–C_{phenyl}$	2.121(4)	2.129(2), 2.134(2)	2.129(2), 2.138(2)	—
Sn–Cl	2.4052(11), 2.3959 (11)	2.4438(6), 2.4608(6)	2.4339(7), 2.4580(7)	—
$∠N_{imine}–M–O_{phenoxido}$	175.58(12), 170.41 (12)	177.57(8), 178.99(8)	177.76(9), 179.06(9)	175.2(4), 175.3 (4)
$∠O_{methoxido/water}–M–C_{phenyl}$	169.29(14)	—	—	178.9(3)
∠Cl–Sn–Cl	96.40(4)	99.91(3)	99.74(3)	—
M⋯Sn	2.9632(6)	3.2471(5)	3.2209(5)	—
M⋯M′	—	4.6909(5)	4.7616(5)	—
Displacement of M from the least squares N_2O_2 plane	0.106	0.014	0.011	0.078
Displacement of Sn from the least squares O_2Cl_2 plane	0.241	0.054	0.070	—
M⋯π(Ph)	—	3.545	3.458	—

2.278 Å) relative to **2** and **3** (avg. 2.433 and 2.432 Å, Table 1), also to shorter Sn–Cl lengths (avg. 2.401 Å in **1**, against 2.453 Å in **2** and 2.446 Å in **3**) and to a proximity between the metals (see Table 1). The binding of the $\{SnPhCl_2\}$ cluster in **1** resulted in a slight lengthening of the $Co^{III}–O_{phenoxido}$ and the Co^{III}–C, and in a shortening of the Co^{III}–N bond distances, as compared to **1A**.

Compound **1** is engaged with three four-membered rings of type MO_2Sn (only one in **2** and **3**), as well as one five-membered MC_2N_2 and two six-membered MC_3NO rings (Fig. 2).

One of the most interesting findings of the current study is the high affinity of the transition metal ion toward a phenyl group from the Sn(IV) unit. In compounds **1** and **1A** the group shifted from Sn^{IV} to Co^{III} and the isolation of an unusual organocobalt(III)–organotin(IV) compound (**1**) was thus achieved. In complexes **2** and **3** the Ni^{II} and the Cu^{II} metal ions do not abstract the phenyl group from tin but establish a relevant non-covalent metal⋯π interaction, slightly less intense with nickel than with copper (metal⋯centroid distances of 3.545 and 3.458 Å for **2** and **3**, respectively, Table 1).

For structural comparison of our compounds (**1–3** and **1A**) with reported ones, we made several searches on the CSD (Cambridge Structural Database, ConQuest Version 1.18)[15] which reveal that no compound with Co–Ph and Sn–Ph fragments had been reported, and thus complex **1** is the first example. Other searches made for the heterometallic combination of Ni/Cu and Sn–Ph fragments, excluding the C=N moiety, result in only 15 compounds attesting that **2** and **3** also belong to a rare family. Literature survey concludes that

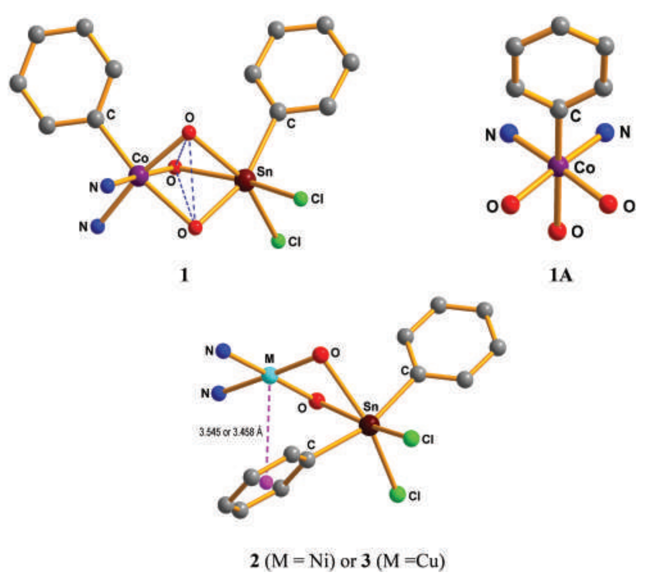


Fig. 2 Geometry around the metal centers in **1–3** and **1A**.

the intermetallic phenyl-group-shifts as observed in **1** are also not common.[19] A more detailed discussion on the structural scarcity of these organometallic compounds is presented in the ESI.†

Flexibility and lability of a phenyl ligand

It can be assumed that the reaction between H_2L, $CoCl_2·6H_2O$ and $[SnPh_2Cl_2]$ initially produces the heterobimetallic inter-

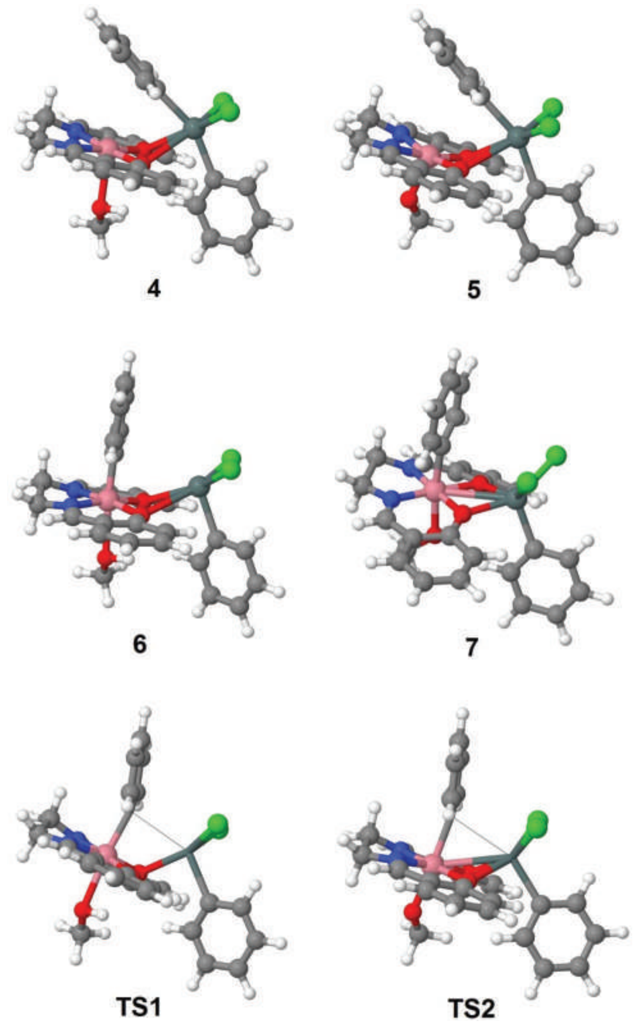


Fig. 3 Calculated equilibrium structures.

mediate compound [{$Sn^{IV}Ph_2Cl_2$}(κO_2N_2-μ-L){Co^{II}(MeOH)}] (**4**)[5m] in which one axial coordination place of Co is occupied by a solvent MeOH molecule and a Ph group from the {$SnPh_2Cl_2$} moiety is leaned toward the other axial site (Fig. 3 and Scheme 2). The cobalt(II) metal in **4** could easily undergo aerobic oxidation followed by deprotonation of the ligated methanol affording [{$Sn^{IV}Ph_2Cl_2$}(κO_2N_2-μ-L){Co^{III}(MeO)}] (**5**). The Ph group shift from the Sn atom to the Co atom could occur either before oxidation (*i.e.* in complex **4**) or after oxidation (*i.e.* in complex **5**).

To clarify this situation, theoretical DFT calculations have been carried out. The transition state **TS1** was found for the Ph group shift in the non-oxidized cobalt(II) complex **4** (Fig. 3). Both initial complex **4** and **TS1** as well as product **6** of this reaction step have the doublet ground spin state while the quartet spin state is clearly higher in energy (Table S2, ESI†). Analysis of the electronic structure in intermediate **6** indicates that the spin density is localized mostly at the {$SnCl_2C_{Ph}$} moiety and it is nearly zero at the Co atom (Fig. 4). Thus, the Ph group shift in **4** induces an intramolecular redox process, and complex **6** should be better described as the cobalt(III) species with a single-electron reduced {$SnPhCl_2$} fragment. The calculated activation energy, $\Delta G_s^{\ddagger}$, of the Ph shift in **4** is 29.5 kcal mol^{-1}. With such a high value of the activation barrier, complex **4** should be efficiently accumulated and easily detected in the reaction mixture, what contradicts the experimental observations.

The oxidized cobalt(III) complex **5** has almost degenerate singlet and triplet spin states, the latter being only 0.6 kcal mol^{-1} lower in energy, while the transition state **TS2** (Fig. 3) corresponding to the Ph shift in complex **5** and product **7** of this step are singlet structures. The calculated activation barrier of the Ph shift in **5** is 15.7 kcal mol^{-1}. Taking into account that the overall **5** → **1** transformation is highly exergonic (ΔG_s = −16.4 kcal mol^{-1}), such a relatively low activation barrier does not permit an efficient accumulation of **5** in the reaction mixture correlating with the experimental data. Thus, in accord with the calculations, the Ph group shift occurs after the oxidation of Co^{II} to Co^{III}. The following isomerization of the Sn coordination sphere in complex **7** (Scheme 2) to intermediate **8** induces the formation of the bridging bond between the OMe ligand and the Sn atoms to produce the final product **1**.

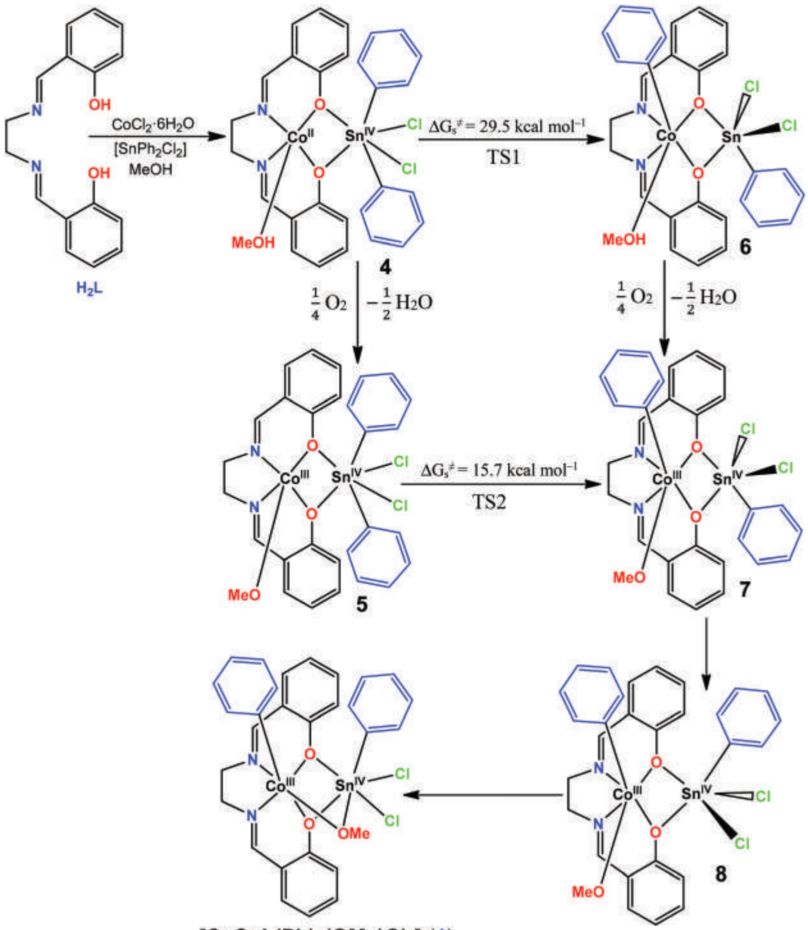
The higher electrophilic character of Co^{III}, in comparison with Ni^{II} and Cu^{II}, promotes the phenyl shift from Sn^{IV} to form the phenyl–Co^{III} complexes **1** and **1A**. The weaker electrophilicity of Ni^{II} and Cu^{II} is not sufficient for such an intermetallic phenyl transfer to occur, resulting nevertheless into a M⋯π(Ph at Sn^{IV}) interaction (M = Ni^{II} or Cu^{II}, complex **2** or **3**) which can represent an intermediate step of the phenyl ligand shift from Sn^{IV} to the 3d-metal ion.

Baeyer–Villiger oxidation of cyclohexanone

The Baeyer–Villiger (B–V) reaction is applied for the catalytic conversion of cyclohexanone into ε-caprolactone (Scheme 3). Possible oxidants used in this reaction include organic peroxyacids, hydrogen peroxide or molecular oxygen combined with a sacrificial aldehyde.[10c,12b,14a,d,i] For example, O_2/benzaldehyde is effective in this reaction.[14a,d,i–k] In our study, experiments using the heterometallic complex **1** as the pre-catalyst, carried out with two different oxidants (aqueous hydrogen peroxide *vs.* dioxygen/benzaldehyde) under the same reaction conditions, revealed that the sacrificial aldehyde method is better than the other one (yields of 78.4% and 27.9%, respectively, in 1,2-dichloroethane after 4 h, Table 2). The study of the influence of other reaction parameters (variations of solvents, temperature, catalyst concentration and time) was also carried.

In a blank experiment (without any catalyst) the yield of ε-caprolactone was only *ca.* 10% (Table 3, entry 1), whereas in the presence of [$SnPh_2Cl_2$], $CoCl_2·6H_2O$, $NiCl_2·6H_2O$, $CuCl_2·2H_2O$ or the Schiff base (H_2L), the observed yields (Table 3, entries 2–6) were also much lower than that for the catalytic system **1**.

As presented in Table 3, solvent screening reveals that **1** is active in 1,4-dioxane (entry 7, Table 3) and especially in 1,2-



Scheme 2 Proposed mechanism for the intermetallic phenyl shift in **1** (for the correct ligand arrangements, see Fig. 3).

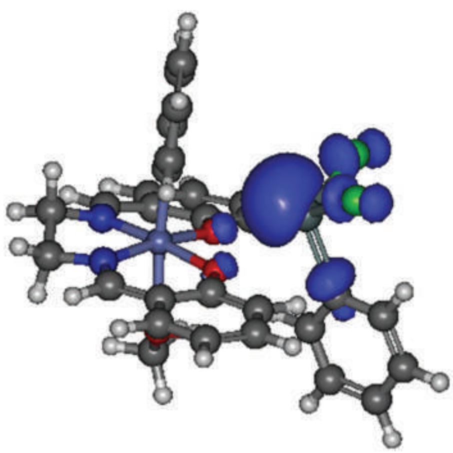
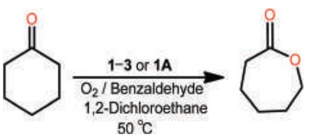


Fig. 4 Spin density distribution in complex **6**.



Scheme 3 Aerobic B–V oxidation of cyclohexanone catalyzed by **1–3** or **1A**.

dichloroethane (1,2-DCE) (entry 9, Table 3). No desired product was detected in other common organic solvents such as dichloromethane (DCM), *N*,*N*-dimethylformamide (DMF), acetonitrile (MeCN) and dimethylsulfoxide (DMSO). Complex **1** is soluble in all the tested solvents. 1,2-DCE has a high resistance to oxidizing agents and typically favours higher cyclohexanone conversion in comparison to other solvents.[12f,j,k,14a]

The amount of catalyst was also optimized under fixed reaction conditions (Table 3, entries 8–10). A slightly lower yield (71.1%, entry 8, Table 3) of ε-caprolactone was obtained with 0.1 mol% of **1** in comparison with that (78.1%, entry 9, Table 3) for 0.2 mol% of **1**, but no significant increase (78.6% with 0.4 mol% of **1**, entry 10, Table 3) in the yield was observed above 0.2 mol% of **1**.

The influence of the temperature on the present catalytic reaction was also investigated. From 25 to 50 °C the production of ε-caprolactone increased (70% *vs.* 78%, entries 11 *vs.* 9, Table 3) but further heating up to 70 °C leads to a decrease to 61% (entry 12, Table 3). The lowest selectivity

Table 2 Baeyer–Villiger oxidation of cyclohexanone with **1** as a precatalyst[a]

Essay	Oxidant	Conversion[b] (%)	Yield[c] (%)	Selectivity[d] (%)	TON[e]	TOF[f] (h^{-1})
1	H_2O_2 (30%)	57.3	27.9	48.7	139.4	34.8
2	O_2/benzaldehyde	92.0	78.1	84.9	390.7	97.7

[a] Reaction conditions (unless stated otherwise): Cyclohexanone, 1 mmol; hydrogen peroxide or benzaldehyde, 2 mmol; dioxygen, 1 atm; pre-catalyst **1**, 0.2 mol% (relative to cyclohexanone substrate); 1,2-dichloroethane, 1.0 mL; reaction temperature, 50 °C; reaction time, 4 h (% of yield, TON determined by GC analysis). [b] Conversion of cyclohexanone. [c] Molar yield (%) based on the cyclohexanone, *i.e.* moles of ε-caprolactone per 100 moles of ketone. [d] Selectivity based on ε-caprolactone per 100 moles of converted cyclohexanone. [e] Turnover number (moles of ε-caprolactone per mole of **1**). [f] TON per hour.

Table 3 Aerobic Baeyer–Villiger oxidation of cyclohexanone with O_2/benzaldehyde[a]

Entry	Precatalyst	Conversion[b] (%)	Yield[c] (%)	Selectivity[d] (%)	TON[e]	TOF[f] (h^{-1})
1	None	16.0	9.6	60.0	—	—
2	$[SnPh_2Cl_2]$	27.8	15.8	56.8	79	19.8
3	$CoCl_2$	47.5	29.2	61.5	146	36.5
4	$NiCl_2$	35.5	18.2	51.3	91	22.8
5	$CuCl_2 \cdot 2H_2O$	48.8	28.4	58.2	142	35.5
6	H_2L	39.1	22.4	57.3	112	28.0
7[g]	**1**	72.4	55.5	76.7	278	69.4
8[h]	**1**	80.6	71.1	88.2	711	178
9	**1**	92.0	78.1	84.9	391	97.7
10[i]	**1**	93.1	78.6	84.4	197	49.1
11[j]	**1**	78.5	69.7	88.8	349	87.1
12[k]	**1**	80.5	61.0	75.8	305	76.3

[a] Reaction conditions (unless stated otherwise): Cyclohexanone, 1 mmol; benzaldehyde, 2 mmol; dioxygen, 1 atm; precatalyst **1**, 0.2 mol% (relative to cyclohexanone substrate); 1,2-dichloroethane, 1.0 mL; reaction temperature, 50 °C; reaction time, 4 h (% of yield, TON determined by GC analysis). [b] Conversion of cyclohexanone. [c] Molar yield (%) based on the cyclohexanone, *i.e.* moles of ε-caprolactone per 100 moles of cyclohexanone. [d] Selectivity based on ε-caprolactone per 100 moles of converted cyclohexanone. [e] Turnover number (moles of ε-caprolactone per mole of **1**). [f] TON per hour. [g] Solvent, 1,4-dioxane. [h] Precatalyst **1**, 0.1 mol%. [i] Precatalyst **1**, 0.4 mol%. [j] Reaction temperature, 25 °C. [k] Reaction temperature, 70 °C.

(75.6%, entry 12, Table 3) was observed at the highest applied temperature (70 °C) due to the decomposition of ε-caprolactone into adipic acid.[14a,b]

Monitoring the catalytic reaction along the time (Fig. 5, Table S4, ESI†) shows that the reaction is fast in the first hour, leading to a yield of 70%. Moreover, the highest yield is reached after 6 h (entry 7, Table S4, ESI†), beyond which no further yield increase is observed.

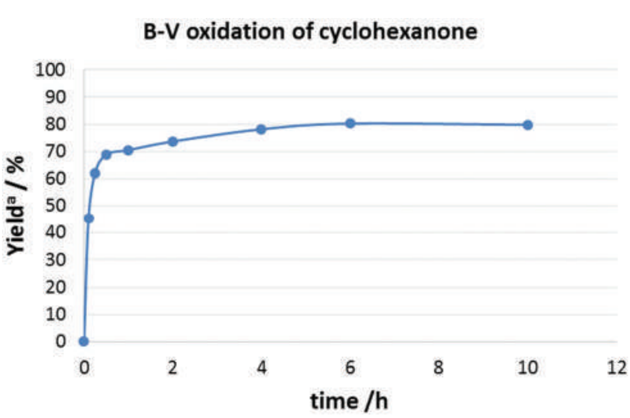


Fig. 5 Aerobic Baeyer–Villiger oxidation of cyclohexanone using different reaction times. [a] Reaction conditions: Cyclohexanone, 1 mmol; benzaldehyde, 2 mmol; dioxygen, 1 atm; precatalyst **1**, 0.2 mol% (relative to cyclohexanone substrate); 1,2-DCE, 1.0 mL; reaction temperature, 50 °C; reaction time, 4 h (% of yield and TON determined by GC analysis).

To verify the catalytic activity with alternative heating modes, we tested the same reaction at the same temperature (50 °C) under micro-wave or ultrasonic irradiation. However, no significant differences in the conversion or yield were observed (Table S5†).

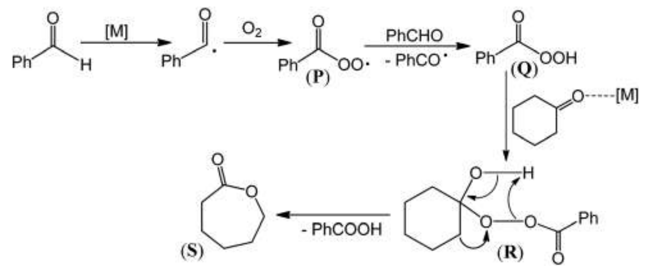
The catalytic activities of **1–3** and **1A** were also compared under the optimized reaction conditions (Table 4). The $Ni^{II}Sn^{IV}$ catalyst **2** leads to the highest yield of ε-caprolactone (83%, Table 4, entry 2), which is slightly higher than that (79%) for the other two heterometallic ($Co^{III}Sn^{IV}$ **1** and $Cu^{II}Sn^{IV}$ **3**) systems. The yield produced by the mononuclear Co^{III} system **1A** is slightly lower than that for the heterometallic $Co^{III}Sn^{IV}$ compound (**1**) (Table 4, entry 4 *vs.* 1).

Oxidation of cyclohexanone was completely inhibited when diphenylamine was used as a radical trap, thus supporting a radical mechanism. This can be postulated as shown in Scheme 4. The metal-promoted reaction of benzaldehyde with molecular oxygen would give the acylperoxyl radical **P**, which would abstract hydrogen from aldehyde to give peroxybenzoic acid **Q**. This would undergo a facile reaction with the metal-activated ketone to give the Criegee-type adduct **R**. Alternatively, **R** can be obtained upon reaction of the radical **P**

Table 4 Aerobic Baeyer–Villiger oxidation of cyclohexanone with O_2/benzaldehyde using different precatalysts[a]

Entry	Precatalyst	Conversion[b] (%)	Yield[c] (%)	Selectivity[d] (%)	TON[e]	TOF[f] (h^{-1})
1	**1**	91.7	78.5	85.6	393	98.1
2	**2**	93.6	83.1	88.7	415	104
3	**3**	93.2	79.1	84.9	391	97.7
4	**1A**	87.8	73.3	83.5	367	91.6

[a] Reaction conditions (unless stated otherwise): Cyclohexanone, 1 mmol; benzaldehyde, 2 mmol; dioxygen, 1 atm; precatalyst, 0.2 mol% (relative to cyclohexanone substrate); 1,2-DCE, 1.0 mL; reaction temperature, 50 °C; reaction time, 4 h (% of yield, TON determined by GC analysis). [b] Conversion of cyclohexanone. [c] Molar yield (%) based on the cyclohexanone, *i.e.* moles of ε-caprolactone per 100 moles of cyclohexanone. [d] Selectivity based on ε-caprolactone per 100 moles of converted cyclohexanone. [e] Turnover number (moles of ε-caprolactone per mole of **1**). [f] TON per hour.



Scheme 4 Possible mechanism of cyclohexanone oxidation with O_2/benzaldehyde catalyzed by metal complexes (**1–3** and **1A**). M = Sn, Co, Ni or Cu centre.

firstly with the ketone, followed by reaction with the benzaldehyde. **R**, upon heterolytic O–O bond cleavage, would convert to the lactone **S** and benzoic acid.[14i–k] In this mechanism, the relevant Lewis acid role of the metal centres is evident from the activation of the ketone and of the benzaldehyde. In some Sn^{II}– and Fe^{III}–porphyrin systems, the formation of a high-valent Sn^{IV}=O or Fe^{V}=O oxido intermediate species, respectively, was proposed,[14a,b] being formed upon oxidation by peroxybenzoic acid. In the present case, for instance for the $Ni^{II}Sn^{IV}$ complex **2** this would lead to a Ni^{IV}(=O)Sn^{IV} species. A dinuclear Ni^{IV} oxido bridged complex has recently been reported,[20] but we have no evidence for such high valent metal species in our study.

Conclusion

Reactions of *N*,*N*′-ethylenebis(salicylaldimine) (H_2L) with a 3d-block metal salt ($MCl_2 \cdot xH_2O$, M = Co, Ni or Cu; x = 2 or 6) and diphenyldichlorotin(IV) [$SnPh_2Cl_2$] in methanol produce the new heterobimetallic 3d metal-Sn complexes [{$SnPhCl_2$}(1κO^2N^2,2κO^2-μ-L)(μ-OMe){CoPh}] (**1**), [{$SnPh_2Cl_2$}(1κO^2N^2,2κO^2-μ-L)Ni] (**2**) and [{$SnPh_2Cl_2$}(1κO^2N^2,2κO^2-μ-L)Cu] (**3**) in an open atmosphere and at room temperature. The mononuclear phenylcobalt(III) compound [Co(κO^2N^2)Ph(H_2O)] (**1A**) was obtained while trying to synthesize the species analogous to **1** in ethanol. Structural analysis reveals a few interesting aspects, such as (i) the Sn^{IV}-to-Co^{III} intermetallic shift of a phenyl ligand to stabilize a novel $Co^{III}Sn^{IV}$ system (**1**), and (ii) a non-covalent M⋯π (Ph) interaction in **2** and **3**, where the phenyl ligand leans over the Ni^{II} or Cu^{II}. To our knowledge, this study discloses in an unprecedented way the flexibility and lability of a phenyl ligand at a Sn^{IV} center in a Schiff base containing metal systems.

The aerobic oxidation of Co^{II} to Co^{III} is a driving force for the stabilization and formation of **1**. Theoretical DFT calculations demonstrate that the Ph ligand shift occurs after the oxidation of Co^{II} to Co^{III}. The lower electrophilic character of Ni^{II} and Cu^{II}, in comparison with Co^{III}, is insufficient for the occurrence of such a transfer, but nevertheless leads to a Ni^{II} (or Cu^{II})⋯π(Ph at Sn^{IV}) interaction (in **2** and **3**), which can be viewed as an intermediate stage of the Sn-to-M phenyl transfer. In addition to the unique composition (organocobalt–organotin) of compound **1**, complexes **2** and **3** also belong to a rare heterometallic family containing the phenyltin(IV) unit, according to CSD searches.

The heterometallic complexes **1–3** efficiently catalyze the B–V aerobic oxidation of cyclohexanone with benzaldehyde as a sacrificial reagent, to produce ε-caprolactone in a selective manner. Radical trap experiments confirm a radical-based mechanism. The metal centres play an important role in the reaction, acting as a Lewis acid to activate the carbonyl group of cyclohexanone (substrate) and promoting the formation of peroxybenzoic acid from the benzaldehyde (sacrificial reductant).

The compounds of this study can contribute to a better understanding of the chemistry of heterometallic and organometallic complexes of 3d and p block metal ions, and further reactions of these types of metal salts and ligand combinations to prepare new heterometallic complexes are underway.

Experimental section

Materials and physical methods

All the reagents and solvents were purchased from commercial sources and used as received. The Schiff base H_2L was synthesized by condensing salicylaldehyde with 1,2-ethylenediamine following a reported procedure.[9] The NMR spectra of H_2L were obtained on a Bruker 400 MHz spectrometer using tetramethylsilane [$Si(CH_3)_4$] as an internal reference. FT-IR spectra were recorded in the region 400–4000 cm^{-1} on a

Bruker Vertex 70 spectrophotometer with samples as KBr disks; abbreviations: s = strong, m = medium, and w = weak. Elemental analyses were performed on a PerkinElmer 2400 II analyzer. Gas chromatographic (GC) measurements were carried out using a FISONS Instruments GC 8000 series gas chromatograph with a FID detector and a capillary column (DB-WAX, column length: 30 m; internal diameter: 0.32 mm). The temperature of injection was 240 °C. The initial temperature was maintained at 80 °C for 1 min, then increased from 10 °C min^{-1} up to 180 °C, and held at this temperature for 1 min. Helium was used as the carrier gas.

Syntheses

Compounds **1–3** and **1A** are prepared following the procedure described below for **1**. However, $NiCl_2·6H_2O$ or $CuCl_2·2H_2O$ was used for the synthesis of **2** or **3**, respectively, instead of $CoCl_2·6H_2O$, while ethanol was used as the solvent instead of methanol for **1A**.

[{SnPhCl$_2$}(1κO^2N^2,2κO^2-μ-L)(μ-OMe){CoPh}] (1). To a methanol suspension (15 mL) of H_2L (0.067 g, 0.25 mmol) was added a methanol solution (2 mL) of $CoCl_2·6H_2O$ (0.060 g, 0.25 mmol) to obtain an orange solution. Subsequent addition of a methanol solution (3 mL) of $[SnPh_2Cl_2]$ (0.086 g, 0.25 mmol) to that reaction mixture produced a dark red solution which was kept at room temperature for slow evaporation. Within 3 d, the formed dark red crystals, suitable for X-ray diffraction analysis, were collected by filtration and washed with cold methanol. Yield: 0.121 g (69%). $C_{29}H_{27}Cl_2N_2O_3CoSn$ (700.08): calcd C 49.75, H 3.89, N 4.00%; found. C 49.61, H 3.93, N 3.93%. IR data (KBr, cm^{-1}): ν(C═N), 1627s.

[Co(κO^2N^2)Ph(H$_2$O)] (1A). Yield: 0.080 g (76%). $C_{22}H_{21}N_2O_3Co$ (420.35): calcd C 62.86, H 5.04, N 6.66%; found. C 62.99, H 4.98, N 6.62%. IR data (KBr, cm^{-1}): ν(HO), 3429br; ν(C═N), 1658s.

[{SnPh$_2$Cl$_2$}(1κO^2N^2,2κO^2-μ-L)Ni] (2). Yield: 0.104 g (62%). $C_{28}H_{24}Cl_2N_2O_2NiSn$ (668.81): calcd C 50.28, H 3.62, N 4.19%; found. calcd C 50.12, H 3.65, N 4.13%. IR data (KBr, cm^{-1}): ν(C═N), 1624s.

[{SnPh$_2$Cl$_2$}(1κO^2N^2,2κO^2-μ-L)Cu] (3). Yield: 0.059 g (35%). $C_{28}H_{24}Cl_2N_2O_2CuSn$ (673.62): calcd C 49.92, H 3.59, N 4.16%; found. C 50.14, H 3.55, N 4.11%. IR data (KBr, cm^{-1}): ν(C═N), 1630s.

Crystal structure determination

X-ray quality crystals of **1–3** and **1A** were immersed in cryo-oil and mounted in a nylon loop and measured at 298 K. Intensity data were collected using a Bruker APEX II SMART CCD diffractometer with graphite monochromated Mo-Kα (λ 0.71073) radiation. Cell parameters were obtained with Bruker SMART[21a] software and refined with Bruker SAINT[21a] on all the observed reflections. Absorption corrections were made by using the multi-scan method (SADABS).[21a] Structures were solved by direct methods by using the SHELXS-2014 package[21b] and refined with SHELXL-2014/7.[21b] Calculations were performed using the WinGX System-Version 2014.1.[21c] Hydrogen atoms attached to carbon atoms were inserted at geometrically calculated positions and included in the refinement using the riding-model approximation. $U_{iso}(H)$ were defined as $1.2U_{eq}$ of the parent carbon atoms for phenyl and ethylene residues and $1.5U_{eq}$ of the parent oxygen atom for the water molecule (in **1A**). The ethylene moieties in **2**, **3** and **1A** were disordered and were modelled by means of PART 1 and PART 2 instructions, leading to occupancies of 59 and 41% in **2**, 58 and 42% in **3** and 49 and 51% in **1A**. Least squares refinements with anisotropic thermal motion parameters for all the non-hydrogen atoms were employed. Crystallographic data are summarized in Table S1 (ESI†). Attempts to solve **1A** in the centrosymmetric orthorhombic *Cmc*21 space group were not successful in view of the disorder of the ethylene moiety which led to unreasonable dimensions.

Computational details

Full geometry optimization of complexes and transition states has been carried out at the DFT/HF hybrid level of theory by using the B3LYP* functional[22] with the help of the Gaussian 09 program package.[23] It was shown that while the popular pure functionals such as BP86 underestimate the relative stability of high spin states and many hybrid functionals with a high contribution of the Hartree–Fock term overestimate the relative stability of high spin states, the modified B3LYP functional (B3LYP*) correctly describes the relative energies of the low and high spin states in a number of transition metal complexes.[22c,d] Since the relative stability of the high and low spin states is a crucial issue for the understanding of the Ph group migration, namely the B3LYP* functional was selected for the calculations. The B3LYP* functional was constructed using the keywords blyp, IOp(3/76 = 1000001500), IOp(3/77 = 0720008500) and IOp(3/78 = 0810010000). The relativistic and pseudo-relativistic Stuttgart pseudo-potentials that describe 10 and 46 core electrons (MDF10 and MWB46) and the appropriate contracted basis sets[24] were employed for the Co and Sn atoms, respectively, whereas the standard basis set 6-31G(d) was applied for all other atoms. Then, single-point calculations were performed on the basis of the equilibrium geometries found using the 6-311+G(d,p) basis set for non-metal atoms. No symmetry operations have been applied for any of the structures calculated.

The Hessian matrix was calculated analytically to prove the location of correct minima (no imaginary frequencies) or saddle points (only one imaginary frequency) and to estimate the thermodynamic parameters, the latter being calculated at 25 °C.

Total energies corrected for solvent effects (E_s) were estimated at the single-point calculations on the basis of gas-phase geometries at the CPCM-B3LYP*/6-311+G(d,p)//gas-B3LYP*/6-31G(d) level of theory using the polarizable continuum model in the CPCM version[25] with methanol as the solvent. The UAKS model was applied for the molecular cavity, and dispersion, cavitation, and repulsion terms were considered. The entropic term in MeOH solution (S_s) was calculated according to the procedure described by Wertz[26a] and Cooper and Ziegler[26b] using eqn (1)–(4)

$$\Delta S_1 = R \ln V^{s}_{m,liq}/V_{m,gas} \quad (1)$$

$$\Delta S_2 = R \ln V^{\circ}_{m}/V^{s}_{m,liq} \quad (2)$$

$$\alpha = \frac{S^{\circ,s}_{liq} - (S^{\circ,s}_{gas} + R \ln V^{s}_{m\cdot liq}/V_{m\cdot gas})}{(S^{\circ,s}_{gas} + R \ln V^{s}_{m\cdot liq}/V_{m\cdot gas})} \quad (3)$$

$$\begin{aligned} S_s &= S_g + \Delta S_{sol} = S_g + [\Delta S_1 + \alpha(S_g + \Delta S_1) + \Delta S_2] \\ &= S_g + [(-12.72\ \mathrm{cal\ mol^{-1}\ K^{-1}}) - 0.32(S_g - 12.72\ \mathrm{cal\ mol^{-1}\ K^{-1}}) \\ &\quad + 6.37\ \mathrm{cal\ mol^{-1}\ K^{-1}}] \end{aligned} \quad (4)$$

where S_g = gas-phase entropy of the solute, ΔS_{sol} = solvation entropy, $S^{\circ,s}_{liq}$, $S^{\circ,s}_{gas}$, and $V^{s}_{m,liq}$ = standard entropies and the molar volume of the solvent in liquid or gas phases (127.2 and 239.9 J mol^{-1} K^{-1} and 40.46 mL mol^{-1}, respectively, for MeOH), $V_{m,gas}$ = molar volume of the ideal gas at 25 °C (24 450 mL mol^{-1}), V°_{m} = molar volume of the solution corresponding to the standard conditions (1000 mL mol^{-1}). The enthalpies and Gibbs free energies in solution (H_s and G_s) were estimated using expressions (5) and (6)

$$H_s = E_s(\text{6-311+G(d, p)}) - E_g(\text{6-31G(d)}) + H_g(\text{6-31G(d)}) \quad (5)$$

$$G_s = H_s - TS_s \quad (6)$$

where E_s, E_g and H_g are the total energies in solution and in the gas phase and the gas-phase enthalpy calculated at the corresponding level. All related calculated total energies, enthalpies, Gibbs free energies (in Hartree) and entropies (in cal mol^{-1} K^{-1}) are included in Table S2† while Cartesian atomic coordinates (Å) of the calculated structures are deposited in Table S3.†

Procedure for the Baeyer–Villiger oxidation studies

In a typical experiment, a mixture of a catalyst precursor (2×10^{-3} mmol, 0.2 mol% relative to cyclohexanone), cyclohexanone (1 mmol), benzaldehyde (2.0 mmol), and an organic solvent (1.0 mL) was mixed in a 5 mL reaction flask with a stirring bar equipped with a condenser and a balloon containing oxygen. The reaction mixture was magnetically stirred at 50 °C for 4 h. To extract the substrate and the organic products from the reaction mixture, acetonitrile (3.0 mL) was added. Then, 50 μL of cycloheptanone (internal standard) was introduced and the mixture was stirred for 10 min. A sample (0.5 μL) was taken from the organic phase and analysed by GC using the internal standard method.

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Inorganic Chemistry
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Syntheses, Structures, Magnetic Properties, and Density Functional Theory Magneto-Structural Correlations of Bis(μ-phenoxo) and Bis(μ-phenoxo)-μ-acetate/Bis(μ-phenoxo)-bis(μ-acetate) Dinuclear $Fe^{III}Ni^{II}$ Compounds

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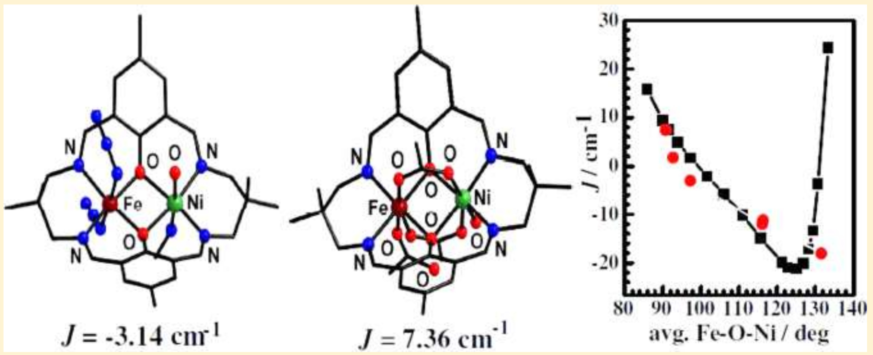
Supporting Information



ABSTRACT: The bis(μ-phenoxo) $Fe^{III}Ni^{II}$ compound $[Fe^{III}(N_3)_2LNi^{II}(H_2O)(CH_3CN)](ClO_4)$ (**1**) and the bis(μ-phenoxo)-μ-acetate/bis(μ-phenoxo)-bis(μ-acetate) $Fe^{III}Ni^{II}$ compound $\{[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-}OAc)]_{0.6}\cdot[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]_{0.4}\}(ClO_4)\cdot 1.1H_2O$ (**2**) have been synthesized from the Robson type tetraiminodiphenol macrocyclic ligand H_2L, which is the [2 + 2] condensation product of 4-methyl-2,6-diformylphenol and 2,2′-dimethyl-1,3-diaminopropane. Single-crystal X-ray structures of both compounds have been determined. The cationic part of the dinuclear compound **2** is a cocrystal of the two species $[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-}OAc)]^+$ (**2A**) and $[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]^+$ (**2B**) with weights of 60% of the former and 40% of the latter. While **2A** is a triply bridged bis(μ-phenoxo)-μ-acetate system, **2B** is a quadruply bridged bis(μ-phenoxo)-bis(μ-acetate) system. Variable-temperature (2−300 K) magnetic studies reveal antiferromagnetic interaction in **1** and ferromagnetic interaction in **2** with J values of −3.14 and 7.36 cm^{-1}, respectively ($H = -2JS_1\cdot S_2$). Broken-symmetry density functional calculations of exchange interaction have been performed on complexes **1** and **2** and also on previously published related compounds, providing good numerical estimates of J values in comparison to experiments. The electronic origin of the difference in magnetic behavior of **1** and **2** has been well understood from MO analyses and computed overlap integrals of BS empty orbitals. The role of acetate and thus its complementarity/countercomplementarity effect on the magnetic properties of diphenoxo-bridged $Fe^{III}Ni^{II}$ compounds have been determined on computing J values of model compounds by replacing bridging acetate and nonbridging acetate ligand(s) by water ligands in the model compounds derived from **2A**,**B**. The DFT calculations have also been extended to develop several magneto-structural correlations in these types of complexes, and the correlations focus on the role of Fe−O−Ni bridge angle, average Fe/Ni−O bridge distance, Fe−O−Ni−O dihedral angle, and out-of-plane shift of the phenoxo group.

■ INTRODUCTION

The phenomenon of magnetic exchange in discrete molecules was discovered/explained by Guha[1a] and Bleaney and Bowers[1b] in the early 1950s while studying the variable-temperature magnetic properties of diaqua-$\mu_{1,3}$-acetate dicopper(II). Over the decades, the magnetic properties of coordination compounds have received tremendous attention. Eventually, a number of experimental[2−7] and theoretical[2a,3d,f,4b−d,8−10] magneto-structural correlations were determined. Moreover, a number

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of single-molecule magnets (SMMs)[11−16] have been developed with the aim of their possible application in molecular spintronics and quantum computing.[17] In fact, a renaissance has been continuing to develop 3d/4f/3d-4f-SMMs with greater blocking temperatures (T_B; the maximum to date is 8.3 K)[15c] and greater energy barriers (U_{eff}; the maximum to date is 556 cm^{-1})[15b] to magnetization reversal.

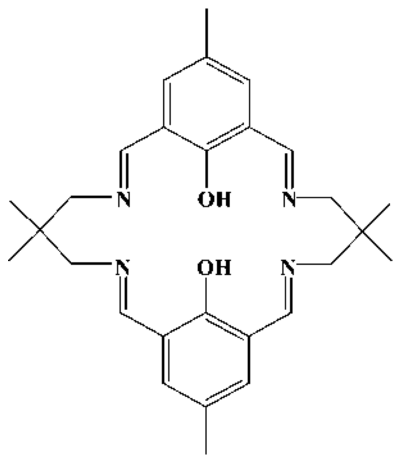
Developing magneto-structural correlations using only experimental means is a nontrivial task because this demands structurally similar or identical complexes with all but one governing parameter being identical. On the other hand, theoretical magneto-structural correlations can be more easily established because it is easily possible in most cases to construct model systems in which one parameter varies, keeping others constant.[3d,f,4b,d,8−10] Even when it is problematic to vary one parameter at a time for some heterobridged systems, theoretical correlations can be established defining new parameters, which are functions of two other parameters.[4c] Density functional theoretical (DFT) methods have been established as valuable tools in developing magneto-structural correlations and modeling the magnetic properties of the full structures of exchange-coupled systems to gain insight into the magnetic coupling mechanism and to analyze various contributions.[3d,f,4b−d,8−10,18]

The first and perhaps the most elegant magneto-structural correlation was determined experimentally in planar dihydroxo-bridged dicopper(II) compounds by Hatfield, Hodgson, and co-workers.[3a] The other types of homometallic systems for which experimental or theoretical magneto-structural correlations have been determined include alkoxo-bridged Cu^{II}_4 of cubane type,[3b] diphenoxo-bridged Cu^{II}_2,[3c,d,8a] roof-shaped dihydroxo-bridged Cu^{II}_2,[8b] monophenoxo-bridged Cu^{II}_2 having axial−equatorial bridging atoms,[3e] dihydroxo/dialkoxo-bridged Cu^{II}_2,[8c,d] linear Cu^{II}_3 having a diphenoxo bridge in a pair of metal ions,[3f] chloro-bridged Cu^{II}_2,[3g] dihalo-bridged Cu^{II}_2,[8e] bis($\mu_{1,1}$-azide) Cu^{II}_2,[8f,g] bis($\mu_{1,3}$-azide) Cu^{II}_2,[8g,h] diphenoxo-bridged Ni^{II}_2,[4a] bis($\mu_{1,1}$-azide) Ni^{II}_2,[8f,9] bis($\mu_{1,3}$-azide) Ni^{II}_2,[8h] bis($\mu_{oximate}$) Ni^{II}_2,[4b] heterobridged μ-phenoxo-$\mu_{1,1}$-azide Ni^{II}_2,[4c] $\mu_{1,3}$-azide Ni^{II}_4,[4d] oxo-bridged Fe^{III}_2,[5a,b] dialkoxo-bridged Fe^{III}_2,[5c,d] asymmetrically phenoxo-, alkoxo-, and hydroxo-bridged Fe^{III}_2,[5e] heterobridged Fe^{III}_2/Fe^{III}_3 having one oxo/hydroxo/alkoxo plus at least one carboxylate/sulfate/phosphate bridging moiety in a pair of metal ions,[5f] Fe^{III}_6 having a μ-hydroxo-bis(μ-carboxylate) or bis(μ-alkoxo)-μ-carboxylate in a pair of metal ions,[5g] bis($\mu_{1,1}$-azide) Mn^{II}_2,[8f] bis(μ-oxo) Mn^{IV}_2,[6a,b] and tris(μ-oxo)/tris(μ-hydroxo)/tris(μ-ethoxide)/tris(μ-chloride)/tris(μ-bromide)/tris(μ-iodide)/bis(μ-oxo)-μ-hydroxo/μ-oxo-bis(μ-hydroxo) Mn^{IV}_2/Cr^{III}_2/V^{II}_2.[6c] While several correlations have been determined in homometallic systems, they are few in heteronuclear systems; experimental correlations[7] in diphenoxo-bridged $Cu^{II}Gd^{III}$ and $Cu^{II}V^{IV}O$ compounds and theoretical correlations[10] in diphenoxo-bridged $Cu^{II}Gd^{III}$ and $Ni^{II}Gd^{III}$ systems are probably the only examples of straightforward magneto-structural correlations in heterometallic systems.

In the midst of the renaissance in the development of SMMs having greater T_B and U_{eff} values, the determination of magneto-structural correlations for scarcely investigated systems, such as FeNi compounds, deserves attention; only a few dinuclear $Fe^{III}Ni^{II}$ complexes have been reported to date and $Fe^{III}Ni^{II}$ complexes having one or more phenoxo/alkoxo/hydroxo bridges are scarce.[19] Therefore, we have been motivated to explore this area. Such compounds are also important from a biomimetic point of view, especially after the determination of the protein crystal structure of Fe−Ni hydrogenase, which has disclosed a unique Fe−Ni heterometallic site.[20] Despite the fact that these dinuclear units are building blocks for several polynuclear FeNi complexes[21] and are important from the perspective of biomimetic chemistry, magneto-structural correlations for this pair have not been developed either experimentally or theoretically.

To isolate $Fe^{III}Ni^{II}$ compounds, we have chosen the mononuclear Fe^{III} compound $[Fe^{III}(H_2L)(H_2O)Cl](ClO_4)_2 \cdot 2H_2O$ as the precursor, where H_2L (Chart 1) is the symmetrical dinucleating Robson type[22] tetraiminodiphenolate macrocyclic ligand in which the dialdehyde counterpart is 4-methyl-2,6-diformylphenol and the diamine counterpart is 2,2′-dimethyl-1,3-diaminopropane.[14e] We anticipated that, on using secondary bridging ligands such as acetate and azide, the values of structural parameters involving the phenoxo bridges would be changed, which should affect the nature/magnitude of the magnetic exchange interaction. Again, the targeted systems having acetate/azide bridges should be interesting examples to explore complementarity/countercomplementarity effects in such complicated systems having so many combinations of magnetic orbitals. With all this in mind, we have isolated the bis(μ-phenoxo) $Fe^{III}Ni^{II}$ compound $[Fe^{III}(N_3)_2LNi^{II}(H_2O)(CH_3CN)](ClO_4)$ (**1**) and the cocrystalline bis(μ-phenoxo)-μ-acetate/bis(μ-phenoxo)-bis(μ-acetate) $Fe^{III}Ni^{II}$ compound $\{[Fe^{III}(OAc)LNi[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-}OAc)]_{0.6} \cdot [Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]_{0.4}\}(ClO_4) \cdot 1.1H_2O$ (**2**). Herein, we report the syntheses, crystal structures, and magnetic properties of **1** and **2** and density functional theoretical modeling of the magnetic properties of **1**, **2**, and related compounds along with DFT magneto-structural correlations.

Chart 1. Chemical Structure of H_2L



■ EXPERIMENTAL SECTION

Materials and Physical Measurements. All of the reagents and solvents were purchased from commercial sources and used as received. The mononuclear iron(III) complex $[Fe^{III}(H_2L)(H_2O)(Cl)](ClO_4)_2 \cdot 2H_2O$ was prepared following the reported procedure.[14e] Elemental (C, H, and N) analyses were performed on a Perkin-Elmer 2400 II analyzer. IR spectra were recorded in the region 400−4000 cm^{-1} on a Bruker-Optics Alpha-T spectrophotometer with samples as KBr disks. Variable-temperature magnetic susceptibility measurements were carried out with a Quantum Design MPMS−XL7 SQUID magnetometer. Diamagnetic corrections were estimated from the Pascal constants. The temperature dependent magnetic contribution of the holder was experimentally determined and substracted from the measured susceptibility data.

Computational Details. There has been a great deal of interest in the evaluation of magnetic exchange couplings using the techniques of quantum chemistry.[3d,f,4b−d,8−10,18] In dinuclear complexes the magnetic exchange interaction between Ni and Fe is described by the spin Hamiltonian

$$\hat{H} = -2J\hat{S}_{Fe}\hat{S}_{Ni} \tag{1}$$

Here J is the isotropic exchange coupling constant and S_{Fe} and S_{Ni} are spins on iron(III) ($S = {}^5/_2$) and nickel(II) ($S = 1$) atoms, where a negative J value corresponds to an antiferromagnetic interaction. To compute the J values, the energies of the high-spin state ($S_T = {}^7/_2$) and one low-spin state ($S_T = {}^3/_2$) were calculated. The energy of the high-spin state can be computed straightforwardly using a single-determinant wave function such as DFT methods. Noodleman's broken-symmetry (BS) approach is used for the computation of the energy of the low-spin state of very large systems, which is derived from a spin-unrestricted reference wave function.[23] This BS approach consists of performing either unrestricted Hartree−Fock (UHF) or density functional theory (DFT) calculations for low-spin open-shell molecular systems in which the α and β densities are allowed to localize on different atomic centers, which are referred to as BS calculations. A detailed description of the computation of the magnetic exchange interaction can be found elsewhere.[18] Noodleman's BS approach and the widely used exchange-correlation functional B3LYP provide a good numerical estimate of the exchange coupling constant in comparison to experiment.[18] Accordingly, all calculations here we have used the B3LYP[24a] functional with the valence triple-ζ quality basis sets (TZV) of Ahlrichs and co-workers.[24b,c] All calculations were performed using the Gaussian 09 suite of programs.[25]

Syntheses of [$Fe^{III}(N_3)_2LNi^{II}(H_2O)(CH_3CN)](ClO_4)$ (1) and {[$Fe^{III}(OAc)LNi^{II}(H_2O)(\mu$-$OAc)]_{0.6}$·[$Fe^{III}LNi^{II}(\mu$-$OAc)_2]_{0.4}$}($ClO_4$)·1.1$H_2O$ (2). **2** was prepared by following a procedure similar to that described below for **1**, except that solid NaOAc was used for **2** instead of solid NaN_3 for **1**.

To a stirred acetonitrile/ethanol (1/1) solution (20 mL) of [$Fe^{III}(H_2L)(H_2O)(Cl)](ClO_4)_2\cdot 2H_2O$ (0.201 g, 0.25 mmol) under N_2 atmosphere were successively added solid NaN_3 (0.065 g, 1 mmol) and $Ni(ClO_4)_2\cdot 6H_2O$ (0.092 g, 0.25 mmol). After 4 h of stirring, the dark red solution was filtered to remove any suspended particles and the filtrate was kept at room temperature for slow evaporation. After 2 days, a dark violet crystalline compound containing diffraction-quality single crystals deposited. The compound was collected by filtration and washed with ethanol. Yield: 149 mg (73%). Anal. Calcd for $C_{30}H_{39}N_{11}O_7ClFeNi$: C, 44.17; H, 4.82; N, 18.89. Found: C, 44.21; H, 4.89; N, 18.85. IR (cm^{-1}, KBr): $\nu(H_2O)$, 3430 w; $\nu(N_3)$, 2048 vs; ν(C═N), 1630 vs; $\nu(ClO_4)$, 1089 vs, 620 w.

Data for **2** are as follows. Yield: 165 mg (80%). Anal. Calcd for $C_{32}H_{43.4}N_4O_{11.7}ClFeNi$: C, 46.78; H, 5.33; N, 6.82. Found: C, 46.45; H, 5.49; N, 6.64. IR (cm^{-1}, KBr): $\nu(H_2O)$, 3466 w; ν(C═N), 1640 s; $\nu_{as}(CO_2^-)$, 1571 m; $\nu_s(CO_2^-)$, 1430 m, 1400 m; $\nu(ClO_4)$, 1091 vs, 621 w.

Crystal Structure Determination of 1 and 2. The crystallographic data for **1** and **2** are summarized in Table 1. Diffraction data were collected on a Bruker-APEX II SMART CCD diffractometer at 296 K using graphite-monochromated Mo Kα radiation (λ = 0.71073 Å). For data processing and absorption correction the packages SAINT[26a] and SADABS[26b] were used. The structure was solved by direct and Fourier methods and refined by full-matrix least squares based on F^2 using the SHELXTL[26c] and SHELXL-97[26d] packages. The compound **2** is a cocrystal of two species, [$Fe^{III}(H_2O)LNi^{II}(OAc)(\mu$-$OAc)]^+$ (**2A**) and [$Fe^{III}LNi^{II}(\mu$-$OAc)_2]^+$ (**2B**), in amounts of about 60% and 40%, respectively (vide infra). Two hydrogen atoms of the coordinated water molecule in **2A** and the hydrogen atoms of the water molecule of crystallization were not located and could not be inserted. On the other hand, the hydrogen atoms in **1** were located from a difference Fourier map. Other hydrogen atoms in **1** and **2** were inserted at calculated positions with isotropic thermal parameters and refined. Using anisotropic treatment for the non-hydrogen atoms and isotropic treatment for the hydrogen atoms, the final refinements converged at R1 values ($I > 2\sigma(I)$) of 0.0418 and 0.0453 for **1** and **2**, respectively.

Table 1. Crystallographic Data for 1 and 2

	1	2
empirical formula	$C_{30}H_{39}N_{11}O_7ClFeNi$	$C_{32}H_{40}N_4O_{11.7}ClFeNi$
fw	815.73	817.89
cryst color	violet	violet
cryst syst	triclinic	monoclinic
space group	$P\bar{1}$	$P2/c$
a (Å)	9.2970(4)	11.7203(7)
b (Å)	10.3736(5)	16.0018(9)
c (Å)	18.8629(9)	22.9976(11)
α (deg)	104.5340(10)	90.00
β (deg)	94.7120(10)	117.610(3)
γ (deg)	101.1640(10)	90.00
V (Å^3)	1711.21(14)	3821.9(4)
Z	2	4
T (K)	293(2)	296(2)
2θ (deg)	2.26−50.98	2.54−51.64
μ (mm^{-1})	1.116	1.004
ρ_{calcd} (g cm^{-3})	1.583	1.421
F(000)	846	1698
abs cor	multiscan	multiscan
index range	$-11 \le h \le 12$	$-12 \le h \le 14$
	$-12 \le k \le 12$	$-19 \le k \le 19$
	$-22 \le l \le 22$	$-28 \le l \le 28$
no. of rflns collected	21105	47133
no. of indep rflns (R_{int})	6325 (0.0511)	7367 (0.0462)
R1^a/wR2^b ($I > 2\sigma(I)$)	0.0418/0.0976	0.0453/0.1168
R1^a/wR2^b (all F_o^2)	0.0655/0.1119	0.0743/0.1360

^{a}R1 = $[\sum||F_o| - |F_c||/\sum|F_o|]$. ^{b}wR2 = $[\sum w(F_o^2 - F_c^2)^2/\sum wF_o^4]^{1/2}$.

■ RESULTS AND DISCUSSION

Syntheses and Characterization. The two macrocyclic heterodinuclear $Fe^{III}Ni^{II}$ complexes [$Fe^{III}(N_3)_2LNi^{II}(H_2O)(CH_3CN)](ClO_4)$ (**1**) and {[$Fe^{III}(OAc)LNi^{II}(H_2O)(\mu$-$OAc)]_{0.6}$·[$Fe^{III}LNi^{II}(\mu$-$OAc)_2]_{0.4}$}($ClO_4$)·1.1$H_2O$ (**2**) were readily obtained in high yield from the reaction mixture of [$Fe^{III}(H_2L)(H_2O)(Cl)](ClO_4)_2\cdot 2H_2O$, $Ni(ClO_4)_2\cdot 6H_2O$, and sodium salt (NaN_3 for **1** and NaOAc for **2**) of the appropriate secondary ligand in a 1:1:4 ratio.

The vibration due to the C═N bond appears at 1630 cm^{-1} for **1** and 1640 cm^{-1} for **2**. The presence of perchlorates in **1** and **2** is evidenced by the appearance of the following infrared bands: a very strong band at 1089 cm^{-1} and a weak band at 620 cm^{-1} for **1**, and a very strong band at 1091 cm^{-1} and a weak band at 621 cm^{-1} for **2**. The water stretchings are observed at 3430 and 3466 cm^{-1}, respectively, for **1** and **2**. A very strong signal at 2048 cm^{-1} in the spectrum of **1** indicates the presence of azide, whereas one $\nu_{as}(CO_2^-)$ vibration at 1571 cm^{-1} and two $\nu_s(CO_2^-)$ vibrations at 1430 and 1400 cm^{-1} in the spectrum of **2** suggest the presence of two types of acetate ligands.[19a] As will be seen below, the crystal structure of compound **2** contains both bridging and unidentate aceate ligands.

Description of the Structures of 1 and 2. Crystal structures of **1** and **2** are shown in Figures 1 and 2, respectively. The structures reveal that both are dinuclear $Fe^{III}Ni^{II}$ compounds derived from the tetraiminodiphenolate macrocyclic ligand $[L]^{2-}$. In both compounds, one N_2O_2 compartment of $[L]^{2-}$ is occupied by a Fe^{III} ion, while the second N_2O_2 compartment is occupied by a Ni^{II} ion.

The metal centers in [$Fe^{III}(N_3)_2LNi^{II}(H_2O)(CH_3CN)](ClO_4)$ (**1**) are doubly bridged by the two phenoxo oxygen atoms. Both metal centers in **1** are hexacoordinated; the fifth and

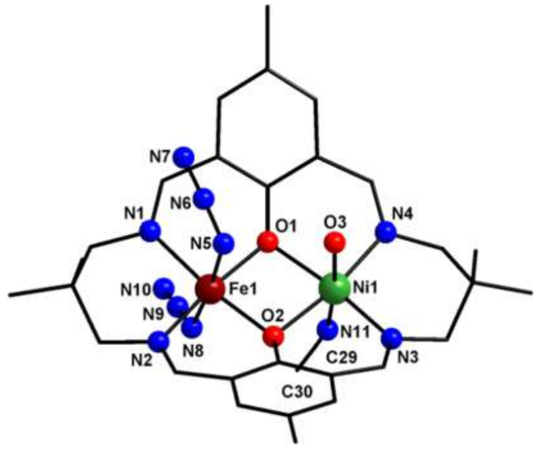


Figure 1. Crystal structure of $[Fe^{III}(N_3)_2LNi^{II}(H_2O)(CH_3CN)](ClO_4)$ (**1**). The perchlorate anion and all hydrogen atoms are omitted for clarity.

sixth coordination positions of the iron(III) center are occupied by two nitrogen atoms of two terminal azide ligands, while the fifth and sixth coordination positions of the nickel(II) center are provided by a water oxygen atom and an acetonitrile nitrogen atom.

The cationic part of the dinuclear compound $\{[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-}OAc)]_{0.6}\cdot[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]_{0.4}\}(ClO_4)\cdot 1.1H_2O$ (**2**) is a cocrystal of the two species $[Fe^{III}(H_2O)LNi^{II}(OAc)(\mu\text{-}OAc)]^+$ (**2A**) and $[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]^+$ (**2B**) with weights of 60% of the former and 40% of the latter. While **2A** is a triply bridged bis(μ-phenoxo)-μ-acetate system, **2B** is a quadruply bridged bis(μ-phenoxo)-bis(μ-acetate) system. The iron(III) and nickel(II) centers in both **2A** and **2B** are hexacoordinated. The two species **2A** and **2B** have the common two N_2O_2 compartments (one for Fe^{III} and the second for Ni^{II}) and the common bis(μ-phenoxo)-μ-acetate bridging core, but they differ in terms of the sixth coordination positions; the sixth coordination positions of iron(III) and nickel(II) in **2A** are occupied by a monodentate acetate ligand (through O6) and a water oxygen atom (O4), respectively, and thus the bridging moiety is bis(μ-phenoxo)-μ-acetate, while the sixth coordination positions of the two metal centers in **2B** are occupied by a bridging acetate ligand (through O4′ to Ni^{II} and through O6′ to Fe^{III}) and thus the bridging core is bis(μ-phenoxo)-bis(μ-acetate).

The hexacoordinated coordination environments for iron(III) and nickel(II) in **1** and **2** are distorted octahedral, in which the basal positions are occupied by the imine nitrogen and phenoxo oxygen atoms.

Selected bond lengths and bond angles of **1** and **2** are given in Tables 2 and 3, respectively. The two iron–phenoxo, iron–imine, nickel–phenoxo, or nickel–imine bond distances in both compounds are very close. Again, in both compounds, the two

Table 2. Selected Bond Lengths (Å) and Bond Angles (deg) in the Coordination Environments of the Metal Centers in 1

Bond Lengths			
Ni1–O1	2.052(2)	Fe1–O1	1.993(2)
Ni1–O2	2.062(2)	Fe1–O2	1.988(2)
Ni1–N3	1.994(3)	Fe1–N1	2.064(3)
Ni1–N4	1.998(3)	Fe1–N2	2.050(3)
Ni1–O3	2.125(3)	Fe1–N5	2.149(3)
Ni1–N11	2.075(3)	Fe1–N8	2.027(3)
		Fe1···Ni1	3.038
Bond Angles			
O1–Ni1–N3	170.45(10)	O1–Fe1–N2	172.87(11)
O2–Ni1–N4	171.18(11)	O2–Fe1–N1	171.89(10)
O3–Ni1–N11	174.51(11)	N5–Fe1–N8	174.97(12)
O3–Ni1–O1	86.41(9)	N5–Fe1–O1	87.31(10)
O3–Ni1–O2	87.61(9)	N5–Fe1–O2	87.47(10)
O3–Ni1–N3	89.71(11)	N5–Fe1–N1	85.50(11)
O3–Ni1–N4	88.58(10)	N5–Fe1–N2	89.91(11)
N11–Ni1–O1	88.16(11)	N8–Fe1–O1	93.23(11)
N11–Ni1–O2	90.80(10)	N8–Fe1–O2	97.55(11)
N11–Ni1–N3	95.54(12)	N8–Fe1–N1	89.49(12)
N11–Ni1–N4	92.25(11)	N8–Fe1–N2	90.11(12)
O1–Ni1–O2	81.03(9)	O1–Fe1–O2	84.35(9)
O1–Ni1–N4	90.79(11)	O1–Fe1–N1	91.27(10)
O2–Ni1–N3	90.09(10)	O2–Fe1–N2	88.97(10)
N3–Ni1–N4	97.84(12)	N1–Fe1–N2	95.06(11)
		Fe1–O1–Ni1	97.39(10)
		Fe1–O2–Ni1	97.22(10)

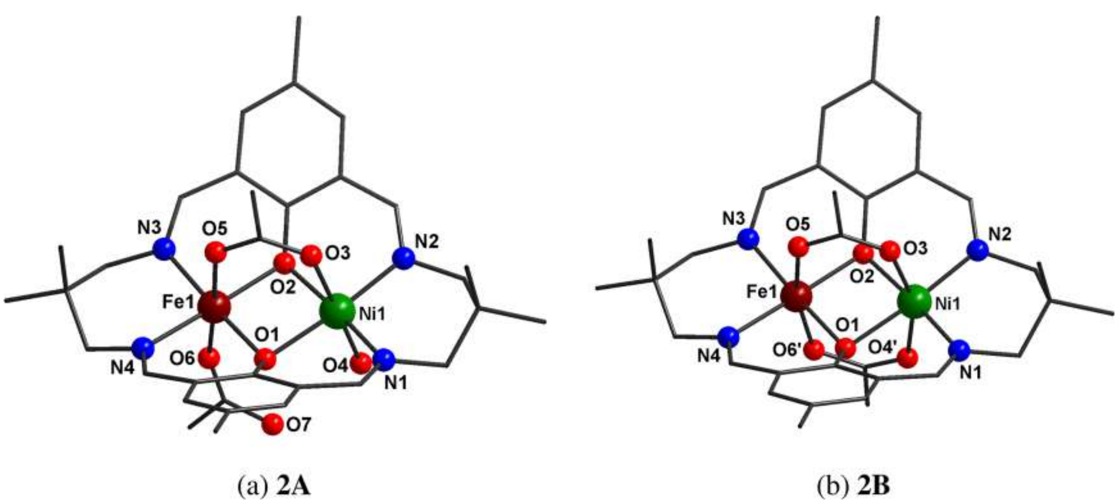


Figure 2. Crystal structure of $\{[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-}OAc)]_{0.6}\cdot[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]_{0.4}\}(ClO_4)\cdot 1.1H_2O$ (**2**; 60% **2A** + 40% **2B**): (a) $[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-}OAc)]^+$ (**2A**); (b) $[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]^+$ (**2B**). The perchlorate anion, water of crystallization, the minor component of the disordered methyl carbon atom of the common bridging acetate, and all of the hydrogen atoms are omitted for clarity.

Table 3. Selected Bond Lengths (Å) and Bond Angles (deg) in the Coordination Environments of the Metal Centers in **2** (**2A** + **2B**)

Bond Lengths			
Ni1−O1	2.054(2)	Fe1−O1	2.003(2)
Ni1−O2	2.057(2)	Fe1−O2	2.005(2)
Ni1−N1	2.028(3)	Fe1−N3	2.050(3)
Ni1−N2	2.006(3)	Fe1−N4	2.054(3)
Ni1−O3	2.055(2)	Fe1−O5	2.049(2)
Ni1−O4/O4′	2.053(6)[a]/2.196(9)[b]	Fe1−O6/O6′	2.002(8)[c]/2.028(13)[d]
		Fe1···Ni1	2.8923(7)
Bond Angles			
O1−Ni1−N2	175.21(11)	O1−Fe1−N3	176.72(11)
O2−Ni1−N1	175.24(11)	O2−Fe1−N4	176.04(11)
O3−Ni1−O4/O4′	176.28(18)[a]/158.6(2)[b]	O5−Fe1−O6/O6′	177.29(15)[c]/165.3(2)[d]
O1−Ni1−N1	88.23(11)	O1−Fe1−N4	88.61(11)
O1−Ni1−O2	87.47(9)	O1−Fe1−O2	90.34(9)
O1−Ni1−O3	85.48(9)	O1−Fe1−O5	85.97(9)
O1−Ni1−O4/O4′	94.46(17)[a]/77.0(2)[b]	O1−Fe1−O6/O6′	96.60(18)[c]/81.7(3)[d]
O2−Ni1−N2	88.81(11)	O2−Fe1−N3	87.74(10)
O2−Ni1−O3	86.04(9)	O2−Fe1−O5	86.34(9)
O2−Ni1−O4/O4′	90.25(17)[a]/81.2(3)[b]	O2−Fe1−O6/O6′	94.5(2)[c]/85.7(4)[d]
O3−Ni1−N1	91.59(12)	O5−Fe1−N3	91.25(11)
O3−Ni1−N2	91.26(12)	O5−Fe1−N4	89.78(11)
N1−Ni1−N2	95.37(13)	N3−Fe1−N4	93.13(12)
N1−Ni1−O4/O4′	92.12(19)[a]/99.8(3)[b]	N3−Fe1−O6/O6′	86.21(19)[c]/100.8(3)[d]
N2−Ni1−O4/O4′	88.57(18)[a]/105.4(3)[b]	N4−Fe1−O6/O6′	89.4(2)[c]/97.9(4)[d]
		Fe1−O1−Ni1	90.96(9)
		Fe1−O2−Ni1	90.81(9)

[a]For O4 (in **2A**). [b]For O4′ (in **2B**). [c]For O6 (in **2A**). [d]For O6′ (in **2B**).

iron−phenoxo bond distances (average 1.990 Å in **1** and 2.003 Å in **2**) are shorter than the two nickel−phenoxo bond distances (average 2.057 Å in **1** and 2.055 Å in **2**). In contrast, the iron−imine bond distances (average 2.057 Å in **1** and 2.052 Å in **2**) are longer than the nickel−imine bond distances (average 1.996 Å in **1** and 2.017 Å in **2**). Of the two axial iron−azide bond distances in **1**, that (ca. 2.027 Å) involving N8 is in the range (ca. 1.990−2.057 Å) of the basal bond distances, while that (ca. 2.149 Å) involving N5 is significantly longer, while both the two axial bond distances (ca. 2.075 and 2.125 Å) involving nickel in **1** are longer than the basal bond distances (1.994−2.062 Å).

As already mentioned, one bridging acetate is common in **2A,B**. For this common acetate bridge, the iron−acetate (ca. 2.049 Å) and nickel−acetate (ca. 2.055 Å) bond distances are very close. However, for the second bridging acetate in **2B**, the nickel−acetate bond distance (ca. 2.191 Å) is significantly longer than the iron−acetate bond length (ca. 2.029 Å).

The iron(III)···nickel(II) separation in **1** (3.038 Å) is longer than that in **2** (2.892 Å). The phenoxo bridge angles in **1** (Fe1−O1−Ni1 = 97.39° and Fe1−O2−Ni1 = 97.22°) are almost identical, as are the two phenoxo bridge angles in **2** (Fe1−O1−Ni1 = 90.96° and Fe1−O2−Ni1 = 90.81°). Clearly, the phenoxo bridge angles in **2** are smaller than those in **1**. The dihedral angles between the two N(imine)$_2$O(phenoxo)$_2$ basal planes in **1** and **2** are 7.24 and 11.1°, respectively, indicating that the bridging moieties in both compounds are slightly twisted.

There exist a few intra-/intermolecular hydrogen bonding interactions in both **1** and **2**, some of which interlink the individual dinuclear units to generate a one-dimensional topology in **2** and a two-dimensional topology in **1** (see the Supporting Information for a description, demonstration (Figures S1 and S2), and geometries of the hydrogen bonds (Table S1)).

Magnetic Properties. dc magnetic susceptibility data were collected for crushed crystalline samples of **1** and **2** at an applied magnetic field of 1 T in the 2−300 K temperature range. The data of **1** and **2** are shown in Figures 3 and 4, respectively, as $\chi_M T$ versus T plots.

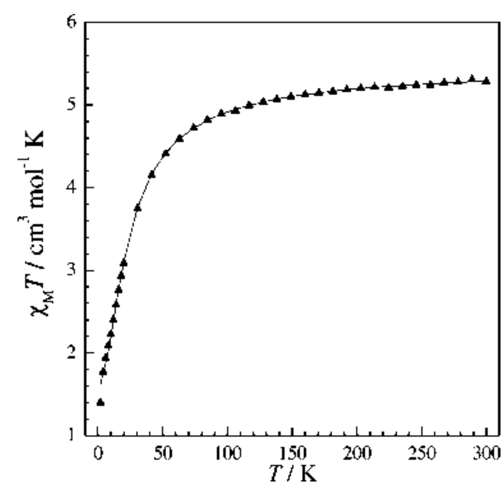


Figure 3. Fitting of the $\chi_M T$ versus T plot of [Fe^{III}(N$_3$)$_2$LNi^{II}(H$_2$O)(CH$_3$CN)](ClO$_4$) (**1**) between 2 and 300 K. The experimental data are shown as black triangles, and the black line corresponds to the theoretical values (see text).

For **1**, the $\chi_M T$ value at 300 K, 5.28 cm^3 mol^{-1} K, is only slightly less than the expected value of 5.38 cm^3 mol^{-1} K for one high-spin Fe^{III} ion ($S = {}^5/_2$) and one high-spin Ni^{II} ion ($S = 1$) with g = 2.0. As the temperature decreases from 300 to 60 K, the $\chi_M T$

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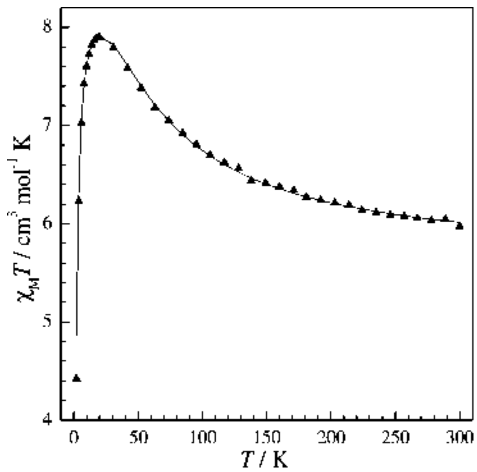

Figure 4. Fitting of the $\chi_M T$ versus T plot of $\{[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-}OAc)]_{0.6}\cdot[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]_{0.4}\}(ClO_4)\cdot1.1H_2O$ between 2 and 300 K. The experimental data are shown as black triangles, and the black line corresponds to the theoretical values (see text).

product decreases slowly to 4.60 cm^3 mol^{-1} K. On further lowering of the temperature, $\chi_M T$ decreases rapidly to 1.40 cm^3 mol^{-1} K at 2 K. Clearly, the $\chi_M T$ versus T profile indicates that the Fe^{III} and Ni^{II} centers in **1** are antiferromagnetically coupled.

In the case of **2**, the $\chi_M T$ value at 300 K, 5.97 cm^3 mol^{-1} K, is greater than the expected value of 5.38 cm^3 mol^{-1} K for one high-spin Fe^{III} ion ($S = {}^5/_2$) and one high-spin Ni^{II} ion ($S = 1$) with $g = 2.0$. As the temperature decreases from 300 K, the $\chi_M T$ product increases steadily to reach a maximum of 7.90 cm^3 mol^{-1} K at 20 K. On further cooling to 2 K, the $\chi_M T$ product decreases rapidly to 4.42 cm^3 mol^{-1} K. The profile in this case clearly indicates ferromagnetic coupling between the two metal centers.

The susceptibility data were fitted using juIX software,[27] in which $H = -2JS_1\cdot S_2$ is taken as the model Hamiltonian for dinuclear systems. To consider the intermolecular interaction which may propagate through the hydrogen-bonding interactions in **1** and **2**, the Weiss constant Θ was taken into consideration in the simulation process. It is also worth mentioning that an average J value was considered for **2A**,**B** and this approximation is highly logical because most of the structural parameters that may govern magnetic properties in **2A**,**B** are identical and the countercomplementarity effect of acetate is rather small (see below). The fittings converged with $J = -3.14$ cm^{-1}, $g_{Fe} = 2.0$, and $g_{Ni} = 2.094$ for **1** and $J = 7.36$ cm^{-1}, $g_{Fe} = 2.0$, $g_{Ni} = 2.2$, and $\Theta = -0.75$ K for **2**. It is relevant to mention that, as compound **2** is a cocrystal of 60% of **2A** and 40% of **2B**, the J value of 7.36 cm^{-1} is actually the sum of the 60% of the J value of **2A** and 40% of the J value of **2B**. In fact, the DFT-computed J value of **2** has been calculated in this way (vide infra, footnotes of Table 4).

Comparison of Structures and Magnetic Properties of 1 and 2 with those of Related Compounds. Previously the $Fe^{III}Ni^{II}$ compound $[Fe^{III}(H_2O)L^XNi^{II}(OAc)(\mu\text{-}OAc)](ClO_4)\cdot2H_2O$ (**3**), derived from a tetraaminodiphenol macrocyclic ligand, H_2L^X (the saturated analogue of a tetraiminodiphenolate macrocycle obtained on 2:2 condensation of 2,6-diformyl-4-methylphenol and 1,3-diaminopropane), has been reported.[19a] The metal centers in **3**, as in **2**, are bridged by a bis(μ-phenoxo)-μ-acetate moiety. Regarding dinuclear $Fe^{III}Ni^{II}$ compounds having a phenoxo/hydroxo/alkoxo bridging moiety, only four other examples are known.[19b−d,28] Three of these are the heterobridged μ-phenoxo-bis(μ-carboxylate) compounds $[Fe^{III}Ni^{II}(BPMP)(\mu\text{-}OPr)_2](BPh_4)_2$ (**4**),[19b] $[Fe^{III}Ni^{II}(BPBPMP)(\mu\text{-}OAc)_2](ClO_4)$ (**5**),[28] and $[Fe^{III}Ni^{II}(IPCPMP)(\mu\text{-}OAc)_2(CH_3OH)](PF_6)$ (**6**),[19c] where BPMP, BPBPMP, and IPCPMP are the anions of 2,6-bis[(bis(2-pyridylmethyl)amino]methyl]-4-methylphenol, 2-bis[{(2-pyridylmethyl)-aminomethyl}-6-{(2-hydroxybenzyl)(2-pyridylmethyl)}aminomethyl]-4-methylphenol, and 2-(N-isopropyl-N-((2-pyridyl)methyl)aminomethyl)-6-(N-(carboxylmethyl)-N-((2-pyridyl)methyl)aminomethyl)-4-methylphenol, respectively. On the other hand, the remaining compound is the heterobridged μ-alkoxo-μ-diphenylphosphate system $[Fe^{III}Ni^{II}(HPTB)\{\mu\text{-}O_2P(OPh)_2\}\{O_2P(OPh)_2\}(CH_3OH)](ClO_4)_2\cdot2H_2O$ (**7**),[19d] where HPTB is the anion of N,N,N',N'-tetrakis(2-benzimidazolylmethyl)-2-hydroxy-1,3-diaminopropane.

As compounds **1**−**3** contain a diphenoxo bridging moiety, parameters such as Fe−O−Ni bridge angles, Fe−O/Ni−O bridge distances, Fe−O−Ni−O dihedral angles, and out-of-plane shifts of the phenoxo group should be considered as the possible governing factors to determine the nature and extent of magnetic exchange interactions. For **2** and **3**, due to the presence of an additional carboxylate bridge, the effect of orbital complementarity or orbital countercomplementarity should also have a role on the magnetic properties. On the other hand, as compounds **4**−**7** contain a monophenoxo (for **4**−**6**) or monoalkoxo (for **7**) bridging moiety, Fe−O−Ni bridge angles and Fe−O/Ni−O bridge distances should be considered as the possible governing

Table 4. Experimental and DFT-Computed J Values along with Some Relevant Structural Parameters of 1, 2, and Previously Published Related $Fe^{III}Ni^{II}$ Compounds

complex[a]		J_{DFT} (cm^{-1})	J_{exp} (cm^{-1})	av Fe−O−Ni (deg)	av M−O (Å)	Fe−O−Ni−O (deg)	τ (Å)[b]	ref
$[Fe^{III}(N_3)_2LNi(H_2O)(CH_3CN)]^+$ (**1**)	bis(μ-phenoxo)	−4.5	−3.14	97.33	2.0262	0.9	30.95	this work
$\{[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-}OAc)]_{0.6}\cdot[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]_{0.4}\}^+$ (**2**)	bis(μ-phenoxo)-μ-acetate; bis(μ-phenoxo)-bis(μ-acetate)	7.23[c] (7.86 for **2A** 6.28 for **2B**)	7.36	90.88	2.0295	4.88	39.05	this work
$[Fe^{III}(H_2O)L^XNi^{II}(OAc)(\mu\text{-}OAc)]^+$ (**3**)[d]	bis(μ-phenoxo)-μ-acetate		1.7	92.8	2.0825	NA[e]	NA[d]	19a
$[Fe^{III}Ni^{II}(BPMP)(OPr)_2]^{2+}$ (**4**)	μ-phenoxo-bis(μ-propionate)	−13.2	−12	116.2	1.989			19b
$[Fe^{III}Ni^{II}(BPBPMP)(OAc)_2]^+$ (**5**)	μ-phenoxo-bis(μ-acetate)	−6.86	NA	118.66	2.026			25
$[Fe^{III}Ni^{II}(IPCPMP)(\mu\text{-}OAc)_2(CH_3OH)](PF_6)$ (**6**)	μ-phenoxo-bis(μ-acetate)	−8.27	−11.2	116.43	2.014			19c
$[FeNi(HPTB)\{\mu\text{-}O_2P(OPh)_2\}\{O_2P(OPh)_2\}(CH_3OH)]^{2+}$ (**7**)	μ-alkoxo-μ-diphenylphosphate	−15.6	−18.1	131.7	2.0185			19d

[a]The formula of only the cationic part is given; see text for the composition of L^1, BPMP, BPBPMP, and HPTB. [b]Out-of-plane shift of the phenoxo group. [c]Calculated from 60% of $[Fe^{III}(H_2O)LNi^{II}(OAc)(\mu\text{-}OAc)]^+$ (**2A**; J_{DFT} = 7.86 cm^{-1}) and 40% of $[Fe^{III}LNi^{II}(\mu\text{-}OAc)_2]^+$ (**2B**; J_{DFT} = 6.28 cm^{-1}). [d]J_{DFT} value has not been computed because structural data are not available (see text). [e]Not available.

factors to determine the nature and extent of magnetic exchange interactions. For these compounds **4**−**7**, orbital complementarity/countercomplementarity of carboxylates/phosphate bridging moieties should also have roles in the magnetic behavior. Of compounds **3**−**7**, the magnetic properties of all but **5** were investigated. The observed J values (of **1**−**4**, **6**, and **7**) along with the structural parameters (of **1**−**7**), which may influence the magnetic properties, have been given in Table 4. While compounds **2** and **3** exhibits ferromagnetic interactions with J values of 7.36 and 1.7 cm^{-1}, respectively, the metal centers in compounds **1**, **4**, **6**, and **7** are antiferromagnetically coupled with J values of −3.05, −12, −11.2, and −18.1 cm^{-1}, respectively.

It has been established in different types of systems that the nature and magnitude of magnetic exchange interactions are very much dependent on the bridge angle.[2,3a−g,4a,5a−d,6a,8a,c−g,9] The general trend in oxo/hydroxo/alkoxo-bridged systems is that interaction becomes less antiferromagnetic as the bridge angle becomes smaller; below a particular angle, the crossover angle, the interaction becomes ferromagnetic.[2,3a−f,4a,5a−d,6a,8a,c,d] The crossover angle as determined from magneto-structural correlations in some systems are 97.5° in planar dihydroxo-bridged Cu^{II}_2,[3a] 95.7° in alkoxo-bridged Cu^{II}_4,[3b] 89° in diphenoxo-bridged Cu^{II}_2,[3d] 97° in diphenoxo-bridged Ni^{II}_2,[4a] and 91.2° in dialkoxo-bridged Fe^{III}_2.[5c] Even in monophenoxo-bridged Cu^{II}_2 compounds having an axial−equatorial mode of the bridging atom, a general trend of angle dependence of the exchange interaction has been observed.[3e] In that case, the ferromagnetic interaction becomes smaller as the bridge angle increases from 126° and the interaction becomes nil when the bridge angle is 142°. The general trend of angle dependence of exchange interaction has also been observed also in oxo-bridged Fe^{III}_2,[5a,b] dioxo-bridged Mn^{IV}_2,[6a] and Fe^{III}_6 compounds having μ-hydroxo-bis(μ-carboxylate)/bis(μ-alkoxo)-μ-carboxylate bridges.[5g] Now, the average phenoxo-bridge angle and the J value in the $Fe^{III}Ni^{II}$ compounds **1**−**3**, all having a common diphenoxo bridging moiety, are 97.33° and −3.14 cm^{-1} for **1**, 92.8° and 1.7 cm^{-1} for **3**, and 90.88° and 7.36 cm^{-1} for **2**, revealing that the well-known general trend of bridge angle dependence of exchange integral is maintained here also. Similarly, although the $Fe^{III}Ni^{II}$ compounds **4**, **6**, and **7** have a monophenoxo bridging moiety, stronger antiferromagnetic (J ranges between −11.2 and −18.1 cm^{-1}) interaction in these compounds in comparison to that in **1**−**3** is probably related to the greater values of Fe−O−Ni bridge angle (range 116.2−131.7°) in the former three compounds.

Although the exchange interactions can be qualitatively rationalized in terms of the bridge angle, it is relevant to know the roles of different parameters in a quantitative way as well as to understand the electronic origin of exchange interaction. To compute the theoretical J values, to understand the role of the aforementioned different factors on magnetic properties, and to determine magneto-structural correlations, DFT calculations have been carried out for compounds **1**, **2**, and **4**−**7** and for model systems. The DFT calculation on **3** could not be done because required structural data are not available in the CSD.[29] It may also be noted that the refinement of the structure of **3** was hampered (R = 0.123) due to the rather inferior quality of diffraction data and severe disordering of the perchlorate anions.[19a] The theoretical studies are described in the following section.

Theoretical Studies. DFT calculations have been performed on the full structures of complexes **1**, **2**, and **4**−**7**, including the counteranion wherever applicable, and the results are summarized in Table 4. Of these six compounds, observed J values for compounds **1**, **2**, **4**, **6**, and **7** are available and it is interesting to note that an excellent agreement with the experimental J values has been obtained in all five of these systems. In this series of complexes, one can classify three types of interactions: (i) moderately weak antiferromagnetic interactions (complexes **1** and **5**), (ii) moderately strong antiferromagnetic interactions (complexes **4**, **6**, and **7**), (iii) moderately weak ferromagnetic interactions (complex **2**). DFT calculations have nicely reproduced this trend; in fact, in all cases an excellent agreement with experimental J values has been detected.

Additionally, our aim has been to understand the electronic origin of the difference in the magnetic coupling (antiferro−ferro) observed between complexes **1** and **2** and then to develop extensive magneto-structural correlations for this class of compounds. To understand the difference in the magnetic coupling strength, we have performed MO analysis and computed overlap integrals using BS empty orbitals which are in many instances shown to be useful in performing qualitative analysis of the coupling mechanism.[30] As there is no crystallographic symmetry in these complexes, there are in principle 10 interactions possible between the magnetic orbitals of Ni(II) and Fe(III). We have computed the overlap integral for complex **1**, and the following interactions are found to be significant: $d_{z^2}|p|d_{yz}$, $d_{z^2}|p|d_{z^2}$, $d_{z^2}|p|d_{x^2-y^2}$, $d_{x^2-y^2}|p|d_{z^2}$, and $d_{x^2-y^2}|p|d_{x^2-y^2}$ (see Figure 5). Since

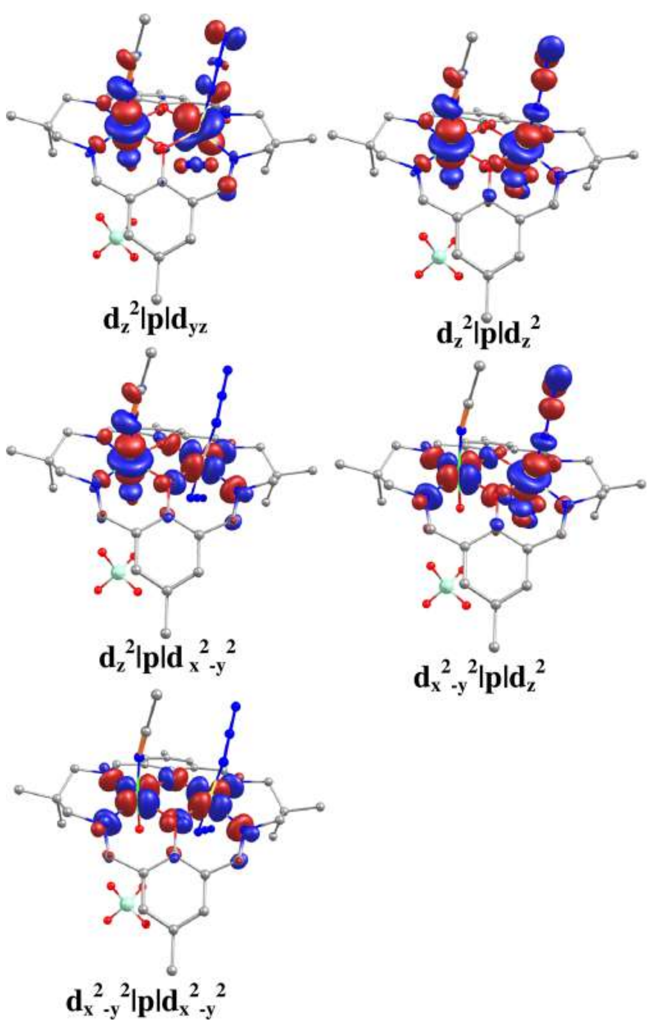

Figure 5. Pair of BS empty magnetic orbitals (for complex **1**) where a significant overlap integral has been computed. The α set on Ni^{II} and β set of orbitals on Fe^{III} are plotted simultaneously for clarification.

the overlap integral is directly proportional to the antiferromagnetic part of the coupling constant, these five dominant interactions lead to antiferromagnetic coupling in complex **1**. It is important to note here that, out of these five interactions, the interactions between the axial orbitals are expected to dominate over the others (d_{z^2}−d_{z^2} and/or $d_{x^2-y^2}$−$d_{x^2-y^2}$). For the species **2A** in complex **2**, significant interactions (see Figure 6) are

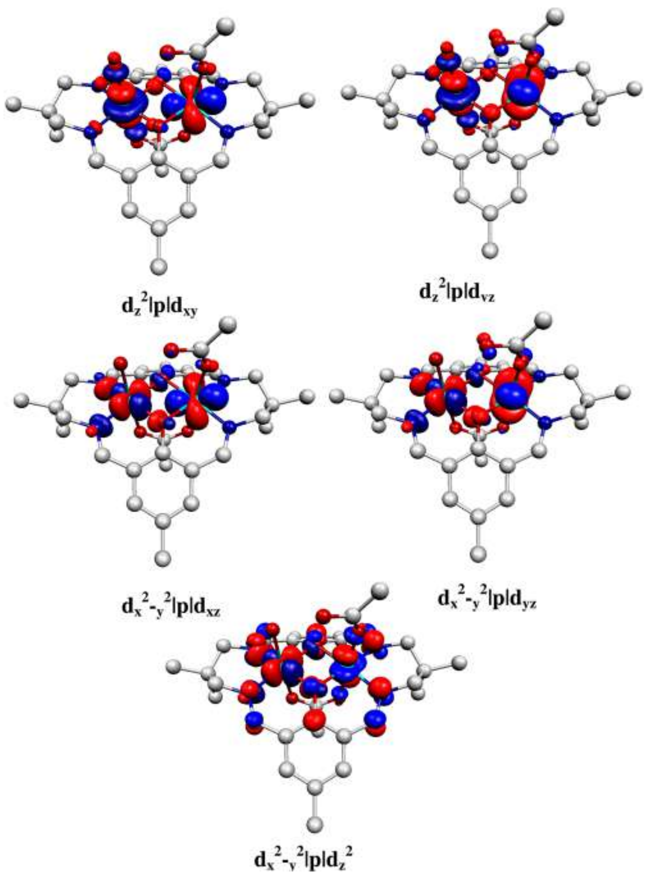

Figure 6. Pair of BS empty magnetic orbitals (for complex **2A**) where a significant overlap integral has been computed. The α set on Ni^{II} and β set of orbitals on Fe^{III} are plotted simultaneously for clarification.

$d_{z^2}|p|d_{xy}$, $d_{z^2}|p|d_{yz}$, $d_{x^2-y^2}|p|d_{xz}$, $d_{x^2-y^2}|p|d_{yz}$, and $d_{x^2-y^2}|p|d_{z^2}$. Although this complex also possesses five significant interactions, the $d_{z^2}|p|d_{z^2}$ and $d_{x^2-y^2}|p|d_{x^2-y^2}$ interactions are found to be less significant and this is likely due to the relatively acute Fe−O−Ni angle observed in complex **2** (90.87° in **2** versus 97.31° in **1**). In addition, it is important to note here that, due to the shorter Fe−Ni distance in complex **2**, some new interactions which are less significant in **1** are noticed ($d_{z^2}|p|d_{xy}$ and $d_{x^2-y^2}|p|d_{xy}$). For the species **2B** in complex **2**, seven strong interactions (see Figure 7) are observed: $d_{z^2}|p|d_{xz}$, $d_{z^2}|p|d_{yz}$, $d_{z^2}|p|d_{z^2}$, $d_{z^2}|p|d_{x^2-y^2}$, $d_{x^2-y^2}|p|d_{xy}$, $d_{x^2-y^2}|p|d_{z^2}$, and $d_{x^2-y^2}|p|d_{x^2-y^2}$. Clearly, the number of interactions is more in **2B** than in **2A**. Out of the seven interactions in **2B**, $d_{x^2-y^2}|p|d_{x^2-y^2}$ is the strongest. Therefore, because of the higher number of strong interactions, **2B** is expected to have a lower positive *J* value than **2A**. It is worth mentioning that the higher number of interactions in **2B** in comparison to those in **2A** arises because of the monodentate (to Fe^{III}) acetate ligand in **2A** versus bridging acetate ligand in **2B**, which is the only structural difference between **2A** and **2B**.

Apart from the differences in the bond angles, additional carboxylates (one for **2A** and two for **2B**) present in **2** may

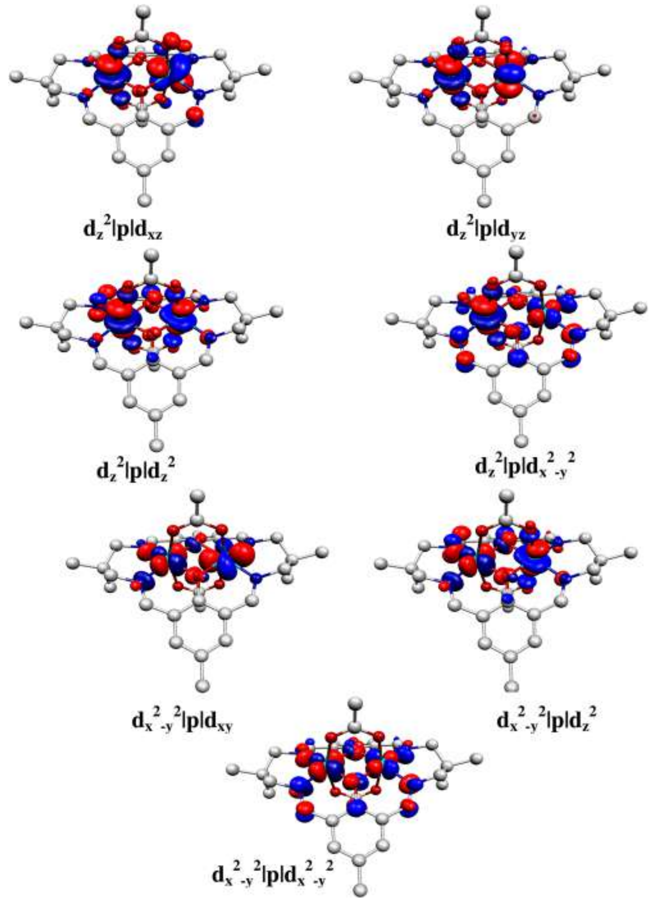

Figure 7. Pair of BS empty magnetic orbitals (for complex **2B**) where a significant overlap integral has been computed. The α set on Ni^{II} and β set of orbitals on Fe^{III} are plotted simultaneously for clarification.

bring forth complementarity/countercomplementarity effects. To understand the complementarity/countercomplementarity effects of the acetate group, calculations have been performed on model complexes constructed from **2A,B**. In the first model, $[Fe^{III}(H_2O)_2LNi^{II}(\mu\text{-}OAc)]^{2+}$, the terminal acetate in **2A** coordinated to Fe(III) ion is modeled as a water molecule, and here a decrease in the ferromagnetic *J* value from 7.9 cm^{-1} (J_{DFT} value of **2A**) to a value of 5.7 cm^{-1} has been observed. When the bridging acetate is also modeled as water molecules, the *J* value further decreases to 5.3 cm^{-1}. A detailed discussion about complementarity/countercomplementarity effects are given in ESI (See Scheme S1 and the relevant discussion in the ESI). Clearly, the terminal acetate has a greater effect on the *J* value in comparison to the bridging acetate and the electronic reasons for this behavior are routed back to the interaction with the magnetic orbitals where the terminal acetate is found to have a stronger antibonding interactions with Fe (see Figure 6) in comparison to the bridging acetate. At this point, it is worth mentioning that the electronic effect of the coligands on exchange interactions is known.[31]

Both acetates in **2B** are also modeled as water ligands sequentially where the computed *J* values are 5.74 and 5.30 cm^{-1}, respectively, in comparison with 6.28 cm^{-1} (J_{DFT} value of **2B**). This suggests a countercomplementarity effect: i.e., the presence of terminal acetate enhances the strength of ferromagnetic *J* in this complex.

The spin density plots of complexes **1** and **2** (**2A** and **2B**) are shown in Figure 8. In both cases the spin density of Fe^{III} is found to be ∼4.2 while that on Ni^{II} is ∼1.7, revealing that spin delocalization is operative in both complexes. In both complexes,

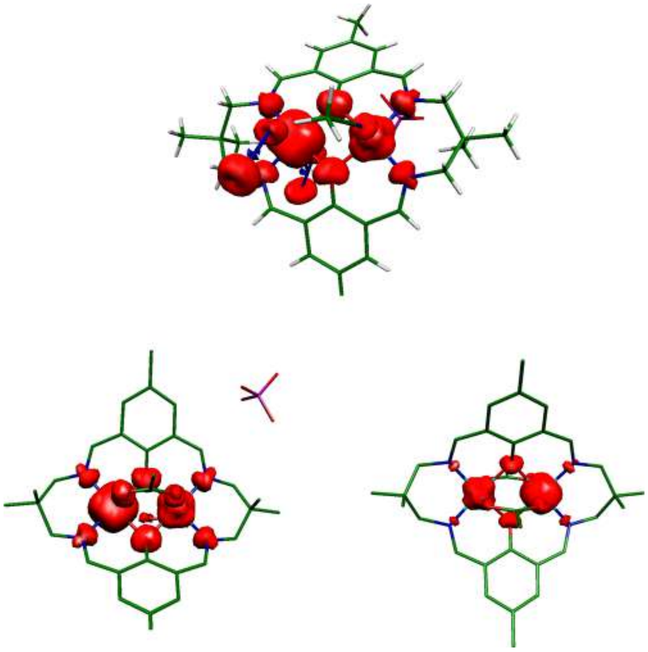

Figure 8. Computed spin density plots for complexes **1** (top) and **2** (bottom; **2A** left, **2B** right).

the two μ-phenoxo oxygen atoms have different magnitudes of spin densities (for high-spin state: 0.12 and 0.14 in **1**, 0.16 and 0.15 in **2A**, and 0.15 and 0.15 in **2B**) and this is likely due to the unsymmetrical M−O bond length observed in these complexes; these differences in the magnitudes are reflected in the computed magnetic coupling.

As magnetic data could be well simulated with the model Hamiltonian $H = -2JS_1 \cdot S_2$, it is evident that Fe^{III} in **1** and **2** is in a high-spin state throughout the temperature range 2−290 K. DFT-computed spin state energies of a model $Fe^{III}Zn^{II}$ compound constructed on replacing Ni^{II} in **2A** by Zn^{II} and t_{2g}−e_g splitting of the Fe^{III} center in **2A** also clearly indicate that Fe^{III} is in a high-spin state and spin crossover is not possible (Figure S3, Supporting Information).

With this, we set out to develop magneto-structural correlations for a diphenoxo-bridged $Fe^{III}Ni^{II}$ dinuclear unit. Since the two μ-phenoxo bridges are the parts of the macrocyclic ligand system and one structural variation at a time demands variation of Fe···Ni and O···O distances, a model complex has been constructed for complex **1** (Figure S4 in the Supporting Information). The calculation of *J* for this model is different from that for the original complex, with the computed *J* being 1.6 cm^{-1}, but we believe that this model is good enough to understand the trend observed among different structural parameters.

J versus Bond Angle Correlation. The correlation on average Fe−O−Ni angle has been developed and is shown in Figure 9. The *J* values decrease with an increase in average Fe−O−Ni angle up to around 125°. This is due to predominant $d_{x^2-y^2}|p|d_{x^2-y^2}$ overlap. After 125°, a steep increase in *J* value is observed. At first instance, this result seems surprising, as we are observing ferromagnetic interactions at large Fe−O−Ni angles. However, such a trend has already been noted in di-μ-oxo Cu(II) complexes.[8c] At larger angles the Ni−O and Fe−O σ^* orbitals gradually lose the major metal d contribution and, at extremely larger angles, a significant electron transfer from these σ^* orbitals to the O−O π^* orbital is taking place, indicating an O−O interaction. It is noted here that at the extreme angle of 133°, the

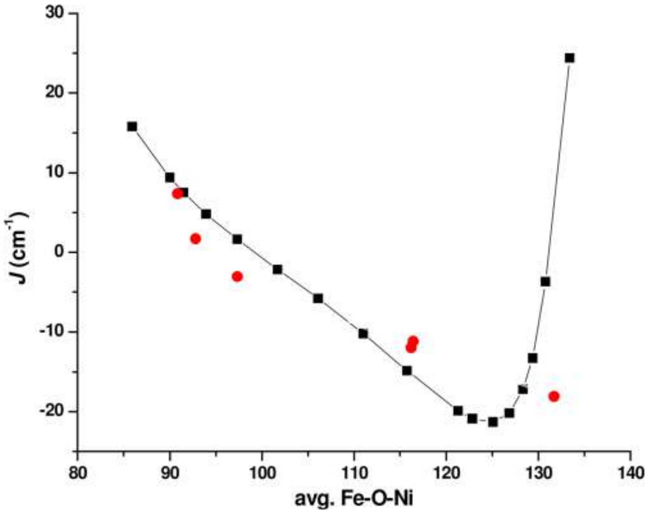

Figure 9. Magneto-structural correlation of average Fe−O−Ni bond angle of the model complex **1A** (as shown in Figure S4 in the Supporting Information) along with the experimental data points (shown by red dots; see Table 4).

O···O distance is merely 1.60 Å, indicating a rather strong O···O interaction. This statement is verified by plotting the spin density of Fe^{III} together with the average spin density on the oxygen atoms, and this clearly indicates that the spin population on the oxygen atoms increases steeply at a larger Fe−O−Ni bond angle (Figure 10a). A steep decrease in Fe^{III} (also Ni^{II}) is accompanied by a steep increase in oxygen, indicating a strong oxygen−oxygen interaction and strong ferromagnetic coupling. The distortion for smaller and larger angles costs ca. 100 kJ/mol for structures up to 120° while much larger angles steeply raise the energy penalty; this is due to O···O interactions being the dominant factor (see Figure S5 in the Supporting Information). Among the correlations developed (see below), the Fe−O−Ni angle is the most important, as this parameter induces changes in *J* values of about 40 cm^{-1} while the changes in magnitude by other parameters are minimal.

J versus Bond Distance Correlation. The average Fe/Ni−O distance correlation developed is shown in Figure 10b (see also Figure S6 in the Supporting Information for experimental points). The *J* value steadily decreases with an increase in the average distance, tending toward less ferromagnetic *J* at greater distances. This is somewhat expected from the strong axial type d orbital overlap detected for this complex; as the distance increases the overlap decreases, leading to a smaller ferromagnetic *J* value at longer bond distances. This correlation, however, is expected to saturate at longer distance, and it is important to note here that the scale of change is much smaller as compared to the bond angle correlation.

J versus Fe−O−Ni−O Dihedral Angle Correlation. The third correlation, *J* versus Fe−O−Ni−O dihedral angle, is shown in Figure 10c (see also Figure S6 in the Supporting Information for experimental points). Initially, the *J* value decreases with a increase in the dihedral angle value and reaches saturation at higher angles. The spin densities on Fe^{III} and Ni^{II} along with different dihedral angles reveal that there is an antagonizing behavior where Fe^{III} spin density increases while Ni^{II} spin density decreases initially while at a higher angle the Fe^{III} spin density steeply decreases. This indicates that, at greater dihedral angles, a stronger delocalization is observed leading to an antiferromagnetic interaction at greater angles (see Table S2, Supporting Information).

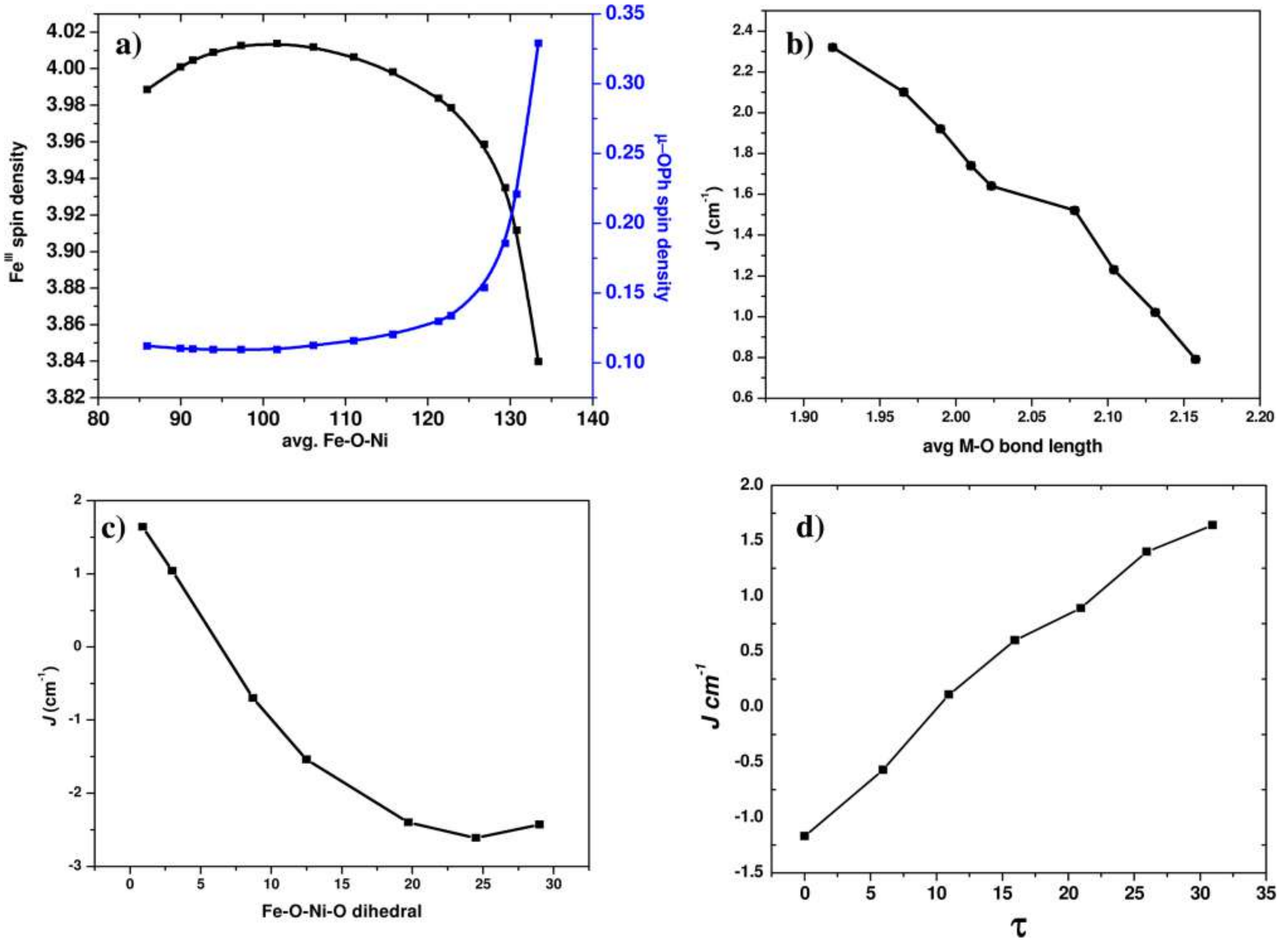

Figure 10. (a) Computed spin densities on the Fe^{III} and μ-phenoxo (average) atoms for various Fe−O−Ni angles. Magneto-structural correlations: (b) J versus average M−O (Fe−O and Ni−O distances); (c) J versus Fe−O−Ni−O dihedral angle; (d) J versus out-of-plane shift (τ) of the phenoxy group. The calculations for these correlations have been performed on the model complex **1A** (Figure S4, Supporting Information).

J versus Out-of-Plane Shift Correlation. The fourth correlation, J versus out-of-plane shift (τ) of the phenoxo group, has also been developed and is shown in Figure 10d (see also Figure S6 in the Supporting Information for experimental points). As we can see from this figure, an increase in the τ value increases the ferromagnetic J value and the molecule exhibits ferromagnetic exchange at higher τ value. The antiferromagnetic contribution to J decreases by increasing the τ parameter because the interaction between the oxygen and the phenyl group in plane orbitals (when τ is set to 0) weakens, leading to weaker overlap between the magnetic orbitals and stronger ferromagnetic exchange, as observed in Figure 10b.

■ CONCLUSIONS

The bis(μ-phenoxo) $Fe^{III}Ni^{II}$ compound $[Fe^{III}(N_3)_2LNi^{II}(H_2O)(CH_3CN)](ClO_4)$ (**1**) and the cocrystalline bis(μ-phenoxo)-μ-acetate/bis(μ-phenoxo)-bis(μ-acetate) $Fe^{III}Ni^{II}$ compound $\{[Fe^{III}(OAc)LNi^{II}(H_2O)(\mu\text{-OAc})]_{0.6}\cdot[Fe^{III}LNi^{II}(\mu\text{-OAc})_2]_{0.4}\}(ClO_4)\cdot 1.1H_2O$ (**2**) are among the very rare examples of $Fe^{III}Ni^{II}$ compounds containing one or more phenoxo/hydroxo/alkoxo bridges. The antiferromagnetic interaction in **1** and ferromagnetic interaction in **2** reveal that our design to produce systems having remarkably different magnetic properties has been successful. Interestingly, the DFT calculated J values are nicely matched with the experimental J values of **1**, **2**, and previously reported μ-phenoxo-bis(μ-carboxylate) or μ-alkoxo-μ-diphenylphosphate $Fe^{III}Ni^{II}$ compounds. Therefore, as no experimental J value is available for a μ-phenoxo-bis(μ-carboxylate) $Fe^{III}Ni^{II}$ system (**5**), the DFT calculated J value (-6.9 cm^{-1}) can be taken as the exchange coupling constant for this compound.

The countercomplementarity effect of carboxylate in heterobridged μ-phenoxo/hydroxo/alkoxo-μ-carboxylate dicopper(II) compounds can be easily understood from the orbital model, which is rather simple due to a single magnetic orbital per metal ion. This idea is usually extended to the heterobridged systems of other metal ions having more than one magnetic orbital. However, in the latter cases, the situation may be complicated due to several possible combinations of magnetic orbitals. Therefore, a clear demonstration of the countercomplementarity effect of an acetate moiety in $Fe^{III}Ni^{II}$ systems from DFT calculations in the present investigation deserves attention, although it has also been shown that the countercomplementarity effect is smaller and not additive. It has also been demonstrated that the countercomplementarity effect of carboxylate is operative via the d_{z^2} orbital. Again, the more prominent role of terminal bridging acetate in comparison to the bridging acetate is an interesting observation, which has been also clarified in terms of MO analyses. The DFT computed magneto-structural correlations of J with several parameters (Fe−O−Ni angle, average Fe/Ni−O distance, Fe−O−Ni−O dihedral angle, and out-of-plane shift of the phenoxo group) of the diphenoxo-bridged $Fe^{III}Ni^{II}$ systems, as determined here, are among the only few correlations

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(theoretical or experimental) in heteronuclear systems and are the first in any type of $Fe^{III}Ni^{II}$ system. The profiles of magneto-structural correlations here have been justified also in terms of structural parameters or in terms of DFT-computed spin densities. The first examples of magneto-structural correlations, as demonstrated here, will be hopefully useful in analyzing the magnetic behavior of $Fe^{III}Ni^{II}$ polynuclear clusters in terms of pairwise interactions and therefore will also be useful in designing $Fe^{III}Ni^{II}$-based SMMs.

With regard to the countercomplementarity effect of carboxylate and the magneto-structural correlations established here, all of the factors are not in the same line to govern anti-ferromagnetic interactions in **1** and ferromagnetic interactions in **2**. However, the electronic origin of the interactions has been nicely understood from MO analysis.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

Text, figures, tables, and CIF files giving crystallographic data for **1** and **2**, a description of the supramolecular topology in **1** and **2**, Figures S1−S5, and Tables S1 and S2. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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Slow magnetic relaxation and electron delocalization in an $S = {}^{9}/_{2}$ iron(II/III) complex with two crystallographically inequivalent iron sites

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The magnetic, electronic, and Mössbauer spectral properties of $[Fe_2L(\mu\text{-OAc})_2]ClO_4$, **1**, where L is the dianion of the tetraimino-diphenolate macrocyclic ligand, H_2L, indicate that **1** is a class III mixed valence iron(II/III) complex with an electron that is fully delocalized between two crystallographically inequivalent iron sites to yield a $[Fe_2]^{V}$ cationic configuration with a $S_t = 9/2$ ground state. Fits of the dc magnetic susceptibility between 2 and 300 K and of the isofield variable-temperature magnetization of **1** yield an isotropic magnetic exchange parameter, J, of $-32(2)$ cm^{-1} for an electron transfer parameter, B, of 950 cm^{-1}, a zero-field uniaxial $D_{9/2}$ parameter of $-0.9(1)$ cm^{-1}, and $g = 1.95(5)$. In agreement with the presence of uniaxial magnetic anisotropy, ac susceptibility measurements reveal that **1** is a single-molecule magnet at low temperature with a single molecule magnetic effective relaxation barrier, U_{eff}, of 9.8 cm^{-1}. At 5.25 K the Mössbauer spectra of **1** exhibit two spectral components, assigned to the two crystallographically inequivalent iron sites with a static effective hyperfine field; as the temperature increases from 7 to 310 K, the spectra exhibit increasingly rapid relaxation of the hyperfine field on the iron-57 Larmor precession time of 5×10^{-8} s. A fit of the temperature dependence of the average effective hyperfine field yields $|D_{9/2}| = 0.9$ cm^{-1}. An Arrhenius plot of the logarithm of the relaxation frequency between 5 and 85 K yields a relaxation barrier of 17 cm^{-1}. © *2011 American Institute of Physics*. [doi:10.1063/1.3581028]

I. INTRODUCTION

Over the past two decades, a number of molecules have been shown to retain their magnetization for finite periods of time upon removal of a magnetic field.[1–3] These complexes, which have come to be known as single-molecule magnets, exhibit such slow magnetic relaxation due to a thermal barrier to spin relaxation that arises due to uniaxial anisotropy acting on a high-spin ground state. This barrier is quantified according to the equation $U = S^2|D|$ for integer spins and $U = (S^2 - 1/4)|D|$ for half-integer spins, where S is the ground state spin and D is the axial zero-field splitting parameter. The discovery of single-molecule magnets has generated much interest from a wide range of scientists, as these species could find potential utility in applications such as high-density information storage, quantum computing, and magnetic refrigeration.

While the relaxation barrier of a single-molecule magnet scales with S and D, the strength of magnetic interactions between paramagnetic centers within the molecule must also be considered. The magnitude of these interactions governs how well separated the spin ground state is from excited states. Indeed, mixing of excited states with the ground state spin can lead to a number of fast relaxation pathways, thereby shortcutting the overall anisotropy barrier. To date, the vast majority of single-molecule magnets feature metal centers coupled through a superexchange mechanism, often mediated through oxide[4,5] or cyanide[6] ligands. However, superexchange, especially through multiatom bridges, is often a relatively weak interaction, such that relaxation can be dominated by pathways involving excited states. As an alternative, double exchange[7] via electron delocalization between mixed-valence transition metal ions,[8–11] is generally a much stronger interaction than superexchange, such that the spin ground state is well-isolated even above room temperature. There is a well documented[12,13] example of a spin 9/2 ground state in a class III mixed-valence iron complex but, unfortunately, the relaxation of the magnetization does not slow down in the absence of an applied magnetic field. A more recent example of a spin 5/2 ground state in a class III mixed-valence vanadium complex has been thoroughly investigated.[14] Hence, the synthesis of multinuclear mixed valent iron complexes

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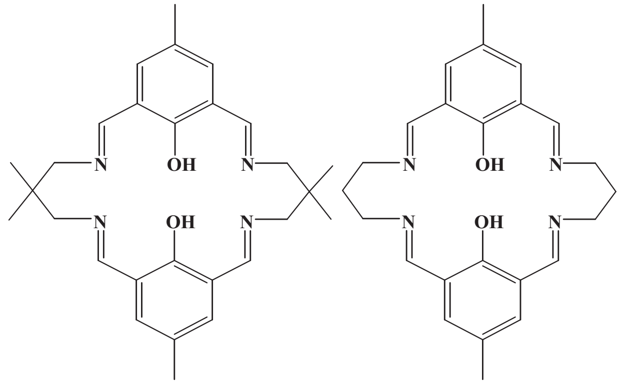

CHART 1. The macrocyclic ligands, H_2L and H_2L^1, used in the preparation of **1**, left, and **2**, right, respectively.

with various ligands is a promising route for the discovery of new complexes exhibiting desirable magnetic properties for applications in fields such as molecular electronics and computing.[15]

Extensive multidisciplinary research into mixed-valence iron complexes has led, through experimental, theoretical, and computational studies,[9–11,16–24] to an enhanced insight into the iron-ligand electron-transfer process and the associated magnetic double exchange mechanism. Further, valence-trapped, class II, and valence-delocalized, class III, mixed-valence iron complexes[8] have been reported in several metalloproteins[25,26] and iron–sulfur proteins.[27–29]

Although several valence-trapped iron(II)-iron(III) complexes have been reported,[17,18] there are only a few examples,[12,13,19] of highly valence-delocalized complexes existing as $Fe^{2.5+}Fe^{2.5+}$ containing complexes, i.e., $[Fe_2]^V$ containing complexes. Only three of these complexes have been characterized by x-ray diffraction, magnetic measurements, and Mössbauer spectroscopy.[12,19] An $S_t = 9/2$ spin ground state has been found in $[(Me_3tacn)Fe(\mu\text{-}OH)_3Fe(Me_3tacn)]_2$, where $Me_3tacn = N, N', N''$-trimethyl-1,4,7-triazacyclononane,[12,13] as shown by the temperature dependence of the molar magnetic susceptibility but, unfortunately, the magnetic relaxation remains too fast at 4.2 K in the absence of an applied magnetic field to observe the magnetic hyperfine splitting in its Mössbauer spectrum. Further, there are only a few complexes exhibiting borderline class II/III behavior.[20] Thus, the design of mixed-valence iron(II)-iron(III) complexes in which the electronic delocalization can be varied from slow to fast, i.e., from valence trapped class II complexes, to partly delocalized class II/III complexes, to completely delocalized class III complexes, is an ongoing challenge.

A fully valence-delocalized $Fe^{2.5+}Fe^{2.5+}$ class III di-iron complex containing two crystallographically *equivalent* iron ions coordinated to the dianionic tetraimino-diphenolate ligand, L^1, see the right portion of Chart 1, has been reported by Nag and co-workers,[19] and it was anticipated that changes in the ligand and, hence, the di-iron coordination environment might yield differing rates of valence-electron delocalization and/or of magnetic relaxation.

By combining synthetic chemistry that yields a new ligand, L, and its associated iron complex, and a microscopic and macroscopic study of its physical properties, we have identified a fully valence-delocalized di-iron complex in which the two iron ions are crystallographically *inequivalent*, with a negative axial zero-field splitting that acts on the $S = {}^9/_2$ ground state to engender slow magnetic relaxation at low temperature. To the best of our knowledge, this complex provides the first example of single-molecule magnetic behavior through double-exchange.

II. EXPERIMENTAL

A. Materials

All the reagents and solvents were purchased from commercial sources and used as received. The mononuclear iron(III) complex, $[Fe(H_2L)(H_2O)Cl](ClO_4)_2 \cdot 2H_2O$, where H_2L is the macrocyclic ligand shown in the left portion of Chart 1, has been prepared by using the method reported[9] earlier, except that 2, 2′-dimethyl-1,3-diaminopropane replaced 1,3-diaminopropane.

B. Synthesis of $[Fe_2L(\mu\text{-}OAc)_2]ClO_4$

$[Fe_2L(\mu\text{-}OAc)_2]ClO_4$, **1**, was synthesized under oxygen-free dry dinitrogen by using standard Schlenk techniques. Solid NaOAc (0.082 g, 1 mmol) and $Fe(ClO_4)_2 \cdot 6H_2O$ (0.063g, 0.25 mmol) were added sequentially to a 25 ml stirred acetonitrile-ethanol (2:3) solution of $[Fe(H_2L)(H_2O)Cl](ClO_4)_2 \cdot 2H_2O$ (0.201 g, 0.25 mmol). After stirring for 2 h, the dark slurry was filtered to remove any suspended particles and the filtrate was cooled to 10 °C a temperature at which a black crystalline precipitate containing diffraction quality single crystals resulted. The crystals were collected by filtration, washed with ethanol, and dried in vacuum. Yield: 148 mg (75%). Anal. Calc. for $Fe_2C_{32}H_{40}N_4O_{10}Cl$: C, 48.79; H, 5.12; N, 7.11. Found: C, 48.70; H, 5.10; N, 7.03. FT-IR (cm^{-1}, KBr): ν(C–H), 2960 w; $\nu(C = N)$, 1625 s; $\nu_{as}(CO_2)$, 1557 m; $\nu_s(CO_2)$, 1404 m; $\nu(ClO_4)$, 1088 vs, 622 w.

C. Physical measurements

The C, H, and N elemental analyses were performed with a Perkin-Elmer 2400 II analyzer. The infrared spectra have been recorded between 400 and 4000 cm^{-1} on a Bruker-Optics Alpha-T spectrophotometer in KBr disks. Absorbance spectra were obtained with a Hitachi U-3501 spectrophotometer.

The Mössbauer spectra have been measured between 5.25 and 310 K in a Janis Supervaritemp cryostat with a constant-acceleration spectrometer, which utilized a rhodium matrix cobalt-57 source and was calibrated at 295 K with α-iron powder. The Mössbauer spectral absorbers contained 90 mg/cm^2 of microcrystalline **1** mixed with boron nitride.

Variable-temperature magnetic studies have been carried out with a Quantum Design MPMS SQUID magnetometer. A crystalline powder sample of **1** was placed in a gel capsule and the crystallites were anchored by adding eicosane into the capsule, taking care that the crystallites were well surrounded by the eicosane. The observed long moments were corrected for the known slightly temperature dependent contribution of the gel capsule and eicosane. A diamagnetic correction of −0.000 350 emu/mol of complex, obtained from tables of Pascal's constants, was applied to the observed molar magnetic susceptibilities. The long moment of **1** has been measured after zero-field cooling to 2 K and subsequent warming to 300 K in a sequence of fields of 0.1, 0.5, 1, and 0.02 T with zero-field cooling between each applied field measurement. The magnetization of **1** was subsequently measured at 2 K between 0 and 7 T. Magnetization data were also collected between 1.8 and 10 K under a range of dc fields. In general, when fitting the magnetization data, several different values of E could be obtained and had little to no effect on the goodness-of-fit, depending only on the input values for E. In addition, often multiple fits of similar quality provided slightly different values of g and D. As such, the average values of these parameters are reported, with the standard deviations given in parentheses. Finally, some fits gave positive values for D, but ac susceptibility measurements demonstrate that D must be negative. Thus, only the magnitude of D was considered when calculating the average value and standard deviation. Ac magnetic susceptibility data have been collected both in a zero dc field between 1.74 and 2.1 K and in a 0.04 mT ac field at frequencies between 1 and 1488 Hz.

Cyclic voltametric measurements have been carried out with a Bioanalytical System EPSILON electrochemical analyzer at a scan rate of 100 mV/s. The concentration of the supporting electrolyte, tetraethylammonium perchlorate, was 0.1 M, whereas that of the complex was 1 mM. The measurements were carried out in acetonitrile solution with a platinum working electrode, a platinum auxiliary electrode, and an aqueous Ag/AgCl reference electrode. The reference electrode was separated from the bulk solution using a tetra- and ethylammonium perchlorate in acetonitrile salt bridge.

D. Crystal structure determination

The single-crystal structure of **1** has been determined at 120 and 293 K with a Nonius Kappa diffractometer equipped with a CCD-area detector, by collecting 641 frames with φ- and ω-increments of one degree with a counting time of 25 s per frame. The crystal-to-detector distance was 30 mm. The reflection data were processed with the Nonius DENZO-SMN[30] programs and corrected for Lorentz polarization, background, and absorption effects. The crystal structure was determined by direct methods, and subsequent Fourier and difference Fourier syntheses, followed by full-matrix least-squares refinements on F^2 using SHELXL-97.[31] All the hydrogen atoms were inserted at calculated positions with isotropic thermal parameters and further refined freely. An anisotropic refinement of the nonhydrogen atoms and an unrestrained isotropic refinement of the hydrogen atoms converged to an R-value of 0.0626 for $I > 2\sigma(I)$ at 120 K. The details of the refinement are given in Tables I and S1 and full details for both the 120 and 293 K structures are given in the crystallographic information files, see the supplementary material.[32]

TABLE I. Crystallographic results for **1**.

Parameter	**1,** 120 K	**1,** 293 K
Empirical formula	$Fe_2C_{32}H_{40}ClN_4O_{10}$	$Fe_2C_{32}H_{40}ClN_4O_{10}$
Formula weight, g/mol	787.83	787.83
Crystal color	Black	Black
Crystal system	Monoclinic	Monoclinic
Space group	$P2/c$	$P2/c$
a, Å	16.094(1)	16.241(3)
b, Å	10.900(1)	11.086(2)
c, Å	21.692(2)	21.795(4)
α, °	90	90
β, °	106.289(3)	105.82(1)
γ, °	90	90
V, Å^3	3652.6(5)	3775.5(9)
Z	4	4
T, K	120(2)	293(3)
2θ	8.30 – 69.90	8.18 – 61.04
μ, mm^{-1}	0.926	0.896
ρ_{calcd}, g cm^{-3}	1.433	1.386
$F(000)$	1636	1636
Scan mode	φ- and ω-scans	φ- and ω-scans
Number of frames	641	641
Scan time per frame, s	25	25
Rotation width, °	1	1
Crystal-detector-dist., mm	30	30
Absorption correction	Multiscan	Multiscan
T_{min}	0.9465	0.9651
T_{max}	0.9552	0.9736
Index ranges	$-25 \leq h \leq 25$	$-23 \leq h \leq 23$
	$-17 \leq k \leq 17$	$-15 \leq k \leq 15$
	$-34 \leq l \leq 34$	$-31 \leq l \leq 31$
Reflections collected	31157	21936
Independent reflections (R_{int})	15974 (0.0263)	11463 (0.0872)
Goodness of fit	1.124	1.005
R_1[a]/wR_2[b] ($I > 2\sigma(I)$)	0.0626/0.1861	0.0804/0.2229
R_1[a]/wR_2[b] (for all data)	0.0858/0.1980	0.2204/0.2850

[a]$R_1 = [\sum||F_o| - |F_c||/\sum|F_o|]$.
[b]$wR_2 = [\sum w(F_o{}^2 - F_c{}^2)^2/\sum wF_o{}^4]^{1/2}$.

III. RESULTS AND DISCUSSION

A. Synthesis and characterization

The $[Fe_2L(\mu\text{-OAc})_2]ClO_4$, **1**, complex, where L is the dianion of the tetraiminodiphenolate macrocyclic ligand, H_2L, see Chart 1, is readily obtained in high yield from the reaction of $[Fe(H_2L)(H_2O)Cl](ClO_4)_2 \cdot 2H_2O$, $Fe(ClO_4)_2 \cdot 6H_2O$, and NaOAc in a 1:1:4 ratio under a dinitrogen atmosphere. The $\nu_{C=N}$ infrared band in **1** appears at 1625 cm^{-1} and the presence of perchlorate is indicated by the appearance of a very strong absorption at 1088 cm^{-1} and a weak absorption at 622 cm^{-1}. The two medium intensity absorptions observed at 1557 and 1404 cm^{-1} can be assigned to the asymmetric and symmetric stretching modes of the bridging acetate ligands, respectively. The positions of the carboxylate stretching modes indicate that the two acetate ligands bridge the iron ions in the same fashion.[19,33]

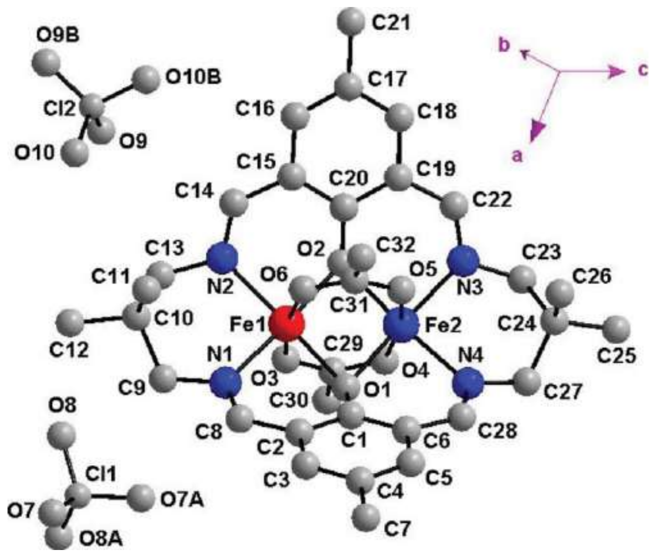

FIG. 1. Crystal structure of $[Fe_2L(\mu\text{-OAc})_2]ClO_4$, **1**. The hydrogen atoms have been omitted for clarity and the two perchlorate anions are each half-occupied.

B. Structure of $[Fe_2L(\mu\text{-OAc})_2]ClO_4$, 1

The crystal structure of **1** is shown in Fig. 1 and selected bond lengths and angles are given in Table II. The structure of **1** reveals a heterobridged bis(μ-phenoxo)bis(μ-acetate) diiron complex containing the tetraiminodiphenolate macrocyclic dianionic ligand, L^{2-}, and two crystallographically inequivalent iron sites, both of which are hexacoordinated by two azomethine nitrogens, two bridging phenolate oxygens, and the two oxygens from the two bridging acetate ligands. The structure of **1** could not be refined in a higher symmetry group with only one crystallographically unique iron site. Even though, the environment of the two iron sites are similar, as described below, the two sites are crystallographically inequivalent with no symmetry element present that can connect the two sites even if refined in other space groups.

The basal plane of the slightly distorted octahedral coordination environment about the two distinct iron sites in **1** consists of N_2O_2 derived from the L^{2-} ligand; two oxygens from the bridging acetate ligands occupy the axial positions. At 120 K, the average deviations from the least-squares N_2O_2 basal planes about Fe(1) and Fe(2) are ±0.002 and ±0.012 Å, respectively, and Fe(1) and Fe(2) are displaced above the N_2O_2 basal plane by 0.063(3) Å toward O(6) and by 0.068(3) Å toward O(5), respectively.

A comparison of the 120 K bond distances and angles about Fe(1) and Fe(2) in **1**, see Table II, reveals that the two coordination environments are very similar. More specifically, the Fe−O(phenoxo) bond distances of 2.016(2) and 2.023(2) Å for Fe(1) and 2.014(2) and 2.020(2) Å for Fe(2) are the *same* within their statistical errors. In contrast, the Fe−N bond distances of 2.122(2) and 2.127(2) Å for Fe(1) and 2.109(2) and 2.121(2) Å for Fe(2) and the Fe−O(acetate) bond distances of 2.032(2) and 2.079(2) Å for Fe(1) and 2.041(2) and 2.071(2) Å for Fe(2) are somewhat *different* within their statistical errors. These small differences result in a just barely significant difference in the summed bond lengths at the two iron sites, see Tables S2 and S3. Further, the *cisoid* angle range of 83.32° to 99.34° for Fe(1) and 83.87° to 99.51° for Fe(2) and the *transoid* angle range of 170.08° to 175.65° for Fe(1) and 169.30° to 175.31° for Fe(2) are virtually identical. Rather similar conclusions may be reached for the

TABLE II. Bond distances, in Å, and bond angles, in deg, for **1** and a comparison with **2**.

	1, 120 K			**2**, 293 K[a]
Fe(1)–O(3)	2.032(2)	Fe(2)–O(4)	2.041(2)	2.047(2)
Fe(1)–O(6)	2.079(2)	Fe(2)–O(5)	2.071(2)	2.061(2)
Fe(1)–O(1)	2.023(2)	Fe(2)–O(1)	2.020(2)	2.028(2)
Fe(1)–O(2)	2.016(2)	Fe(2)–O(2)	2.014(2)	2.036(2)
Fe(1)–N(1)	2.127(2)	Fe(2)–N(4)	2.121(2)	2.142(2)
Fe(1)–N(2)	2.122(2)	Fe(2)–N(3)	2.109(2)	2.139(2)
Fe(1)–Fe(2)	2.6093(6)			2.7414(8)
O(3)–Fe(1)–O(6)	170.08(7)	O(4)–Fe(2)–O(5)	169.30(7)	165.93(7)
O(1)–Fe(1)–N(2)	175.65(7)	O(1)–Fe(2)–N(3)	175.31(8)	175.45(8)
O(2)–Fe(1)–N(1)	175.23(7)	O(2)–Fe(2)–N(4)	174.28(8)	171.27(8)
O(3)–Fe(1)–O(1)	85.74(7)	O(4)–Fe(2)–O(1)	84.88(7)	87.07(8)
O(3)–Fe(1)–O(2)	86.42(7)	O(4)–Fe(2)–O(2)	86.35(7)	82.85(7)
O(3)–Fe(1)–N(1)	97.50(7)	O(4)–Fe(2)–N(4)	98.52(9)	88.46(8)
O(3)–Fe(1)–N(2)	97.91(8)	O(4)–Fe(2)–N(3)	98.65(8)	97.48(8)
O(6)–Fe(1)–O(1)	87.20(7)	O(5)–Fe(2)–O(1)	87.37(7)	84.59(8)
O(6)–Fe(1)–O(2)	87.87(7)	O(5)–Fe(2)–O(2)	87.67(7)	86.63(8)
O(6)–Fe(1)–N(1)	88.62(7)	O(5)–Fe(2)–N(4)	87.97(8)	102.06(8)
O(6)–Fe(1)–N(2)	89.47(7)	O(5)–Fe(2)–N(3)	89.52(8)	90.93(8)
O(1)–Fe(1)–O(2)	99.34(7)	O(1)–Fe(2)–O(2)	99.51(7)	95.17(6)
O(1)–Fe(1)–N(1)	83.70(7)	O(1)–Fe(2)–N(4)	84.00(8)	85.04(7)
O(2)–Fe(1)–N(2)	83.32(7)	O(2)–Fe(2)–N(3)	83.87(7)	85.31(7)
N(1)–Fe(1)–N(2)	93.43(8)	N(3)–Fe(2)–N(4)	92.39(8)	95.17(8)
Fe(1)–O(1)–Fe(2)	80.39(6)	Fe(1)–O(2)–Fe(2)	80.71(6)	84.83(6)

[a]Data obtained from Dutta *et al.* (Ref. 19).

293 K structure. These structural parameters all indicate that the coordination environments about the two crystallographically distinct Fe(1) and Fe(2) sites are structurally very similar, but, as will be noted below, the differences, especially those observed for the bridging acetate ligands, are enough to lead to two *significantly different* sets of Mössbauer spectral hyperfine parameters for the Fe(1) and Fe(2) sites at 9.1 K and below.

The Fe $\cdots$ Fe nonbonding distances of 2.609(1) and 2.601(1) Å observed at 120 and 293 K, respectively, for **1** are shorter than the distance of 2.7414(8) Å observed[19] at 293 K in $[Fe_2L^1(\mu\text{-OAc})_2]ClO_4$, **2**, where L^1 is the dianion of the macrocyclic ligand shown in Chart 1. These Fe $\cdots$ Fe nonbonding distances are remarkably short as compared with those observed[22] in the related mixed-valence complexes containing the dianionic L^1 ligand. For example, the Co $\cdots$ Co distances[22] in $[Co^{III}Co^{II}L^1Br_2(MeOH)_2]Br_3$ are 3.12(2) and 3.16(1) Å at 283 and 303 K[22] and the Mn $\cdots$ Mn distance[22] in $[Mn^{III}Mn^{II}L^1Cl_2Br]$ is 3.168(3) Å at 295(1) K. It should also be noted that at 295 K a very short Fe $\cdots$ Fe distance of 2.50(1) or 2.509(6) Å, obtained by EXAFS and single crystal x-ray structural measurements, respectively, has been reported[19] for the fully delocalized $[Fe_2L^2(\mu\text{-OH})_3](ClO_4)_2 \cdot 2MeOH \cdot 2H_2O$, **3**, complex, where L^2 is N,N',N''-trimethyl-1,4,7-triazacyclononane. Hence, the rather short Fe $\cdots$ Fe nonbonding distance of 2.609(1) Å found in **1** at 120 K might well be expected to favor full electron delocalization.

Because the two iron cations in **2** occupy crystallographically equivalent sites, it is highly probable that at least one of the eleven 3*d* electrons is equally shared by the two iron cations and, thus, **2** can be described as a class III $Fe^{2.5+}Fe^{2.5+}$ binuclear complex. In contrast, in **1** the two iron cations occupy crystallographically inequivalent sites, and hence no immediate conclusion can be drawn solely from the structural results about the distribution of the eleven 3*d* electrons between the two crystallographically inequivalent iron cationic sites in **1**. Thus, in order to better understand the electronic properties, we have undertaken a detailed magnetic and Mössbauer spectral study of **1**.

The changes in the crystal structure of **1** between 120 and 293 K are discussed in detail in the supplementary material,[32] where a comparison of the structure of **1** and **2** can also be found.

C. Magnetic properties

The temperature dependences of χ_MT and $1/\chi_M$ of **1** measured in an applied field of 0.02 and 1 T are shown in Fig. 2. The inverse molar magnetic susceptibility, $1/\chi_M$, measured at 0.02, 0.1, 0.5, and 1 T exhibits perfect linear Curie–Weiss law behavior between 2 and 300 K and yields a Curie constant, *C*, of 11.19(6) emu K/mol, a Weiss temperature, θ, of 0.14(25) K, and a corresponding effective magnetic moment, μ_{eff}, of 9.45(3) μ_B per mole; the quoted errors have been obtained from the standard deviation of the parameters obtained at the four applied fields. This perfect linear Curie–Weiss behavior with a very small Weiss temperature indicates

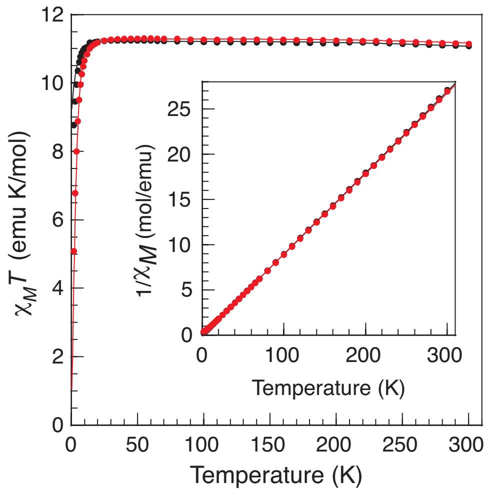

FIG. 2. The temperature dependence of $\chi_M T$ of **1** measured in an applied dc field of 0.02 T, black, and 1 T, red. The black and red solid lines are a fit below 20 K with $g = 1.91$ and 1.92 and $D_{9/2} = -1.04$ and -0.76 cm^{-1}, and between 20 and 300 K with $B = 950$ cm^{-1}, $J = -32(2)$ cm^{-1}, and $g = 1.90(1)$. Inset: The corresponding temperature dependence of $1/\chi_M$ of **1** with a linear Curie-Weiss law fit between 2 and 300 K. The results obtained at 0.02 and 1 T are superimposed upon each other.

an almost perfect paramagnetic behavior and the absence of any long-range magnetic order between 2 and 300 K. The observed μ_{eff} is indicative of a spin 9/2 ground state in full agreement with a fully delocalized electron in a mixed-valence iron(II)-iron(III) binuclear complex.[34] However, the μ_{eff} of 9.45(3) μ_B per mole is smaller than the expected 9.95 μ_B for $S = 9/2$ and corresponds to a g value of 1.90(1). At the four applied fields, the product χ_MT in **1** exhibits only a small decrease, from ca. 11.2 to 11.1 emuK/mol, between 20 and 300 K. The decrease in χ_MT below 20 K results from the combined zero-field splitting and Zeeman splitting of the magnetic states and fits of χ_MT with only uniaxial zero-field splitting at applied fields of 0.02, 0.1, 0.5, and 1 T yields $g = 1.91(1)$ and $D_{9/2}$ between -0.76 and -1.04 cm^{-1} or $D_{9/2} = -0.90(14)$ cm^{-1}. In contrast, with the previous magnetic measurements[35,36] on **2**, no cusp was observed at $\sim$25 K. In order to investigate this apparent difference in the magnetic behavior of **1** and **2**, magnetic susceptibility measurements have been carried out on a sample of **2** anchored in eicosane in an applied field of 0.1 and 1 T, see Fig. S1 in the supplementary material.[32] The cusp previously observed at 25 K in an applied field of 0.5 T was not observed. We conclude that this cusp probably originated either from a poor anchoring of the powder sample or from a paramagnetic impurity.[37]

The analysis of the magnetic susceptibility of **1** and **2** between 20 and 300 K is based on the Hamiltonian for an exchanged-coupled symmetric binuclear complex in the presence of valence delocalization,

$$\boldsymbol{H} = -2J({}^{A}\boldsymbol{S}_A \cdot {}^{A}\boldsymbol{S}_B \mathbf{O}_A + {}^{B}\boldsymbol{S}_A \cdot {}^{B}\boldsymbol{S}_B \mathbf{O}_B) + B\mathbf{T}_{AB}. \qquad (1)$$

In the first term, the isotropic Heisenberg exchange term, $\mathbf{O}_A$ and $\mathbf{O}_B$ are the occupation operators and ${}^{A}\boldsymbol{S}_A$ and ${}^{B}\boldsymbol{S}_A$ and ${}^{A}\boldsymbol{S}_B$ and ${}^{B}\boldsymbol{S}_B$ are the spin operators when the transferable electron is on the A or B site, respectively. The second, $B\mathbf{T}_{AB}$, term expresses the mixing of the states $|S_A, S_B, S_{AB}\rangle^A$

and $|S_A, S_B, S_{AB}\rangle^B$ where $\mathbf{T}_{AB}$ is the transfer operator, and B is the electron transfer parameter, see the supplementary material.[32] Within this model χ_MT is given by Eq. S3. Both the temperature dependence of the molar magnetic susceptibility, χ_M, and of the product, χ_MT, have been fitted. Because of the known high correlation[35,36] between J and B and the weak temperature dependence of χ_MT between 20 and 300 K, it is virtually impossible to simultaneously determine J and B. In all cases, positive J values lead to poor fits. Additional details concerning the analysis of the temperature dependence of the magnetic susceptibility and the fits are given in the supplementary material,[32] where Tables S4 and S5 summarize the most significant fits. The quoted errors are the statistical errors and the absence of an error indicates that the parameter was constrained to the value given.

It is clear that fits of χ_M and χ_MT lead to similar or insignificantly different results, see Tables S4 and S5. The excellent fits indicate that the iron binuclear complexes in **1** and **2** have a spin 9/2 ground state as expected in the presence of an isotropic antiferromagnetic exchange with a small negative J value and a large electron transfer parameter, B. In other words, the isotropic antiferromagnetic exchange is dominated by the ferromagnetic double exchange. For complex **1** at a fixed $B = 950$ cm^{-1}, $J = -32(2)$ cm^{-1} and $g = 1.91(1)$. For **2** at a fixed $B = 940$ cm^{-1}, $J = -65(5)$ cm^{-1} and $g = 2.02(1)$.

In the fits with fixed J values, the double-exchange parameter, B, was always found to be smaller than 950 cm^{-1}, the value expected from the energy of the intervalence charge transfer band, a reduction that is systematically[14] observed in the analysis of double-exchange mixed valence binuclear complexes and has been attributed to a neglect of vibronic coupling[11,13,38] between the electronic and nuclear motions in the complex. In the case of **1**, there is such a small variation in χ_MT between 20 and 300 K that it is not possible to include this vibronic coupling and fit an additional parameter.

The 2 K magnetization of **1** is shown in the inset to Fig. 3. The absence of saturation and the small magnetization of only 7.38 Nβ at 7 T are indicative of the presence of zero-field splitting. The effective spin Hamiltonian describing the magnetic anisotropy of the binuclear complex **1** is given by

$$H_{\text{eff}} = D_{9/2}[S_{t,z}^2 - S_t(S_t+1)/3 + E_{9/2}(S_{t,x}^2 - S_{t,y}^2)/D_{9/2}], \tag{2}$$

where $S_t = 9/2$. To further investigate the magnetic anisotropy in **1**, variable-temperature magnetization data have been collected in a range of dc fields. The resulting plot of the reduced magnetization is shown in the main portion of Fig. 3, which reveals the presence of a series of nonsuperimposable isofield curves that are indicative of an anisotropic ground state. To quantify the extent of the zero-field splitting, the magnetization was fit by using ANISOFIT 2.0 (Ref. 39) to obtain the axial and transverse zero-field splitting parameters, $D_{9/2} = -0.89(6)$ cm^{-1} and $|E_{9/2}| = 0.1(1)$ cm^{-1}, respectively, and $g = 1.954(8)$, a g value that agrees rather well with the value of 1.91(1) obtained from the fits of the temperature dependence of χ_M and χ_MT. The anisotropy in **1** likely results from the presence of an orbital contribution to the moment in the complex composed of two iron ions and one delocalized electron. By using the relationship,[12,40] $D_{\text{Fe}} = 2.22D_{9/2}$, between the binuclear complex zero-field splitting parameter and the local zero-field splitting parameters at each iron, one obtains $D_{\text{Fe}} = -2.0(1)$ cm^{-1}, the average local zero-field splitting parameter. In conclusion, the ground state of **1** is characterized by a total spin, $S_t = 9/2$, and in the 10-fold multiplet, whose degeneracy is removed by the crystal field splitting, because $D_{9/2}$ is negative, the $m_t = \pm 9/2$ substate is the ground state. Unfortunately, in the absence of oriented single crystal studies, it is not possible to determine the orientation of the magnetic anisotropy axis.

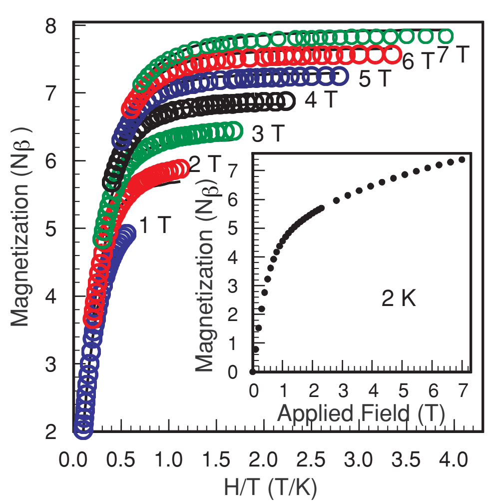

FIG. 3. The low-temperature magnetization of **1** obtained at the indicated applied dc fields. The black lines represent fits of the magnetization. Inset: the dc magnetization of **1** measured at 2 K.

The observed 0.89(6) cm^{-1} magnitude of $D_{9/2}$ found in **1** is smaller than those observed previously in related binuclear delocalized mixed-valence iron compounds. The previous analyses of the magnetic susceptibility of **2** reported[35,36] $|D_{9/2}| = 1.6$ and 3 cm^{-1}, where the determination of these values may possibly have been affected by the presence of the artificial cusp observed at 25 K. The present analysis of χ_MT of **2** between 2 and 20 K yields $D_{9/2} = -0.70(5)$ cm^{-1} and $g = 2.02$, see Fig. S1. $|D_{9/2}|$ values of 1.8(2) and 1.7 cm^{-1} were observed[12,19] in **3** and $[Fe_2(\mu\text{-}O_2CAr^{Tol})_4(4\text{-}^tBuC_5H_4N)_2]X$, where O_2CAr^{Tol} is 2,6-di(p-tolyl)benzoate and X is PF_6^- or Otf^-, respectively.

Because of the $S = {}^9/_2$ ground state and the negative axial zero-field splitting of $D = -0.89(6)$ cm^{-1} obtained for **1**, the ac magnetic susceptibility of **1** has been measured in order to probe for slow magnetic relaxation. Indeed, variable-frequency ac susceptibility measurements, see Figs. 4 and S2, reveal a strong temperature dependence of both the in-phase, χ_M', and out-of-phase, χ_M'', susceptibilities. Cole–Cole plots of χ_M' vs χ_M'' were then fitted with the generalized Debye model for a solid to obtain the relaxation times at the various temperatures, see Fig. S3.[41] For a single-molecule magnet, the relaxation time, τ, should follow an Arrhenius law, where τ increases exponentially with decreasing temperature. Thus, a plot of lnτ vs $1/T$ should be linear, with a slope equal to the energy of the relaxation barrier, U_{eff}. Indeed, such a plot obtained for **1** reveals a linear relationship, see the inset to Fig. 4, and a linear least-squares Arrhenius law fit yields

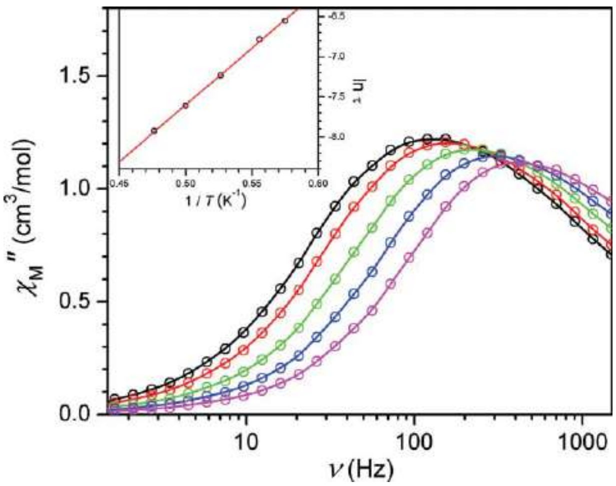

FIG. 4. The variable-frequency out-of-phase ac magnetic susceptibility of **1**, measured at 1.74 K, black, 1.8 K, red, 1.9 K, green, 2.0 K, blue, and 2.1 K, magenta. Inset: An Arrhenius plot of the relaxation time. The solid red line corresponds to a linear fit with $U_{\text{eff}} = 9.8$ cm^{-1}.

$U_{\text{eff}} = 9.8$ cm^{-1} and $\tau_0 = 4.2 \times 10^{-7}$ s. To the best of our knowledge, **1** represents the first example of a single-molecule magnet based on the magnetic behavior of a mixed valence binuclear complex with double-exchange mechanism. The corresponding ac-susceptibility studies for **2** are shown in Fig. S4 and indicate that, because no peak is observed in χ_M'' with increasing frequency between 1.74 and 2.1 K, U_{eff} is too small to be determined at these temperatures. The realization of single-molecule magnetic behavior at more practical temperatures requires a well-isolated spin ground state. Indeed, with exchange constants of $B = 950$ cm^{-1} and $J = -32$ cm^{-1}, the spin ground state of **1** lies $\sim$700 cm^{-1} below the first excited $S = 7/2$ state and could thus support a relaxation barrier well above room temperature if the appropriate bridging ligands can be found to replace the acetate bridging ligands.

All the error bars quoted up to this point are the statistical error bars given by the different fits of the magnetic data. In the presence of a small D value and, hence, of a small orbital contribution to the magnetic moment, a g value slightly larger than 2 is expected for **1**, as is observed for **2**. It is likely that the smaller than 2 values of g obtained from the different fits result from inaccuracies in the mass of the sample, of the gel capsule, and of the eicosane, and in the diamagnetic correction estimated from the Pascal's constants. Although the C, H, and N analysis of **1** is very good and does not point to the presence of an impurity, the presence of a small amount of a diamagnetic impurity cannot be excluded. Hence, the best estimates of the zero-field uniaxial D parameter and g, including experimental inaccuracies, are $-0.9(1)$ cm^{-1} and 1.95(5).

D. Mössbauer spectral study

A detailed iron-57 Mössbauer spectral study of **1** has been undertaken because this spectroscopy probes the microscopic electronic and magnetic properties of each iron in the binuclear complex in the absence of an applied magnetic field and within a timescale of $\sim 5 \times 10^{-8}$ s, the Larmor precession

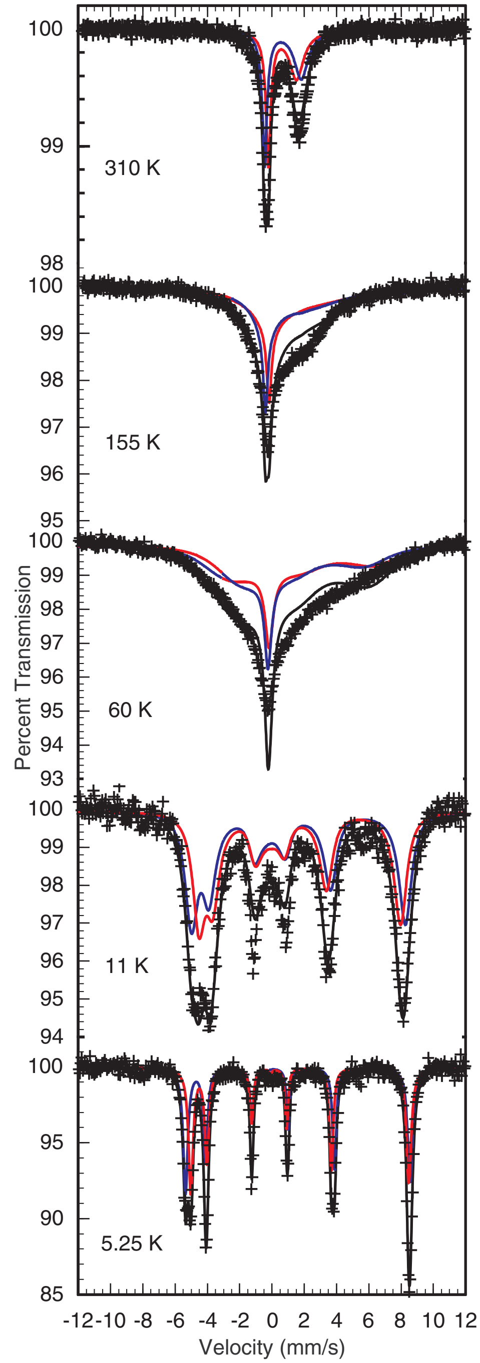

FIG. 5. The iron-57 Mössbauer spectra of **1** obtained at the indicated temperatures. The black solid line is the result of the fit described in the text and the supplementary material (Ref. 32) The red and blue solid lines are the two spectral components in the fit and are tentatively assigned to Fe(1) and Fe(2), respectively.

time of the iron-57 nuclear magnetic moment in the presence of a hyperfine field. The iron-57 Mössbauer spectra of **1** have been measured between 5.25 and 310 K and selected spectra are shown in Figs. 5 and S6.

It is clear from Fig. 5 that at 5.25 K, the Mössbauer spectrum of **1** exhibits two narrow line magnetic sextets, indicating that both iron sites are experiencing a static hyperfine field on the iron-57 Larmor precession time of $\sim 5 \times 10^{-8}$ s. The 5.25 K spectral parameters are given in Table III. As the tem-

TABLE III. Mössbauer spectral parameters for **1**.[a]

T, K	$Area$[b] (%ε)(mm/s)	Fe site	δ,[c] mm/s	$e^2Qq/2$,[d] mm/s	H, T	ν, MHz
310	3.163	2	0.650(5)	1.8	27	514(10)
		1	0.680(5)	2.0	27	514(10)
295	3.611	2	0.660(5)	1.8	27	444(10)
		1	0.690(5)	2.0	27	444(10)
225	7.115	2	0.670(5)	1.8	27	201(5)
		1	0.700(5)	2.0	27	201(5)
155	12.927	2	0.700(5)	1.8	28.5	94.7(5)
		1	0.720(5)	2.0	25.5	94.7(5)
85	23.272	2	0.725(5)	1.8	28	45.3(1)
		1	0.745(5)	2.0	26	45.3(1)
60	28.976	2	0.73(5)	1.8	33	43.5(1)
		1	0.76(5)	2.0	29	43.5(1)
30	33.792	2	0.735(5)	1.8	36	33.7(1)
		1	0.765(5)	2.0	32	33.7(1)
20	35.680	2	0.738	1.8	39.8(1)	22.6(1)
		1	0.768	2.0	32.1(1)	22.6(1)
15	36.606	2	0.74	1.8	40.7(1)	16.7(1)
		1	0.77	2.0	36.8(1)	16.7(1)
11	36.623	2	0.74	1.8	41.1(1)	8.59(5)
		1	0.77	2.0	38.6(1)	8.59(5)
9.1	34.683	2	0.74	1.8	41.3(1)	4.41(5)
		1	0.77	2.0	39.1(1)	4.41(5)
7	35.061	2	0.74	1.8	41.3(1)	1.86(5)
		1	0.76	2.0	39.6(1)	1.86(5)
5.25	36.907	2	0.743(1)	1.816(3)	43.06(1)	0
		1	0.773(1)	2.003(4)	41.68(1)	0

[a]The linewidth was constrained to 0.22 mm/s, the linewidth observed at 5.25 K. Estimated errors are given in parentheses. The absence of an error indicates that the parameter was constrained to the value given.
[b]The statistical error is $\pm$0.002 (%ε)(mm/s).
[c]The isomer shifts are referred to 295 K α-iron powder.
[d]The asymmetry parameter, η, and the angle, θ, for both sites were found equal to 0.12(2) and 8(2)° at 5.25 K and were constrained to these values at all temperatures. The error bars on η and θ were determined from fits that gave a $\chi^2 = 1.2$ times 1.36, i.e., the best obtained χ^2 shown in Fig. 5.

perature increases the narrow line sextets begin to broaden as is shown in Fig. S6 by the spectra obtained at 7 and 11 K. At 15 and 20 K the Mössbauer spectra of **1** exhibit very broad magnetic sextets. Between 30 and 155 K, the spectra are very broad and exhibit a line shape profile characteristic of a relaxation of the hyperfine field. At 225, 295, and 310 K, the spectra are broad asymmetric doublets that, again, indicate relaxation of the hyperfine field.

The ground state spin 9/2 of **1** observed in the dc magnetic measurements could result either from the presence of one high-spin iron(III) and one high-spin iron(II) ions or from the magnetic double exchange mechanism described in Sec. III C. The isomer shifts of 0.743(1) and 0.773(1) mm/s observed for **1** at 5.25 K are both too high to be assigned to high-spin iron(III) and too low to be assigned to high-spin iron(II) in a pseudooctahedrally coordinated complex, and hence a far more acceptable assignment is to two crystallographically distinct iron ions that experience electron delocalization such that their average valence is 2.5, i.e., a $[Fe_2]^V$ binuclear unit. For comparison, it should be noted that a single isomer shift of 0.841(2) mm/s was observed[19] at 1.8 K for the $[Fe_2L^1(\mu\text{-OAc})_2]ClO_4$, **2**, complex and assigned to a fully valence-delocalized $[Fe_2]^V$ binuclear complex. However, in contrast to the single sextet observed[19] in **2**, two distinct isomer shifts and hyperfine fields are clearly observed and are required to obtain a valid fit of the 5.25 to 11 K spectra of **1**, a requirement that is in full agreement with the two inequivalent iron sites found in its crystallographic structure. Specifically, the different intensities and line widths of the Mössbauer spectral absorptions at −5.5 and +8.5 mm/s can only be explained by the use of two sextets. Because the differences in the crystallographic environments of the two iron sites are rather small, fits of the 5.25 K spectrum were attempted with several alternative fitting models including models with a reduced number of adjustable parameters or iron sites. All these alternative fits lead to χ^2 values that are at least twice as large as that of 1.36 shown in Fig. 5; the alternative models are briefly described in the supplementary material.[32]

At this point, it should be noted that these sextets do *not* correspond to long-range magnetic order, an order that is absent as is indicated by the perfectly linear Curie-law behavior observed between 2 and 300 K for the magnetic susceptibility of **1**, see Fig. 2. In contrast, the 5.25 K static hyperfine field results from slow relaxation of the hyperfine field in the $|S_t, m_t> = |9/2, \pm 9/2>$ ground state of **1**. Note that there is no difference in the Mössbauer spectrum for the $m_t = +9/2$ and $m_t = -9/2$ states. The observation[19] of the 4.2 K hyperfine field in **2** results not from the "easy orientation of the internal magnetic field relative to the principal axes of the electric field gradient" as stated in Ref. 19 but from slow relaxation of the hyperfine field in the same $|S_t, m_t> = |9/2, \pm 9/2>$ ground state, a ground state that is *not* compatible with the earlier reported[36] positive value of $D_{9/2}$. In contrast, in the case of the binuclear complex studied[12] by Ding *et al.*, a complex whose ground state is $|S_t, m_t> = |9/2, \pm 1/2>$, with a positive $D_{9/2}$, no hyperfine field is observed at 4.2 K in the absence of an applied magnetic field. A relaxation path within the electronic multiplet ground state is discussed below.

The hyperfine fields of 43.06(1) and 41.68(1) T observed at 5.25 K in **1** could be characteristic of two high-spin iron(II) ions but are also completely consistent with the presence of two intermediate valence $Fe^{2.5+}$ cations. The assignment to intermediate valence $Fe^{2.5+}$ cations is preferred both in view of the similar hyperfine field of 43.38(1) T reported[19] at 1.8 K for **2** and the magnetic properties discussed in Sec. III C. Similar hyperfine field values of ∼47 T have also been observed[12] in related spin 9/2 binuclear complexes and an estimate of their local-spin expectation values has been reported. If the component of the local magnetic hyperfine tensor, A_z, is in the usual range[12,19,42,43] of 16 to 21 T, the observed hyperfine fields indicate that the local-spin expectation values, $\langle S_z \rangle$, are in the range of 2.1 to 2.7. Hence, using the relationship $\langle S_z \rangle = \frac{1}{2}\langle S_{tz} \rangle$, the corresponding total spin expectation value, $\langle S_{tz} \rangle$, for **1** is in the range of 4.2 to 5.4, a range that agrees with its $S_t = 9/2$ ground state.

Finally, the 5.25 K spectrum clearly shows that the quadrupole interactions at the two iron sites in **1** have the same sign and the best fit yields values for $e^2Qq/2$ of 1.816(3) and 2.003(3) mm/s with the same asymmetry parameter, η, of 0.12(2). These $e^2Qq/2$ values are similar to the value of 2.088(4) mm/s reported[19] at 1.8 K for **2** and the values reported for several delocalized di-iron(II/III) compounds.[12,19,38] The angle, θ, between the principal axis of the electric field gradient tensor and the hyperfine field was refined to 8.17(5)° for both sites. The set of fitted hyperfine parameters may not be unique but all the refined values are reasonable and consistent with the observed structure of **1**. Hence, the 5.25 K spectrum presented herein differs essentially from the 1.8 K spectrum obtained for **2** by the presence of two sextets instead of one.[19] As noted above, the magnetically split 5.25 K spectrum does *not* result from long-range magnetic ordering, an ordering that is not observed in the magnetic susceptibility measurements but from slow relaxation of the hyperfine field in the $|S_t, m_t\rangle = |9/2, \pm 9/2\rangle$ ground state. The very narrow line width of 0.220(2) mm/s observed at 5.25 K confirms that there is neither broadening through any relaxation process nor experimental broadening from the spectrometer, which typically yields line widths of 0.23 mm/s for α-iron powder. The assignment of the two sextets to the two iron sites has been tentatively made as described in the supplementary material.[32]

At 7, 9.1, and 11 K, it is also possible to fit the Mössbauer spectra with two sextets with somewhat broadened line widths as a result of the onset of relaxation of the hyperfine field on the Mössbauer time scale of $\sim 5 \times 10^{-8}$ s. The dramatic increase in the spectral line width with increasing temperature above 11 K is equally characteristic of some relaxation of the hyperfine field taking place within the 10-fold multiplet $S = 9/2$ electronic ground state of **1**, the only state to be considered because the $S = 7/2$ multiplet is situated at ~ 700 cm^{-1} above the ground state.

Electronic relaxation between levels is usually caused by time-fluctuating interactions between the electronic spin and its environment, i.e., with other spins through dipole–dipole coupling, lattice vibrations or phonons coupled to the orbital moment and then to the spin through the spin-orbit interaction. The most likely relaxation mechanism in **1,** studied herein, and in **2,** studied in Ref. 19, should be the intermolecular dipole–dipole interaction and the one phonon, direct, and two-phonon, Raman, processes.[44] In the development of the relaxation interaction in powers of the spin operators, the dominant terms are those in S_+ (or S_-) and $S_+{}^2$ (or $S_-{}^2$), linking states with $\Delta m = \pm 1$ and $\Delta m = \pm 2$, respectively, terms that have an oscillator strength much higher than, for instance, terms in $S_+{}^9$ (or $S_-{}^9$). Hence, the relaxation within the two states of the $|9/2, \pm 1/2\rangle$ doublet is likely to be much faster than within the two states of the $|9/2, \pm 9/2\rangle$ doublet. One can add that "elastic" relaxation between degenerate or quasi-degenerate levels, requiring no energy, is more efficient than "inelastic" relaxation between nondegenerate levels, where the "bath" has to provide the energy difference.

The occurrence and origin of the hyperfine field in paramagnetic binuclear compounds depends on the relative values of the relaxation time or times between the electronic levels, τ_R, and the Mössbauer Larmor precession time, τ_L, $\sim 5 \times 10^{-8}$ s. In the hypothesis where relaxation is "fast," i.e., $\tau_R << \tau_L$, between all the electronic levels of the binuclear compound, then the hyperfine field is given by

$$H_{\mathrm{hf}}(\mathrm{T}) = A_{\mathrm{hf}}\langle S_z \rangle_{\mathrm{T}} = \tfrac{1}{2} A_{\mathrm{hf}}\langle S_{tz} \rangle_{\mathrm{T}},$$

where $\langle S_z \rangle_{\mathrm{T}}$ and $\langle S_{tz} \rangle_{\mathrm{T}}$ are the local and total spin expectation values, respectively, at temperature, T, and A_{hf} is the main component of the magnetic hyperfine coupling tensor. In zero-applied magnetic field at low temperature, where only the electronic ground-state doublet of the binuclear complex is populated, one can then understand the very different 4.2 and 5.25 K spectra observed in Ref. 12 and herein, i.e., a quadrupole doublet-type and a fully developed sharp sextet-type magnetic spectrum, respectively. In Ref. 12, the ground state of the complex is the $|9/2, \pm 1/2\rangle$ doublet, which is degenerate in zero-applied field with $\Delta m = \pm 1$. All the conditions are fulfilled for fast relaxation, and hence, as observed, $H_{\mathrm{hf}} = 0$ because $\langle S_{tz} \rangle = 0$ within this ground-state doublet. In **1**, the ground-state doublet is $|9/2, \pm 9/2\rangle$ and "slow" relaxation within this ground-state doublet is expected because $\Delta m = \pm 9$, and the observed spectrum is due to a "slow relaxation" superposition of hyperfine fields corresponding to $|m_t = +9/2\rangle$ and $|m_t = -9/2\rangle$, which have opposite sign and give identical spectra. Because $\langle S_{tz} \rangle = \pm\, 9/2$ at low temperature, the saturated hyperfine field is given by $H_{\mathrm{hf}} = 9/4\ A_{\mathrm{hf}}$, according to the expression given above.

As temperature increases, in the case of **1** from 5 to 20 K, the excited $|\pm m_t>$ levels, with $|m_t| < 9/2$, gradually become populated and relaxation between different m_t states takes place. A possible relaxation path linking the $|+9/2\rangle$ and $|-9/2\rangle$ states, *via* relaxation mechanisms that allow only $\Delta m = \pm 1$ transitions, is pictured in Fig. 6. "Inelastic" relaxation

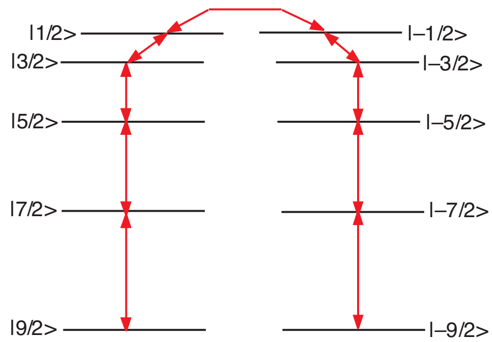

FIG. 6. Possible relaxation path shown as red arrows between the |9/2> and |−9/2> states via excited states within the 10-fold multiplet ground state of **1**.

may occur between the $|9/2\rangle$ and $|7/2\rangle$, $|7/2\rangle$ and $|5/2\rangle$, ... states, but no direct process is allowed between the $|+9/2\rangle$ and the $|-9/2\rangle$ states. Hence, as the temperature increases, rather "fast" relaxation is expected to occur independently between the lowest occupied states of the two "branches" with $m_t > 0$ and $m_t < 0$, a "fast" relaxation that yields a decrease in the observed hyperfine field. A special kind of average over the electronic levels, taking into account that there is "fast" relaxation between the levels in each "branch," but "slow" relaxation between the branches is now introduced and noted by $\langle\langle \ldots \rangle\rangle$. With this notation, $H_{\mathrm{hf}}(\mathrm{T}) = \frac{1}{2}A_{\mathrm{hf}}\langle\langle S_{tz}\rangle\rangle$, where the Boltzmann average is taken over only one of the two "branches". As the temperature increases above 20 K, then the red relaxation path sketched in Fig. 6 may become rather "fast," i.e., the relaxation time between $|9/2\rangle$ and $|-9/2\rangle$, between $|7/2\rangle$ and $|-7/2\rangle$ ... becomes of the order of τ_L and reversal of the hyperfine field occurs, yielding a broadening of the lines and finally a collapse of the magnetic hyperfine structure above 225 K. Of course, the actual situation may be rather more complex, because the crossover between the two regimes with and without hyperfine field reversal is not expected to be well defined and both processes may coexist in a given temperature range. So any fit of the spectra to a relaxation lineshape may be only approximate, but should provide estimates for the average hyperfine field and relaxation time.

The spectra obtained between 5.25 and 310 K have been fitted with a relaxation model[20,44–48] described in the supplementary material[32] and the results of these fits are given in Table III. In the resulting fits shown in Figs. 5 and S6, the effective hyperfine field is reasonably well defined at 60 K and below. Above 60 K, the hyperfine field and the relaxation rate cannot be simultaneously refined because they are highly correlated. Hence, above 60 K, the hyperfine field was kept constant as indicated in Table III. The temperature dependence of the average hyperfine field between 5.25 and 310 K is shown in Fig. 7, where the solid line is the weighted average hyperfine field calculated from the Boltzmann population of the m_t levels of the ground-state 10-fold multiplet for $|D| = 0.9$ cm^{-1} assuming $A_{\mathrm{hf}} = 21$ T. The magnitude of the D parameter is in excellent agreement with the range of values between 0.74 and 1.04 cm^{-1} obtained from the dc-magnetic measurements, see above. Above 60 K, the theoretical Boltzmann average predicts a virtually constant hyperfine field of 27 T, as has been used in the fits.

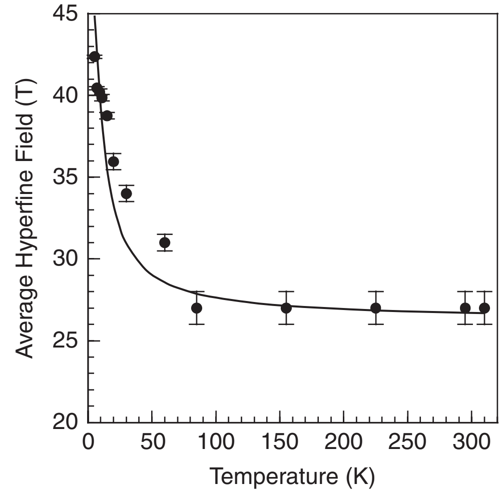

FIG. 7. The temperature dependence of the observed average hyperfine field, circles, and of the average hyperfine field calculated, solid line, for a Boltzmann population of the left or right branch of the $|m_t\rangle$ states shown in Fig. 6.

An Arrhenius plot of the average relaxation frequency in **1** between 5.25 and 310 K is shown in Fig. 8 and a linear fit between 5.25 and 85 K yields an activation energy of 17 cm^{-1}, an energy that is in agreement with the maximum theoretical relaxation barrier of $(S^2 - \frac{1}{4})|D| = 18$ cm^{-1}and the effective energy barrier of 9.8 cm^{-1} for magnetization reversal. Above 155 K, the relaxation process becomes very complex and, as expected, faster because all the m_t sublevels are occupied and provide a fast relaxation pathway.

The temperature dependence of the spectral absorption area of **1** is shown in Fig. S5 and has been fittted with the Debye model[49] for a solid, a model that is perhaps a rather oversimplified approximation for **1**. The resulting Debye temperature, Θ_D, is 125(1) K, a value that is similar to those observed[50] in other organometallic and coordination complexes. The value observed for **1** is relatively small and clearly indicates a soft crystalline lattice that will easily couple to the spin system to provide vibrational energy. The temperature

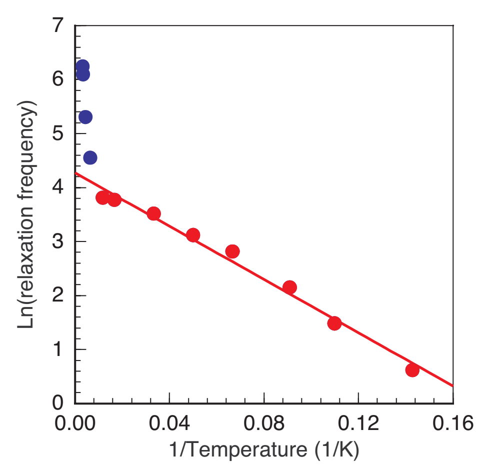

FIG. 8. An Arrhenius plot of the logarthim of the relaxation frequency in **1**, where the red and blue points correspond to low and high temperature regimes.

dependence of the average isomer shift in **1** is shown in Fig. S7 and discussed in the supplementary material.[32]

E. Electronic spectra

The electronic spectrum of **1** measured at 295 K exhibits both a sharp absorption at 353 nm with $\varepsilon = 13\,120$ $M^{-1}cm^{-1}$, an absorption that arises from an intraligand transition and a shoulder at 430 nm with $\varepsilon = 6530$ $M^{-1}cm^{-1}$, an absorption that arises from a phenolate to iron ion charge transfer transition.

An intense absorption is also observed in **1** at 1053 nm with $\varepsilon = 850$ $M^{-1}cm^{-1}$, an absorption that arises from an intervalent charge transfer transition, see Fig. 9. It should be noted that in **2** this absorption is observed[19] at 1060 nm and is more intense with $\varepsilon = 1250$ $M^{-1}cm^{-1}$. In contrast, both the 758 nm position and the $\varepsilon = 2400$ $M^{-1}cm^{-1}$ intensity of the intervalent charge transfer absorption in **3** is significantly different from those observed in **1** and **2**.[19] The intervalent charge transfer absorption observed in **1** has been measured in several noncoordinating solvents such as chloroform, dichloromethane, and nitrobenzene. In all these noncoordinating solvents the position, intensity, and line width at half-maximum, $\Delta\nu_{1/2}$, are virtually identical, see the upper absorption lines in Fig. 9. Both this noncoordinating, solvent independent, spectral behavior and the observation that the experimental $\Delta\nu_{1/2}$ of 3263 cm^{-1} is much smaller than the $\Delta\nu_{1/2} = 4684$ cm^{-1} line width predicted by Hush, confirm that complex **1** is a fully electron delocalized class III complex.[8,9,11,19] Although the intervalent charge transfer absorptions observed for **1** in acetonitrile and acetone are similar to those observed in noncoordinating solvents, in coordinating solvents such as DMF or DMSO the spectral profile has changed because of solvolysis, see the lower lines in Fig. 9.

Clearly, complex **1** remains fully delocalized both in noncoordinating solvents and in acetonitrile and acetone. The degree of electron coupling, H_{AB}, in delocalized complexes is given by $1/2\nu_{max} = 4748$ cm^{-1}, where ν_{max} is the energy of the intervalent charge transfer absorption band. Further, because $\nu_{max} = 10B$, where B is the double exchange parameter,[12,19] $B = 950$ cm^{-1} for **1**. A similar influence of the solvent on the intervalent charge transfer absorption has also been observed for **2**.[19] In the case of **2**, the experimental $\Delta\nu_{1/2}$ is 3980 cm^{-1}, the predicted $\Delta\nu_{1/2}$ is 4668 cm^{-1}, H_{AB} is 4717 cm^{-1}, and B is 943 cm^{-1}.

Although the spectra of **1** and **2** indicate that they are both valence-delocalized even in solution, the $\Delta\nu_{1/2}$, H_{AB}, and B values indicate that the extent of delocalization is slightly greater in **1** than in **2**. For the valence-delocalized complex **3**, the B and H_{AB} values obtained from an electronic spectral study are 1319 cm^{-1} and 6596 cm^{-1}, respectively.[19] It is relevant to mention that for **3** the $B = 1300$ cm^{-1} value obtained from a Mössbauer spectral analysis is almost identical to the 1319 cm^{-1} value obtained from the electronic spectral study.[12]

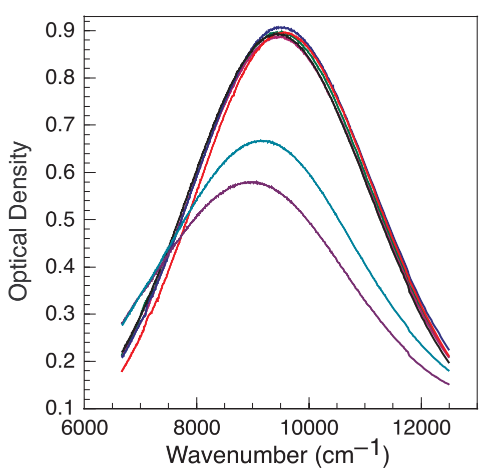

FIG. 9. Intervalent charge transfer absorption band in **1** measured in chloroform, blue; dichloromethane, red; nitrobenzene, black; acetone, green; acetonitrile, magenta; dimethylformamide, cyan; and dimethylsulfoxide, purple. The first five solvents lead to very similar absorption bands.

F. Electrochemical studies

The cyclic voltametric measurements of complex **1** have been carried out in acetonitrile at 298 K in a dinitrogen atmosphere by using platinum as the working electrode. The cyclic voltamogram, obtained between potentials of 0.9 and −0.5 V, is shown in Fig. 10. The one electron reduction $[Fe_2]^V \rightarrow Fe^{II}Fe^{II}$ takes place quasi-reversibly at $E_{1/2} = -0.19$ V with $\Delta E_p = 105$ mV, but, in contrast, the oxidation at $E_{p,a} = 0.70$ V is irreversible. It may be noted that in the cathodic sweep a weak peak is observed at 0.20 V. When the cyclic voltametric measurements are carried out between 0.3 and −0.5 V, prior to the $[Fe_2]^V \rightarrow Fe^{II}Fe^{II}$ reduction at −0.19 V no electrochemical response is observed at 0.20 V, see Fig. S8. In contrast, as is shown in Fig. S9, this wave appears when the measurement is carried out between 0.0 and 1.0 V. Clearly in **1**, following the oxidation of $[Fe_2]^V$ to $Fe^{III}Fe^{III}$, a chemical reaction occurs. It is logical to conclude that the iron(III) containing $[Fe_2L(\mu\text{-OAc})_2]^{2+}$ moiety generated will be highly unstable because of the very close proximity of the two iron(III) ions and, as a result of coulombic repulsion, one of the iron(III) ions will be detached from the macrocyclic ring and the species that remains is then reduced at 0.20 V.

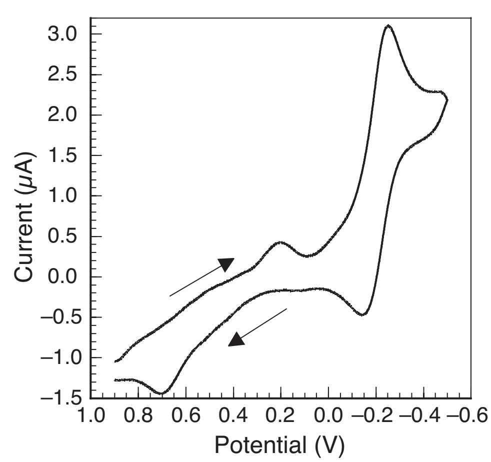

FIG. 10. Cyclic voltamogram of **1** measured in acetonitrile.

IV. CONCLUSIONS

To the best of our knowledge, $[Fe_2L(\mu\text{-OAc})_2]ClO_4$, **1**, is the first example of a fully valence-delocalized $Fe^{2.5+}Fe^{2.5+}$ or $[Fe_2]^V$ binuclear complex in which the electron delocalization takes place between two crystallographically *inequivalent* iron sites. Further, **1** exhibits both a spin 9/2 ground state as a consequence of double exchange and single molecule magnetic behavior in agreement with a negative axial zero-field splitting parameter.

Although from the aspect of connectivity and chemical bonding the two iron sites in **1** seem rather similar, they must be crystallographically inequivalent because there is no symmetry operation that connects the two iron sites. Further, the coordination environments at the two iron cations are sufficiently different that the two sites exhibit significantly different Mössbauer-spectral hyperfine parameters, albeit parameters that are characteristic of a class III fully electron-delocalized mixed valence $[Fe_2]^V$ complex. Between 5.25 and 11 K the two spectral components exhibit slightly different isomer shifts and different quadrupole interactions and hyperfine fields; both sets of parameters are intermediate between those expected of high-spin iron(II) and high-spin iron(III) ions.

The absorption spectrum of **1** exhibits a strong intervalence charge transfer band at 1053 nm, a band that corresponds to an electron transfer parameter, B, of 950 cm^{-1}.

The magnetic properties indicate that **1** is paramagnetic with an $S = 9/2$ ground state, a spin-state that results from a combination of magnetic double exchange coupling through electron delocalization in the $[Fe_2]^V$ binuclear complex. The analysis, between 2 and 300 K, of the ensemble of magnetic properties of **1**, i.e., the temperature dependence of the dc-magnetic susceptibility, the low-temperature field dependence of the dc magnetization, and the variable-frequency ac magnetic susceptibility, yields the following parameters, including the experimental inaccuracies, an isotropic exchange coupling parameter, J, of $-32(2)$ cm^{-1} for a double-exchange parameter, B, of 950 cm^{-1}, a uniaxial zero-field parameter, $D_{9/2}$, of $-0.9(1)$ cm^{-1}, a rhombic zero-field parameter, $|E_{9/2}|$, of $0.1(1)$ cm^{-1}, and $g = 1.95(5)$, and an effective magnetic relaxation barrier, $U_{\rm eff}$, of 9.8 cm^{-1} and an attempt time of relaxation, τ_0, of 4.2×10^{-7} s.

The analysis of the temperature dependence of the average hyperfine field between 5.25 and 30 K yields a $|D_{9/2}| = 0.9$ cm^{-1} in excellent agreement with the magnetic measurements and the analysis of the relaxation frequency of the hyperfine field between 5.25 and 85 K yields an activation energy of 17 cm^{-1}. In view of the D value, a maximum theoretical relaxation barrier of $(S^2 - 1/4)|D| = 18$ cm^{-1} is expected. Hence, both magnetic and Mössbauer spectral measurements give reasonable estimates of the relaxation barrier of the spin 9/2 binuclear complex.

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Synopsis of Past and Current Research

A. Research Interest:

Inorganic and Organometallic Chemistry, X-ray Crystallography, Crystal Engineering and Supramolecular Chemistry, Noncovalent Interactions, Catalytic and Magnetic Studies of Multinuclear (Homo and Hetero) Metal Complexes, Clusters and Metal Organic Frameworks.

B. Glossary of Research Work:

Syntheses under open and inert atmosphere, structure determination by XRD, analyses of molecular and supramolecular structures (crystal engineering), and investigating important properties such as magnetic, catalytic, electrochemical, and spectroscopic of inorganic and organometallic complexes/compounds have always been my favorite exploring area since beginning of PhD.

Synthesizing versatile metal complexes under variable conditions has been my expertise. Most of my works are related to the Schiff base metal compounds. Designing targeted mono to polymeric system containing one (monometallic) or more metal ions (heterometallic) have been my skill. I have worked with many types of metal ions such as s, d, p and f-block metal ions (see publication list). Synthesis skills are evident from the isolation and characterization of mixed valence species (*e.g.*, $Fe^{III}Fe^{II}$, $Co^{III}Co^{II}$, $Cu^{II}Cu^{I}$ and $Sn^{II}Sn^{IV}$) and unusual heterometallic species ($Cu^{II}Na^{I}$, $Cu^{II}K^{I}$, $Cu^{II}U^{VI}$, $Fe^{III}Ni^{II}$, $Cu^{II}Sn^{II}$, $Cu^{II}Sn^{IV}$, $Co^{III}Sn^{IV}$, $Co^{III}Dy^{III}$ *etc*.). Some of which have been investigated in magnetism and catalysis, resulting in several important results (see below).

C. Noticeable/Important Outcomes

(i) Published as research articles

1. $Cu^{II}\cdots Cl$ semicoordination bond in supramolecular heterometallic $\{Cu^{II}\}\{Sn^{IV}\}$ cocrystals (*Organometallics* 2023, *42*, 2672–2683).
2. Ion-pair-assisted tetrel bonds in heterometallic $\{Ni^{II}Sn^{II}\}\{Sn^{IV}\}$ and $\{Ni^{II}Sn^{II}\}\{Sn^{II}\}$ complex salts (*Cryst. Growth Des.*, 2022, *22*, 341–355).
3. Field-supported single molecule magnetism (SMM) in mixed valence $Co^{II}Co^{III}_2$ species (*Molecules*, 2021, *26*, 1060).
4. Flexibility and lability of a phenyl ligand in hetero-organometallic 3d metal-Sn(IV) Compounds (*Dalton Trans.*, 2017, *46*, 13364–13375).
5. Application of benzene-sulfonate and benzene-carboxylate copper polymers in cyclohexane oxidation (*Dalton Trans.*, 2016, *45*, 13957–13968).
6. Structural diversity of heterometallic copper(II)/nickel(II)–tin(II/IV) compounds (*Cryst. Growth Des.*, 2016, *16*, 3777, *Dalton Trans.*, 2017, *46*, 13364 and *Dalton Trans.*, 2016, *45*, 17929).
7. Example of unusual pair of phenoxido and acetato bridged heteronuclear $\{Co^{III}Dy^{III}\}$ single molecule magnets with two slow relaxation branches (*Dalton Trans.*, 2016, *45*, 7510).
8. Introduction of sulfonated Schiff Base metal complexes (Cu^{2+} and Sn^{4+}) as potential anticancer agents (*J. Inorg. Biochem.*, 2017, *174*, 25–36 and *J. Inorg. Biochem.*, 2016, *162*, 83).
9. Catalytic application of several metal organic frameworks (MOFs) in various important organic reactions (*CrystEngComm*, 2016, *18*, 1337–1349, *Cryst. Growth Des.*, 2016, *16*, 1837 and *Cryst. Growth Des.*, 2015, *15*, 4185).
10. Two rare $Fe^{III}Ni^{II}$ heterometallic complexes of contrasting magnetic properties (*Inorg. Chem.*, 2013, *52*, 12881).
11. First example of a fully delocalized $[Fe_2]^{V}$ system in spite the presence of two crystallographically inequivalent iron centers (*J. Chem. Phys.*, 2011, *134*, 174507).
11. First example of a single molecule magnet (SMM) resulted from double exchange (*J. Chem. Phys.*, 2011, *134*, 174507).
12. A new type of hydrate isomerism in coordination chemistry (*Eur. J. Inorg. Chem.*, 2009, 4887).

13. Structural resemblance of sodium(I), lithium(I) and magnesium(II) with 3d metal ions in terms of the similarity in cocrystallization process (*Cryst. Growth Des*., 2009, *9*, 3603 and *CrystEngComm*, 2010, *12*, 4131).
14. Rare examples of cocrystallized double-decker-triple-decker topology (*Cryst. Growth Des*., 2009, *9*, 3603 and *CrystEngComm*, 2010, *12*, 4131).
15. An interesting example of [2×1+1×4] cocrystal of 3d and 5f metal ions resulted from C–H···π and C–H···O interactions (*CrystEngComm*, 2010, *12*, 470).

(ii) Published as review articles

1. Copper tetrazole compounds: Structures, properties and applications (*Coord. Chem. Rev*., 2024, *504*, 215604).
2. Urea and thiourea based coordination polymers and metal-organic frameworks: Synthesis, structure and applications (*Coord. Chem. Rev.*, 2022, *453*, 214314).
3. Platinum and palladium complexes with tetrazole ligands: Synthesis, structure and applications (*Coord. Chem. Rev.*, 2021, *446*, 214132).
4. Biological Evaluation of Azo-and Imino-Based Carboxylate Triphenyltin(IV) Compounds (*Eur. J. Inorg. Chem.*, 2020, *2020*, 930–941).
5. Metal–Tin Derivatives of Compartmental Schiff Bases: Synthesis, Structure and Application (*Coord. Chem. Rev.*, 2019, *395*, 1–24).

Future Research Plan

Title: *OxyGenius: Investigating Oxygen Binding, Activation, Reduction, and Evolution through H_2O Oxidation*

Plan: Oxygen is the highest abundant element on the earth, composing about 49.5% of the total mass of the Earth's crust, waters and atmosphere. Most living beings (*e.g.*, plants, animals, and fungus) need oxygen for cellular respiration, which extracts energy by breaking down food. Most of the mass of living organisms is oxygen as a component of water, the major constituent of lifeforms. It is unnecessary to emphasize that oxygen is one of the most essential and primary elements for the survival of our living world.

Molecular oxygen (O_2) is a ground-state triplet with two unpaired electrons distributed in two doubly degenerate π^* HOMOs. It is thermodynamically a powerful oxidant but due to spin conservation, its reaction with ground-state singlet molecules is not kinetically favourable. In contrast to its unfavourable one-electron reduction to superoxide (O_2^-), the formation of peroxide (O_2^{2-}) either by two-electron reduction of O_2 or by one-electron reduction of superoxide (O_2^-) are thermodynamically favourable. These reduction processes, involving the addition of electrons to the antibonding π^* orbitals, lead to the elongation of the O–O bond.

In biological systems, the function of oxygen transport is carried out by some enzymes called oxygenases. So far, two types of oxygenases are recognized: (i) monooxygenases or mixed function oxidase, which transfer one oxygen atom to the substrate, and reduce the other oxygen atom to water, and (ii) dioxygenases or oxygen transferases, which incorporate both atoms of molecular oxygen (O_2) into the product(s) of the reaction. Most oxygenases contain either a metal, usually iron, or an organic cofactor, usually flavin. Metalloprotein molecules, such as myoglobin, hemoglobin, hemerythrin, and hemocyanin, bind and transport molecular oxygen (O_2). The oxygen binding properties of these bioproteins have inspired researchers to synthesize new species (*e.g.*, metal complexes) which can manipulate such an important phenomenon. Investigations revealed that several transition metals are capable to bind O_2, forming dioxygen complexes, and some of them can bind it reversibly. Metal oxo species can be generated from the O–O bond cleavage after complexation whereas the metal-catalyzed reduction of the coordinated O_2 can lead to the formation of metal hydroperoxo compounds.

Not only in the cellular respiration, the O_2 binding is also important in corrosion processes as well as in industrial chemistry. The reversible binding of O_2 to metal complexes has been applied to purify oxygen from air. Investigations and investments in the binding and activation of O_2 can lead to the advancements of bio-related processes, such as O_2 delivery and O_2 storage.

The reduction of O_2 by metal catalysts is a key half-reaction in fuel cells, being one of the major important current advancing fields. The oxygen reduction reaction (**ORR**) in a typical fuel cell generates environment-friendly products such as water ($O_2 + 4H^+ + 4e^- = 2H_2O$) or peroxide ($O_2 + 2H^+ + 2e^- = H_2O_2$). As compared to a thermal engine, the advantages of fuel cells are high efficiency, no environmental pollution, and unlimited sources of reactants. Among the existing fuel cells, the proton exchange membrane fuel cell (PEMFC) has been developed for use in vehicles, portable electronics, and combined heat and power (CHP) systems due to its simplicity, low working temperature, high power density, and quick start-up. At the anode of a PEMFC, H_2 is oxidized (**HOR**) to produce electrons and protons ($H_2 \rightarrow 2H^+ + 2e^-$) that are transferred through an external circuit and the proton exchange membrane, respectively to the cathode where oxygen is reduced (**ORR**) by them to produce water. However, both the anode and the cathode electrodes consist of highly dispersed Pt-based nanoparticles on carbon black to promote the rates of the **HOR** and **ORR**. However, Pt is a scarce and expensive metal, therefore, replacing it with an abundant and cheap metal would be advantageous. Thus, extensive investigations on oxygen reduction reaction (**ORR**) with cheaper metals (particularly, redox active ones, V, Mn, Fe, Co or Cu) would bring innovations towards solving the current energy related issues.

Similarly, the reverse reaction of the O_2 reduction, *i.e.*, the H_2O oxidation (**WOR**) or H_2O splitting ($2H_2O = 2H_2 + O_2$), producing pure H_2 and O_2, is another important and essential reaction in many technological applications including fuel cells, solar energy production, and catalysis. It has been a research focus for its key role in solving the energy crisis and environment problem. **Electrochemical water splitting**, being a promising approach as it is sustainable and pollution-free, occurs in a water electrolyzer consisting of a hydrogen evolution reaction (**HER**) cathode and an oxygen evolution reaction (**OER**) anode. However, due to the distinctly different catalytic mechanisms, active **HER** catalysts are often found to be poor **OER** catalysts, and vice versa. Another major obstruction in achieving large-scale production of H_2 and O_2 *via* the latter splitting method is the expense and the limited efficiency of current catalysts. Hence the development of a universally active water-splitting catalyst based on earth-abundant materials is of key interest and will be a significant innovation.

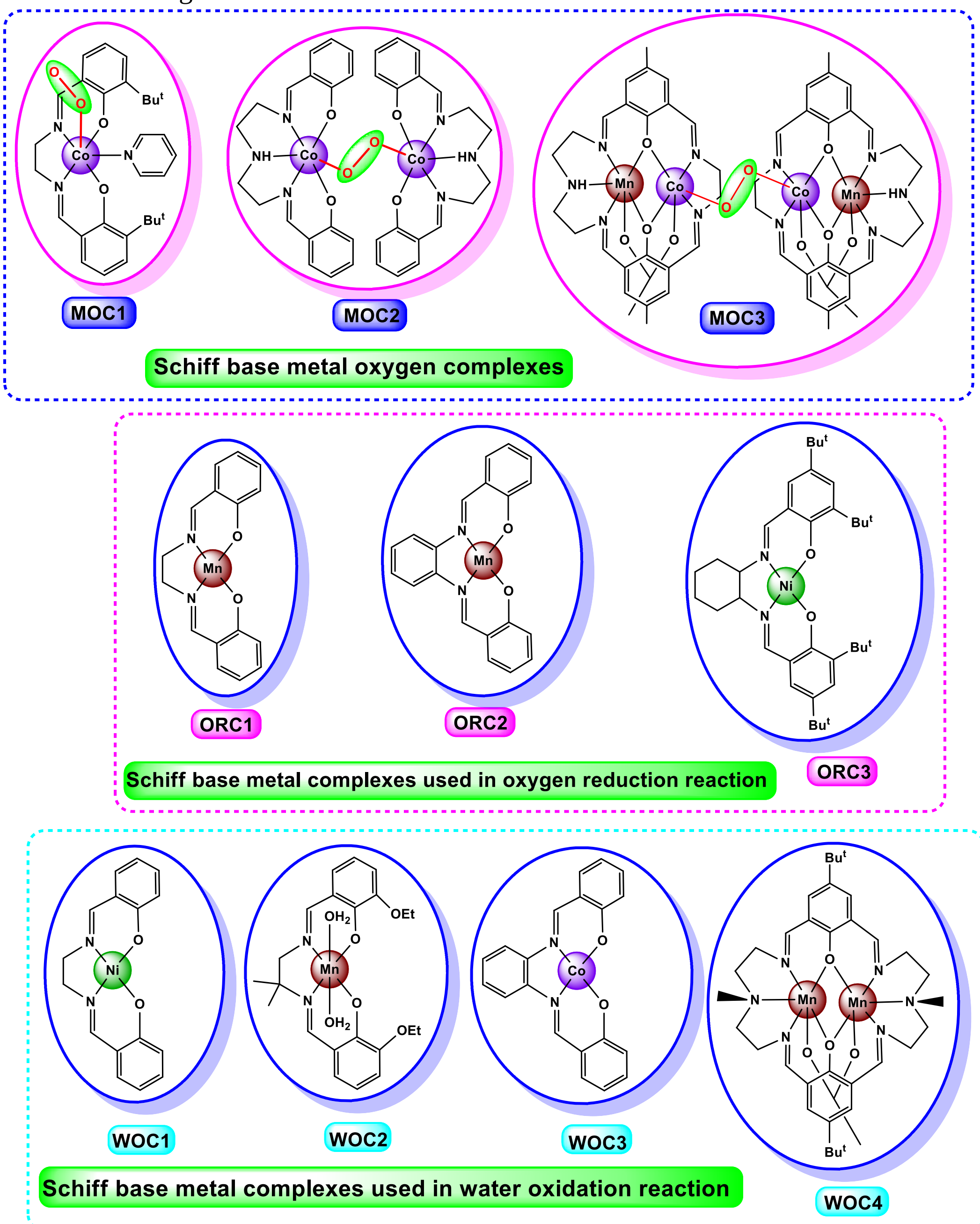

Figure 1. Examples of structurally characterized compartmental Schiff base metal oxygen complexes (**MOC1−3**) and such complexes used in oxygen reduction (**ORR**) or water oxidation (**WOR**) reactions. **MOC** = metal oxygen complex, **ORC** = oxygen reduction complex and **WOC** = water oxidation complex.

On other hand, there have been an extensive growth in the coordination chemistry of salen type Schiff base metal complexes with their significant applications in advanced chemistry such as sorption, optics, conductivity, molecular magnetism, catalysis and biology including O_2 binding and activation.

Salen-analogues (compartmental Schiff bases) coordinate a transition metal ion through imine-N and phenolic-O atoms, (Figure 1), to generate very stable metal complexes. Such complexes can also offer an excellent possibility of binding O_2. However, only a few of them were investigated with oxygen binding and even fewer (*e.g*., **MOC1-3**, Figure 1) were structurally identified. In addition, some of such complexes (*e.g.*, **ORC1-3**, Figure 1) have been investigated in **ORR** and some (*e.g.*, **WOC1-4**, Figure 1) were investigated as catalysts for electrolytic **WOR**.

Industrial, biological and environmental significances of O_2 binding/activation, **ORR** and **WOR**, associated to my expertise in designing/studying salen-type Schiff base metal complexes (both acyclic and macrocyclic) and in catalytic oxidation studies have inspired me to propose such a plan/project, with the aim to make a significant contribution towards finding solutions to the current precarious environmental issues and energy related challenges.

Experimentally, this plan will deal mainly with (*i*) the synthesis of homo or heterometallic systems (of transition metals, mainly redox active ones, V, Mn, Fe, Co or Cu) of few suitable compartmental Schiff bases, (*ii*) investigation approaches to explore the capabilities of the synthesized metal complexes towards O_2 binding/activation and **ORR** and (*iii*) exploring the catalytic capabilities of the designed compounds in **WOR/OER**.

Teaching Interests and Philosophy

A. Subjects of Interests for Teaching (Non-practical): Basic and Advanced Inorganic Chemistry including Radioactivity and Atomic Structure, Chemical Periodicity, Chemical Bonding and Structure, Acid-Base reactions, Bonding and Physical Properties, Chemistry of s-, p-, d- and f-block Elements, Precipitation and Redox Reactions, Organometallics Chemistry, Electrochemistry, Bioinorganic Chemistry, Magneto Chemistry and Other Advanced Chemistry Subjects including Catalysis (Homogeneous and Heterogeneous), Metal-Organic Frameworks and Noncovalent Interactions. Basic Organic Chemistry (e.g., Stereochemistry, Organic Reaction Mechanism, Spectroscopy (UV-vis, Fluorescence, IR, NMR, EPR, Mössbauer, Mass, Rotational, Vibrational and Raman), Principles of Electrochemical and Spectral Analysis, and Analytical Separation etc.)

B. Subjects of Laboratory Teaching (Practical): Acid-Base Titrations, Qualitative and Quantitative Inorganic Syntheses and Characterizations (Analyses), Analytical and Instrumental Estimations, Qualitative Analysis of Organic Compound, Detection of Different Organic Functional Groups, Organic Syntheses, Purifications and Characterizations and Reactions and other Advanced Techniques.

C. Interest of Training Instruments: Advanced instruments such as single crystal and powder XRD, GC, NMR, CV, FT-IR, UV, Fluorescence Spectrometer etc.

D. Teaching Methodology (Philosophy)

1. Commitment to Pupils and Pupil Learning

i) Giving consistently values the individuality of learners and always works towards providing meaningful and relevant experiences to promote and enhance students' learning.
ii) Working for extensive understanding and appreciation of diversity and equity related to student-teacher interactions and student-student interactions.
iii) Providing sophisticated challenges and conceptual frameworks to encourage students to engage in divergent thinking.
iv) Providing modifications, accommodations and alternative experiences based on student strengths and needs.

2. Leadership and Community

i) Engaging with ease in professional conversations to learn with, and from, my Associate Teacher, colleagues, students, and others in the school learning community.
ii) Assume diligently my professional role and duties as defined.

3. Ongoing Professional Learning

Consistently taking the initiative to familiarize myself with current programs, technologies, and instructional practices to enhance student learning.

4. Knowledge (Planning, Implementing, Differentiating and Assessing)

i) Setting up an excellent classroom environment and selecting appropriate resources.

ii) Describing what the students will know and be able to do by selecting appropriate expectations and refining them where necessary.

iii) Including an introduction that highly motivates learners; teaching/learning strategies are varied and clearly support the development of the content.

iv) Content of teaching will be extensively detailed while concepts, facts and skills will be logically sequenced.

v) Relating assessment directly to the expectation(s) with mastery.

vi) Assessing independently, and in-depth, learners' prior learning experiences and needs.

vii) Creating a succinct consolidation that reviews the content developed in the lesson and selecting an application task which allows students to apply the content with ease.

viii) Motivating students through an introductory activity and using teaching/learning strategies to stimulate a high degree of active student involvement.

ix) Where appropriate integrates technology seamlessly and strategic use of resources, for examples, a) films, website, articles, etc.; b) technology e.g., overhead, data projector, computer, SMART board, lab equipment, etc. (if applicable); c) handouts (worksheets, templates, rubrics, checklists, etc.).

x) Where appropriate consolidation of learning insightfully clarifies student understanding of the content; students engage with understanding in application task with ease.

xi) Employing a variety of assessment strategies to improve student learning; using appropriate recording device to monitor and track student learning.

5. Management & Communication Practices

i) Using superior communication practices to enhance the delivery of the lesson.

ii) Engaging expertly with students through using a variety of questioning techniques which encourage a range of levels of thinking.

iii) Consistently promoting a risk-free learning environment which encourages an elevated level of pupil participation and responsibility (student/teacher and student/student).

iv) Demonstrating an excellent understanding of classroom management strategies and consistently applying these strategies to attain the school's expectations.

v) Persistently recognizing and reinforcing appropriate behaviors.